

A Geometric–Fractal Tetrahedral Model of Matter and Space: From Atomic Bonds to Cosmic Structures

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Abstract

We propose a minimal geometric model of space–time—the *Methane Metaverse* (MMU)—in which physics emerges from a tetrahedral lattice with four orthogonal degrees (w_1, w_2, w_3, w_4). The single rule is observational: an external observer measures only the projection onto w_1 ; energy stored in (w_2, w_3, w_4) lowers this projection and appears as time dilation. From this, Maxwell transport along lattice edges (light at speed c and a fixed edge impedance Z_0), continuum elasticity consistent with GR, and quantum envelope equations follow without introducing new free scales.

We demonstrate: (i) internal spring constants and the 21 cm hyperfine scale from a one-step estimate (Appendix Appendix 2); (ii) a clean GR→Newton limit via the projection/clock-rate factor $\zeta = d\tau/dt$ (Appendix Appendix 6); (iii) light as a pure w_2 – w_3 oscillation with $Z_{0,\text{int}} = (P_3/P_2)/\sqrt{2\alpha}$ and its SI calibration (Appendix Appendix 3); (iv) Schrödinger as the one-band and Dirac as the two-band projection of internal oscillations (Appendix Appendix 8), including a geometric derivation of Laguerre radial modes from the proton’s hexagonal “chair” (Appendix Appendix 7); (v) isotope shifts (H/D/T) and the muonic scaling pattern, plus decay as S -node barrier flips with Eyring–Kramers statistics (Appendix Appendix 11); (vi) quark triads (fractional charge, color, confinement) and the neutron as either (u, d, d) or a $(p+e^-)_S$ composite with the same energy bookkeeping (Appendix Appendix 10); and (vii) lepton staging (e, μ, τ), Koide as a geometry identity, hydrogenic lines across $e \rightarrow \mu \rightarrow \tau$, and parameter-free ($g-2$) family ratios (??).

The model yields compact formulas, parameter transparency, and multiple falsifiable checks (impedance, mode speeds, spectral ratios). The appendices contain full derivations; the main text keeps the logic simple enough for a broad physics readership.

1 Introduction

Why another microscopic picture? General Relativity and quantum theory excel in their domains but meet awkwardly at small scales and high energies. The MMU aims *not* to replace either theory, but to supply a common geometric substrate that makes their core structures natural: a tetrahedral lattice with four orthogonal axes (w_1, w_2, w_3, w_4) and one observational principle—“you only ever measure the w_1 projection.” Internal energy stored in (w_2, w_3, w_4) reduces that projection, which an external observer reads as a slower clock. This single mechanism unifies kinematic and “gravitational” dilation.

What the observer actually sees. Edge transport on the lattice locks w_2 and w_3 orthogonally, reproducing Maxwell’s equations along an edge with phase velocity c and an intrinsic wave impedance determined by geometry and α (Appendix Appendix 3). In the slow/weak limit, the w_1 -projection yields the familiar Newtonian bookkeeping: $E_{\text{kin}} = \frac{1}{2}mv^2$ and an inverse-square potential from locality and isotropy (Appendix Appendix 6). In the strong/rapid limit, the same projection rule recovers relativistic time dilation and a GR-consistent continuum stiffness (Appendix Appendix 1).

What we *derive* with almost no knobs to turn.

- **Electromagnetism from geometry.** Light is a massless w_2 – w_3 oscillation; its amplitude ratio defines a scale-free edge impedance $Z_{0,\text{int}} = (P_3/P_2)/\sqrt{2\alpha}$ that SI-calibrates to the vacuum impedance (Appendix Appendix 3).
- **Quantum envelopes.** Projecting fast internal modes yields Schrödinger (one band) and Dirac (two bands) with minimal coupling emerging from edge phases (Appendix Appendix 8); the hydrogenic Laguerre radial functions arise from the proton’s hexagonal microstructure (Appendix Appendix 7).
- **Atomic and nuclear scales.** Internal spring constants fix the 21 cm line at the right scale without QED machinery (Appendix Appendix 2); isotope shifts across H/D/T fall out with the correct mass and finite-size hierarchy, and muonic systems scale with μ^3 as expected (??).
- **Decay and stability.** S -node flips produce exponential decay with an Eyring–Kramers rate; a two-point fit (neutron, tritium) yields a simple $k(Q) = k_0 e^{\beta Q}$ law and explains the stability of ^1H and ^2H (??).
- **Hadronic and leptonic structure.** A triad at an S -node gives fractional charges, color, and linear confinement; the neutron appears equivalently as (u, d, d) or a $(p+e^-)_S$ composite with consistent energy accounting (Appendix Appendix 10). The same geometry stages (e, μ, τ) , proving Koide and predicting hydrogenic lines and $(g-2)$ family ratios without extra parameters (??).

Who should read this and why. For the *physicist*: the model is falsifiable. It predicts a specific internal impedance, fixed node reflection factors, mode speeds, and parameter-free spectral ratios; the appendices supply full algebra and unit-checked numerics. For the *interested reader*: the main text shows how far simple geometry and a single projection rule can go, with diagrams and minimal formulas; technical proofs live in the appendices referenced above.

Information-Theoretic Framing

In the MMU, only the w_1 -projection is observable. This projection acts as an information bottleneck: (i) the clock-rate factor $\zeta = d\tau/dt$ is the geometric compression of internal (w_2, w_3, w_4) dynamics into the observed time axis, lowering the effective information rate; (ii) the edge impedance Z_0 controls the ratio of stored vs. transported energy and thus the “impedance” of information flow along edges; (iii) quantized internal modes define discrete codebooks whose w_1 -projections appear as atomic spectral lines. This ties the quantitative

MMU predictions (21 cm, isotope and Lamb shifts, $(g-2)$ ratios) to information flow under geometric constraints, aligning the work with the scope of *Entropy*.

2 The $w_1w_2w_3w_4$ System

The Methane Metauniverse describes physical space as an elastic tetrahedral lattice whose nodes possess four orthogonal degrees of freedom. These are denoted by the axes w_1 , w_2 , w_3 , and w_4 . They are not identical to the Cartesian coordinates of three-dimensional space, but form an extended vector space in which both classical and quantum-theoretical effects can be represented geometrically.

Physical Meaning of the Axes

- w_1 – **measurable projection:** This axis represents the component perceived as the passage of time or measurable energy in physical space. All physical quantities that we measure are projections of the state vectors onto w_1 .
- w_2 – **electric/energetic displacement:** w_2 describes the displacement of a node in terms of an electric or general energy displacement. In atomic systems, this axis represents the electric component of an electromagnetic wave.
- w_3 – **magnetic/spin-induced displacement:** Orthogonal to w_2 , w_3 describes the magnetic or spin-induced displacement. It is phase-shifted by 90° relative to w_2 and together they form the plane of the electromagnetic oscillation.
- w_4 – **torsional mass anchor:** This axis is perpendicular to the w_2 – w_3 plane and represents the torsion or mass center of the system. w_4 is crucial for coupling internal oscillations to mass and gravitational effects.

Projection onto w_1

In the MMU, every physical state is described by a vector in the $w_1w_2w_3w_4$ space. The directly measurable quantities arise by projecting this vector onto w_1 . The length of this projection depends on the magnitude of the components in w_2 , w_3 , and w_4 . The larger the displacement in these orthogonal axes, the smaller the projection onto w_1 – a geometric mechanism that in the model describes time dilation and energy reduction.

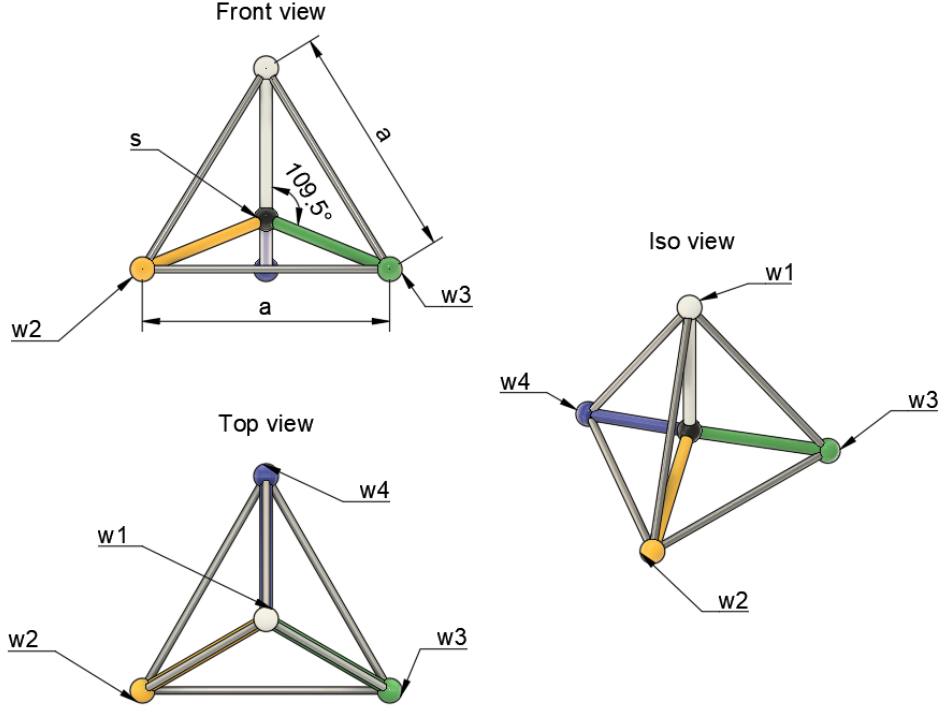


Figure 1: Base tetrahedron of the MMU with the four orthogonal axes w_1 , w_2 , w_3 , and w_4 . **Front view:** vertical arrangement with w_1 pointing upward. **Top view:** projection onto the w_2 – w_3 plane, showing w_4 to the left. **Iso view:** spatial perspective with all axes visible. Inside the tetrahedron the axes enclose the characteristic angle of 109.5° , identical to the bond geometry of the methane molecule. Axis colors: yellow = w_2 , green = w_3 , blue = w_4 , white = w_1 . The central point s marks the tetrahedron node, while the edge length a sets the natural geometric scale of the system.

3 The Counter-Tetrahedron and Its Stabilizing Effect

In the Methane Metauniverse, each fundamental unit of the space lattice consists not only of a tetrahedron but of a *tetrahedron-counter-tetrahedron pair*. The counter-tetrahedron is geometrically mirrored to the base structure and oriented in the opposite direction. Its vertices are positioned so that they touch the midpoints of the faces of the original tetrahedron.

Role of the Counter-Tetrahedron

The counter-tetrahedron fulfills two key functions:

1. **Mechanical stabilization:** It acts as a counterweight, distributing the forces acting on the node evenly in all directions. Any movement of a node in the tetrahedron is *immediately* mirrored by a corresponding movement in the counter-tetrahedron. This response occurs directly via the shared node connecting both structures, so oscillations always occur as paired motions. This prevents local displacements from distorting or destabilizing the lattice.

2. **Symmetry preservation:** Due to its mirrored arrangement, the counter-tetrahedron ensures that the lattice has the same geometric stiffness in all directions. Any torsion or oscillation is executed synchronously but oppositely in both structures, resulting in a perfect balance of spatial forces. This is a prerequisite for the isotropic properties of physical space in the MMU.

Fixed Nodes in the Lattice

The MMU assumes a *fixed space lattice* whose nodes are anchored in place. Physical processes – such as light propagation or particle motion – arise from oscillations and torsions along the edges, not from movement of the nodes themselves. This perspective fundamentally differs from relativity theory, in which space is treated as a smooth, deformable continuum.

Time and Anti-Time

In the MMU, *time* is not treated as an absolute dimension but as the result of projecting a state vector onto the w_1 axis. This projection is determined by the geometry of the tetrahedron and by the oscillatory state of its central node.

The *node* itself plays a special role: it is the reference point where the projection onto w_1 vanishes, i.e. where time is locally zero. From this timeless point, oscillations extend outward along the four axes. Later, this property allows us to explain how two distant regions of the lattice can remain correlated: since both share the same zero-time reference at their nodes, entanglement over large distances arises naturally.

Anti-time originates from the mirrored orientation of the counter-tetrahedron. While the “normal” tetrahedron produces projections onto $+w_1$, the counter-tetrahedron produces projections onto $-w_1$. In a perfectly symmetric lattice, these contributions cancel on average, so that only the differences are measurable. This construction provides a natural geometric explanation for the observed arrow of time.

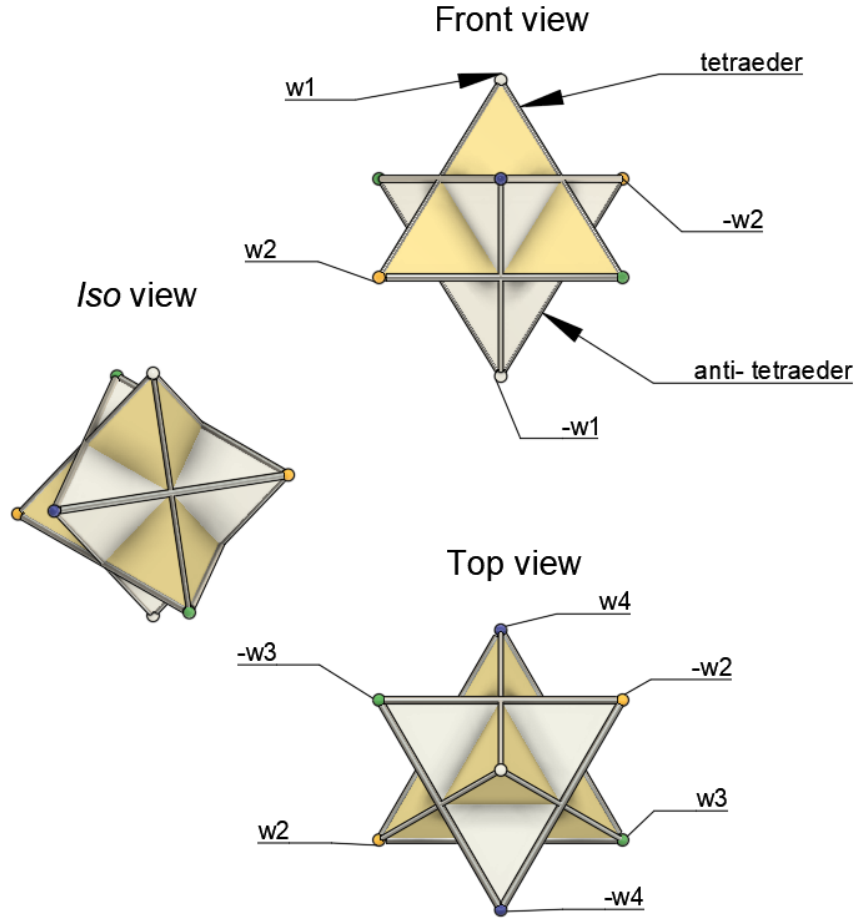


Figure 2: **Tetrahedron-counter-tetrahedron pair** in the MMU. Yellow tetrahedron and mirror-oriented gray counter-tetrahedron share the same central node and all edge lengths. Axes w_1 (white), w_2 (yellow), w_3 (green), and w_4 (blue) mark orthogonal degrees of freedom. **Front view:** w_1 vertical, projection onto $+w_1$ or $-w_1$. **Top view:** view along w_1 onto the w_2 - w_3 plane, w_4 perpendicular. **Iso view:** interlocking (stella octangula) illustrates the mirrored orientation.

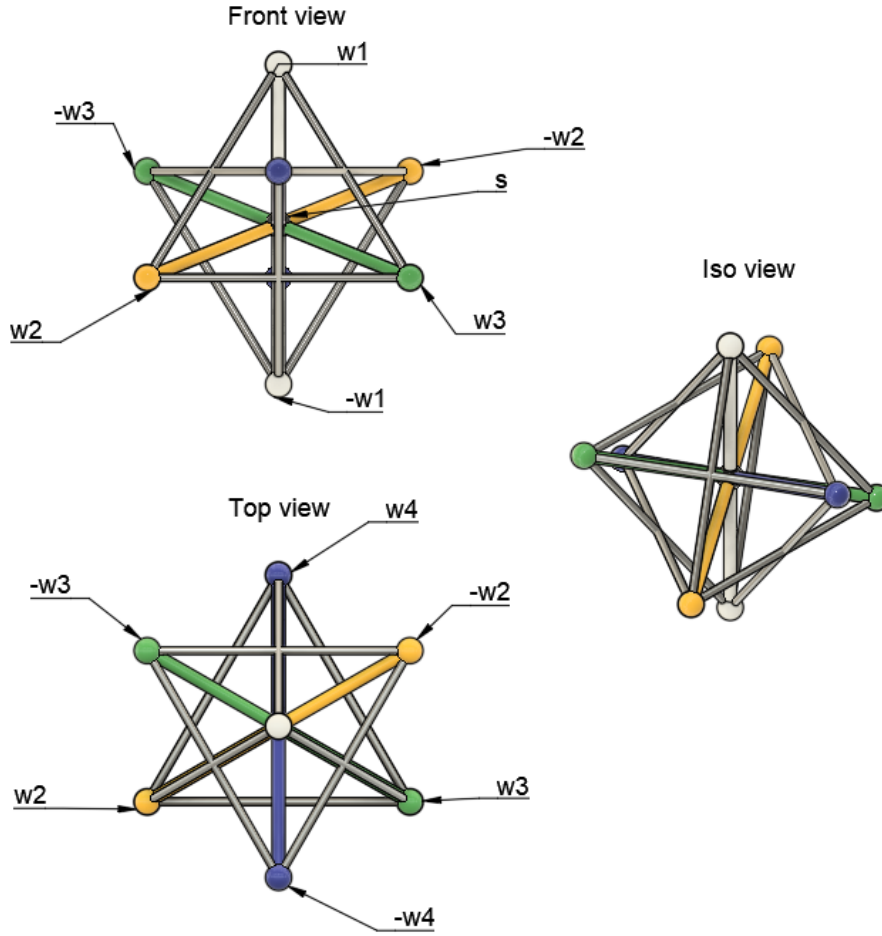


Figure 3: Representation of a tetrahedron-counter-tetrahedron pair in the MMU with the four axes w_1 , w_2 , w_3 , w_4 and their corresponding mirrored axes $-w_1$, $-w_2$, $-w_3$, $-w_4$. **Front view:** shows the vertical arrangement with w_1 at the top and $-w_1$ at the bottom. The node is marked as s . **Top view:** view from above onto the w_2 - w_3 plane and the mirrored axes $-w_2$, $-w_3$. **ISO view:** spatial perspective showing both tetrahedra as interlocked structures. The opposite orientation of the counter-tetrahedron produces projections onto $-w_1$ (anti-time), which in the MMU is responsible for symmetric stabilization and the preservation of spatial isotropy.

4 Fractal Structure of the MMU

The Methane Metauniverse assumes that space is not homogeneous and continuous, but is composed of discrete, geometrically precisely defined units: tetrahedron-counter-tetrahedron pairs, repeated across all scales. This self-similarity – the *fractal character* – is a central structural principle of the model.

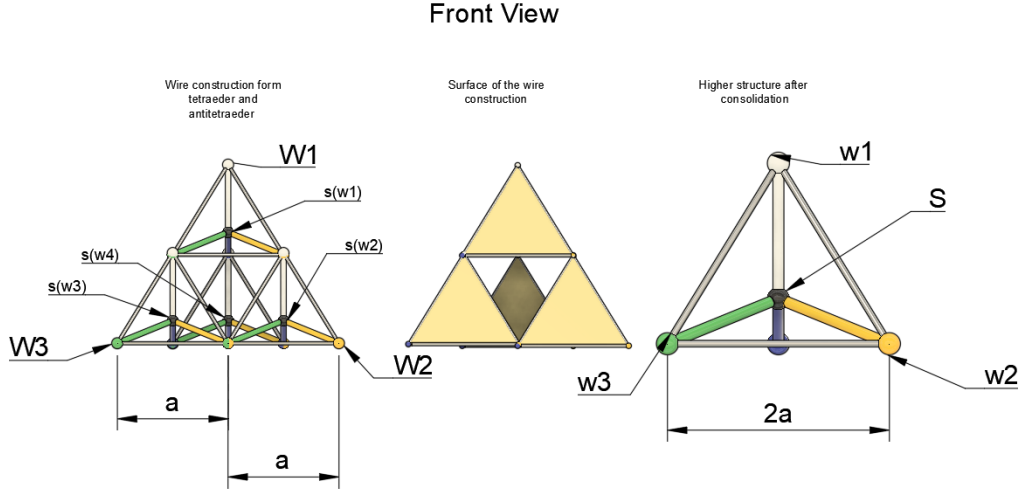


Figure 4: Formation of a higher-level tetrahedron in the MMU. **Left:** wireframe combination of four tetrahedra, forming the internal lattice scaffold with base length a . The four surfaces $s(w_1)$, $s(w_2)$, $s(w_3)$, and $s(w_4)$ emerge as characteristic planes of the construction. **Center:** projection onto the w_2 - w_3 plane, showing the triangular subdivision generated by these surfaces. **Right:** consolidation into a larger tetrahedron with doubled edge length $2a$, in which the four surfaces intersect to define a *new effective node* S . This new node S represents the structural continuation of the lattice at the next hierarchical level. Axis colors: yellow = w_2 , green = w_3 , blue = w_4 , white = w_1 . This figure illustrates how local tetrahedral surfaces merge into a new node, establishing the recursive growth principle of the MMU lattice.

Hierarchical Nesting

Each spatial unit forms the basis for larger structural levels:

- **Planck scale:** smallest lattice units, with edge lengths equal to the Planck length ℓ_P .
- **Atomic scale:** several base cells connect to form stable clusters with atomic properties.
- **Macroscopic scale:** the same geometric rules determine the arrangement of matter in crystals, molecules, and planetary structures.
- **Cosmological scale:** on the largest scales, galaxy clusters appear as projections of the fractal tetrahedral architecture.

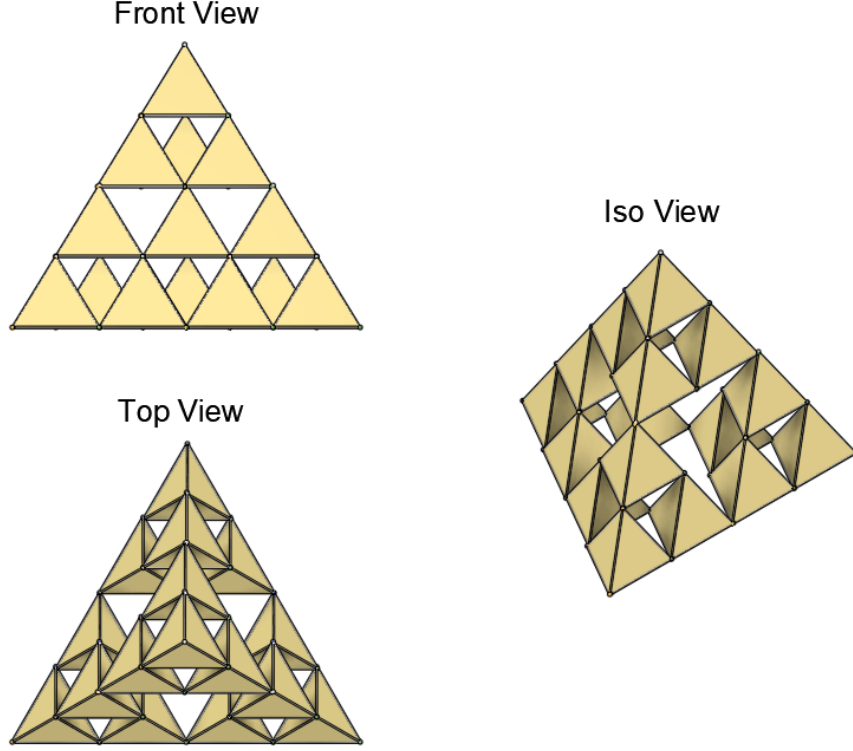


Figure 5: Fractal tetrahedral lattice in the MMU. **Front view:** hierarchical subdivision of the base tetrahedron into smaller self-similar units. **Top view:** projection showing the nested arrangement of sub-tetrahedra in the w_2 – w_3 plane. **Iso view:** three-dimensional perspective highlighting the recursive self-similarity of the structure. This fractal growth principle expresses the MMU assumption that identical geometric rules apply at all scales, from the quantum to the cosmological domain.

Matter–Antimatter Balance

At each fractal level, matter and antimatter are distributed so that their geometric and physical effects cancel on average. Matter and antimatter appear as mirror-image tetrahedra whose motions – as described in the previous chapter – are directly coupled via the shared node. This paired structure ensures that the lattice preserves its shape and symmetry even when excited by external forces.

Communication via Nodes

The fractal structure is not only a geometric nesting but also a *dynamic coupling*. Every movement, torsion, or oscillation at a node is instantly transmitted to the next fractal level via its connections. This creates a kind of *resonance network* in which local processes can influence all scales – from quantum fluctuations to large-scale gravitational patterns.

This direct and immediate coupling between two nodes or between nodes in different fractal levels can manifest in physical space as *entanglement*. In the MMU, entanglement is defined as:

A state in which two or more nodes of the space lattice are coupled through shared geometric anchoring and synchronous projection onto w_1 , so that a

change in the state vector at one node immediately produces a correlated change at the other nodes – regardless of spatial separation.

This definition emphasizes that in the MMU, entanglement is not understood as “spooky action at a distance,” but as a direct consequence of the fixed lattice structure, in which all nodes are mechanically and geometrically connected through the fractal arrangement.

4.1 The Origin of Time Vectors in the Node

In the MMU model, the origin of all time vectors lies at the exact geometric center of the node. From here, the two opposite time directions extend as projections onto the $+w_1$ and $-w_1$ axes. At this point itself, the projected time is *zero*, since the two directions exactly cancel each other.

Physically, this means that the node is a region of *temporal symmetry*, in which no directed time evolution is measurable. This state makes it possible for information or quantum-coupled states to be projected into opposite time directions without any local time difference arising.

From this property arises a natural explanation for *fractal entanglement* in the MMU: Since in every node the time difference vanishes, states can be coupled regardless of their spatial separation as if they were located at the same point. This could be the basis for entanglements remaining stable across cosmological distances and for the symmetry between matter and antimatter tetrahedra being preserved even in widely separated regions of space.

Isotropy and Stability

The fractal structure ensures that space has the same properties in all directions (isotropy), and that disturbances are not confined locally but instead spread and dampen throughout the entire network. The stability of the space lattice is therefore a direct consequence of the fractal symmetry.

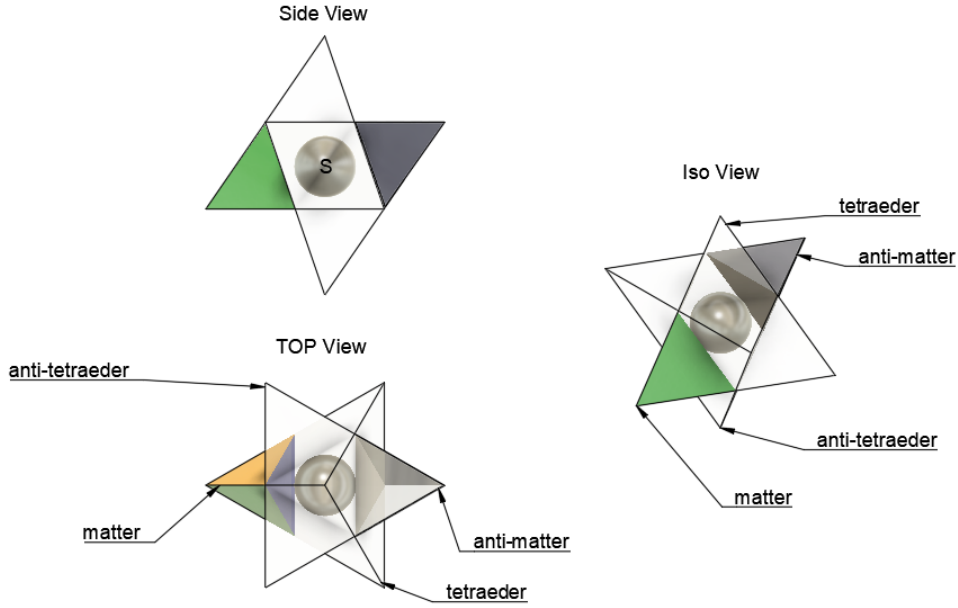


Figure 6: Representation of a matter–antimatter pair in the MMU. **Side view:** view of the central node S with a matter tetrahedron (colored) and a mirror-oriented antimatter tetrahedron (gray) on the opposite side. **Top view:** top-down view of the nested structure, where matter and antimatter tetrahedra are directly coupled via the shared node. **Iso view:** spatial representation of the two tetrahedra with marked faces, illustrating their orientation and axis relationships. Any movement or torsion of a node in the matter tetrahedron is immediately mirrored and synchronously executed in the antimatter tetrahedron – communicated through the shared node.

5 The Fixed Space Lattice of the MMU

The Methane Metauniverse is based on a *fixed, immovable space lattice* whose structure consists of tetrahedron–counter-tetrahedron pairs. The nodes of this lattice are fixed in position and form the reference points for all physical processes. Unlike models with deformable space, the lattice itself is not in motion; all dynamics arise within the fixed geometry.

Information Storage in the Cells

Each smallest lattice cell – defined by a tetrahedron–counter-tetrahedron pair – carries fundamental physical information:

- **Time:** represented by the projection of the local state vector onto the w_1 axis (and w'_1 for anti-time).
- **Mass:** linked to torsional displacement along w_4 .
- **Energy:** described by oscillation amplitudes along w_2 and w_3 .

- **Torsion:** stored twist determining the coupling between mass and energy degrees of freedom.

These parameters are locally defined but remain in constant interaction with neighboring cells.

Motion as Energy Transfer

Since the nodes are anchored in space, no physical displacement of the lattice itself can occur. Instead, motion and propagation – such as of particles or waves – are realized through the *transfer of energy and torsion along the lattice edges*:

- Energy flows along edges as oscillatory excitation in w_2 – w_3 .
- Torsion is transmitted via w_4 couplings between nodes.
- Time information is conveyed through synchronized projection onto w_1 .

A “moving” object in the MMU is therefore not a traveling material point, but a pattern of continuous energy transfer through the fixed lattice.

The Observer’s Point of View

The interpretation of physical laws in the MMU depends on the position of the observer relative to the lattice.

- **Inside the tetrahedron:** The observer perceives the inner oscillatory structure directly. Here, relativistic effects emerge naturally: energy and mass appear as two aspects of the same torsion–displacement system, summarized by Einstein’s relation $E = mc^2$.
- **Outside the lattice:** The observer does not resolve the inner dynamics, but only sees the effective projection of motion onto w_1 . This corresponds to the Newtonian view of kinetic energy as $E = \frac{1}{2}mv^2$, where motion is described without reference to internal torsional storage.

Thus, Newton’s and Einstein’s formulations are unified within the MMU: they are complementary perspectives on the same underlying tetrahedral lattice—one from within, one from outside. The geometric projection principle provides the bridge between both regimes.

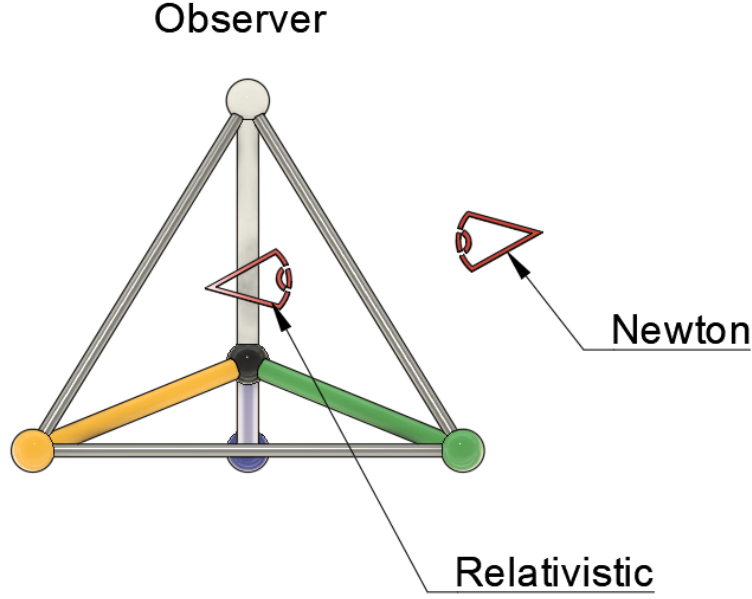


Figure 7: The observer's point of view in the MMU. **Inside the tetrahedron (Relativistic):** the observer perceives the internal oscillatory structure directly, where mass and energy are unified through torsion and displacement, consistent with Einstein's relation $E = mc^2$. **Outside the lattice (Newtonian):** the observer sees only the projected motion onto w_1 , which corresponds to Newton's kinetic energy expression $E = \frac{1}{2}mv^2$. Thus, the MMU unifies the Newtonian and Einsteinian descriptions as complementary perspectives of the same underlying tetrahedral lattice.

See Appendix Appendix 6 for the MMU reduction from GR^* to Newton's law via projection and the weak-field expansion.

A rigorous derivation of the Schrödinger and Dirac equations from internal MMU oscillations is given in Appendix Appendix 4.

6 Static Forces and Tetrahedral Growth in the MMU

Static Forces in the MMU

In the MMU, the stability of each tetrahedral cell is maintained by a balance between internal spring forces and the constraints of the external lattice.

External lattice forces. Tetrahedral cells are never isolated: each node is embedded in and coupled to the surrounding lattice. From an external viewpoint, the resulting forces can be interpreted either as energy projections onto the measurable axis w_1 , or, in the macroscopic limit, by Newtonian relations. In this sense, an outside observer perceives the same underlying structural tension as ordinary kinetic or potential energy,

$$E = \frac{1}{2}mv^2,$$

without explicit reference to the hidden torsional storage.

Hence, the MMU provides a natural unification of relativistic and Newtonian descriptions: the same static equilibrium appears internally as relativistic spring tension, and externally as Newtonian energy projection.

A short calculation of the external spring forces is given in Appendix Appendix 1.

Internal spring forces. In the MMU lattice, each internal axis w_2, w_3, w_4 carries an effective spring stiffness—Coulomb (k_C), spin/torsion (k_S), and gravitation (k_G)—set by the curvature of the corresponding fundamental potentials. For a characteristic microscopic length a one obtains

$$k_C \equiv \frac{2}{4\pi\epsilon_0} \frac{e^2}{a^3}, \quad k_S \equiv \frac{\hbar c}{a^3}, \quad k_G \equiv \frac{2Gm^2}{a^3}.$$

All three scale as a^{-3} , hence their *ratios* are universal:

$$\frac{k_C}{k_S} = 2\alpha, \quad \frac{k_G}{k_S} = 2\left(\frac{m}{m_{\text{Pl}}}\right)^2, \quad \frac{k_C}{k_G} = \alpha\left(\frac{m_{\text{Pl}}}{m}\right)^2,$$

with $\alpha = e^2/(4\pi\epsilon_0\hbar c)$ the fine-structure constant. Under Planck normalization $m = m_{\text{Pl}}$ this reduces to $k_C/k_S = 2\alpha$, $k_G/k_S = 2$, $k_C/k_G = \alpha$. These identities fix the relative weights of internal energies and thus the projection onto w_1 .

Short pointer. The explicit determination of (k_2, k_3, k_4) is given in Appendix Appendix 2. With these constants, the 21 cm hyperfine frequency follows *without QED machinery* from a simple one-step estimate.

Tetrahedral Growth Driven by w_4

Beyond static equilibrium, the MMU allows hierarchical growth of tetrahedral cells.

Role of w_4 . The torsional axis w_4 acts as a *growth driver*. By storing additional torsional displacement, it expands the effective edge length of the tetrahedron, producing a larger cell with higher edge length at the next structural level. This process corresponds to the appearance of a new effective node S .

Role of w_2 . At the same time, the elastic displacement along w_2 acts like a *spring under tension*. As the tetrahedron grows, this spring is further stretched, increasing the effective stiffness of the oscillatory system. The direct result is a shift of eigenfrequencies toward higher values.

Physical meaning. Thus, tetrahedral growth is not a simple geometric scaling, but a dynamical process in which torsional input from w_4 enlarges the structure, while elastic displacement from w_2 retunes the oscillatory spectrum. Higher-level tetrahedra therefore support oscillations at higher frequencies, providing a natural geometric explanation for the hierarchical spectrum of matter, from atomic resonances up to macroscopic and cosmological scales.

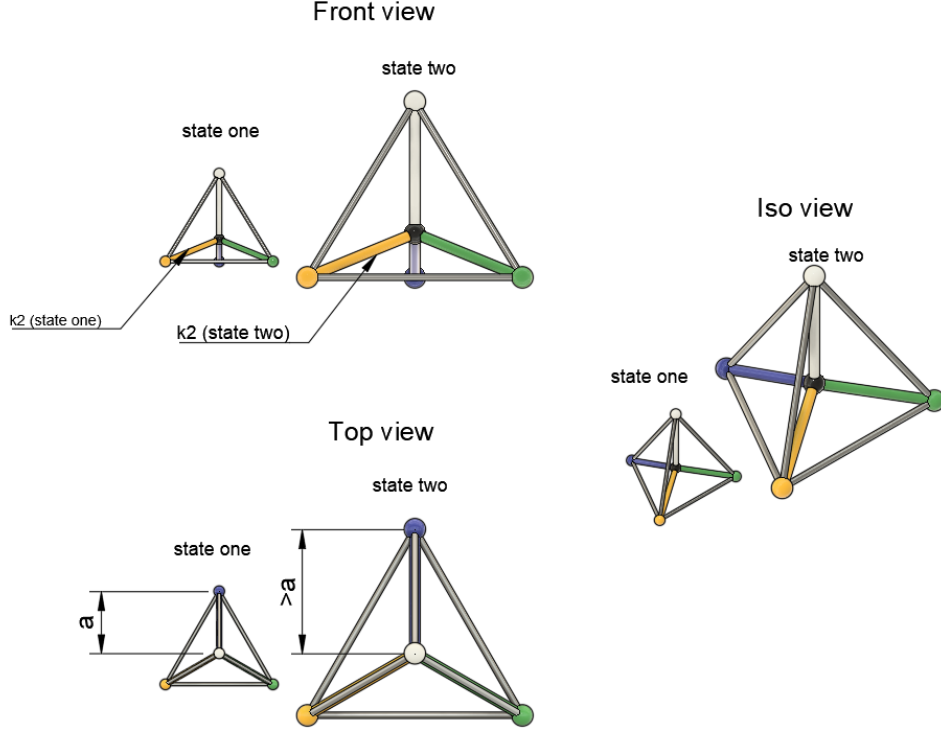


Figure 8: Growth of a tetrahedron in the MMU driven by torsional displacement along w_4 . **Front view:** comparison between the initial configuration (*state one*) with edge length a and the expanded configuration (*state two*) with greater height and effective stiffness k_2 . **Top view:** enlargement of the tetrahedron leads to an increase in edge length and to a retuning of the oscillatory spectrum toward higher frequencies. **Iso view:** spatial comparison of the two states, illustrating how torsion from w_4 enlarges the structure while the elastic spring along w_2 is further stretched. This figure demonstrates the MMU principle that hierarchical growth of tetrahedra is accompanied by a frequency shift: larger structures support higher-frequency oscillations due to the combined action of w_4 (growth driver) and w_2 (spring tension).

7 Dynamic Processes in the Prime Tetrahedron

The prime tetrahedron is the fundamental oscillation unit of the MMU. In its structure, the four orthogonal axes w_1 , w_2 , w_3 , and w_4 are not only geometric directions, but also degrees of freedom for physical processes. All dynamic processes in the MMU – from the propagation of electromagnetic waves to the coupling of spin and mass – can be described as combined oscillations, torsions, and projections within this tetrahedron.

Radial Electric Excitation (w_2)

Displacements along w_2 represent electrical or more generally energetic disturbances. They occur radially and can be propagated through the lattice as wave packets. Coupling to w_3 occurs with a 90° phase shift, producing a rotating field component.

Magnetic and Spin-Induced Displacement (w_3)

The w_3 axis is orthogonal to w_2 and together they form the plane of electromagnetic oscillation. An electric excitation on w_2 induces, through the lattice coupling, a magnetic response on w_3 . Together, they generate a cyclic motion corresponding to Larmor precession.

Torsional Mass Anchor (w_4)

Displacements along w_4 store torsion and couple the internal oscillation state to the mass properties of the system. When w_4 is modulated by combined excitations in w_2 and w_3 , a coupled oscillation of wave transport and torsion arises, periodically shifting the center of mass.

Coupled Modes and Projection onto w_1

The combined displacement in w_2 , w_3 , and w_4 produces a resultant motion whose projection onto w_1 gives the measurable change in time or energy. The amplitude of this projection depends directly on the strength of the lateral and torsional components – this is, in the MMU, the geometric mechanism behind time dilation, frequency shift, and energy change.

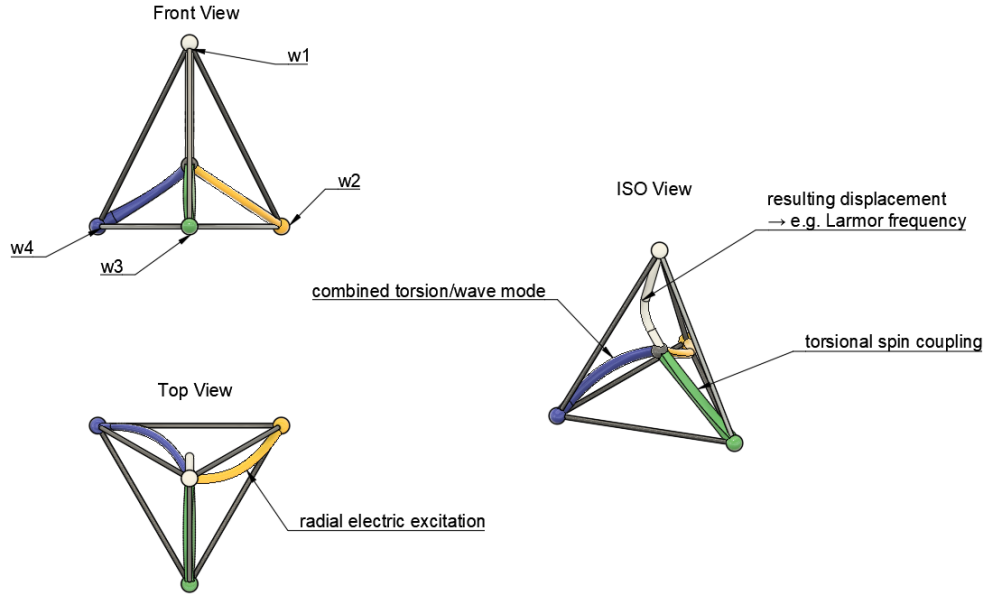


Figure 9: Dynamic processes in the prime tetrahedron. **Front view:** shows the arrangement of the axes w_1 , w_2 , w_3 , w_4 and the radial electric excitation along w_2 . **Top view:** illustrates the 90° phase relationship between w_2 and w_3 . **ISO view:** depicts the coupled torsion/wave motion and the resulting displacement, e.g. corresponding to the Larmor frequency. The combination of torsional spin coupling and radial wave excitation is a key mechanism for the internal dynamics of the MMU.

Spin in the MMU

In the Methane Metauniverse, *spin* is not regarded as a mysterious property explainable only in quantum mechanics, but as a real geometric motion within the prime tetrahedron. Spin arises from a combined torsional and rotational motion of the w_2 – w_3 oscillation plane around the w_4 axis. The direction of this rotation determines the sign of the spin:

- **Spin $+1/2$:** rotation of the w_2 – w_3 plane clockwise (viewed along w_4).
- **Spin $-1/2$:** rotation of the w_2 – w_3 plane counterclockwise.

Physically, spin in the MMU describes the angular momentum resulting from the cyclic conversion between electric (w_2) and magnetic (w_3) displacement. Coupling to w_4 ensures that spin is directly linked to the mass and torsion component of the node. This geometric view makes it clear why spin is quantized: only certain discrete rotation directions are compatible with the fixed tetrahedral structure.

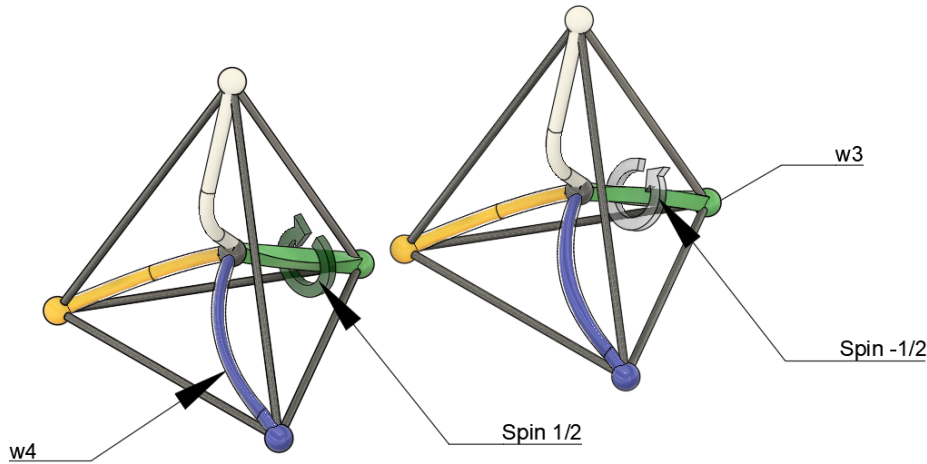


Figure 10: Representation of the two possible spin directions in the MMU’s prime tetrahedron. Left: spin $+1/2$, rotation of the w_2 – w_3 plane clockwise. Right: spin $-1/2$, rotation in the opposite direction. The w_4 axis acts as the rotation axis and mass anchor, while w_2 (yellow) and w_3 (green) form the electromagnetic oscillation plane.

Precession in the MMU

In the Methane Metauniverse, *precession* denotes the rotation of the oscillation plane (w_2 – w_3 plane) around the w_1 axis. This motion corresponds to Larmor precession in quantum mechanics, but in the MMU it has a clearly defined geometric and dynamic cause: the combined displacement in w_2 (electric/energetic component) and w_3 (magnetic/spin-related component) generates a torque about w_1 . This torque is additionally influenced by coupling to w_4 , which acts as an elastic torsion spring.

Extended View: Influence of w_4

- w_4 stores the torsional energy generated by oscillation in w_2 and w_3 .

- Changes in the tension of w_4 act as a restoring force on w_2 and w_3 , modulating the precession frequency.
- Thus, precession is not purely a geometric effect, but a coupled oscillation–torsion mechanism.

Coupled Precession Mechanism

1. **Excitation:** an oscillation on w_2 induces a phase-shifted motion on w_3 .
2. **Torsion:** both axes stretch w_4 like a spring.
3. **Precession:** the resulting feedback from w_4 alters the angular momentum.

Appendix reference. Rigorous proofs are provided in Appendix Appendix 5 (geometric Larmor with $k_{23} = g/2$ and spectral corrections incl. Lamb shift via $\ell_{UV}, \sigma_{\text{geom}}$), Appendix Appendix 6 (projection factor $\zeta = d\tau/dt$ unifying kinematic and gravitational time dilation), and Appendix Appendix 9 (Zeeman with correct Landé factors and the parameter-free $c_1 = 1/(4\pi)$ yielding the Schwinger term $\alpha/(2\pi)$).

Transmission of Torsion and Strain in the MMU Lattice

In the Methane Metauniverse (MMU), the internal oscillations of a primordial tetrahedron are directly linked to the edges of the lattice. Any motion within the w_2 – w_3 – w_4 axes generates:

- **Torsion** — a twisting of the edge along its longitudinal axis.
- **Strain** — an elongation or compression along the edge direction.

These mechanical impulses are *transferred to the adjacent edges without any time delay at the node* and propagate along these edges through the lattice at the speed of light c . Thus, the entire space lattice acts as an elastic, four-dimensionally coupled transmission network in which energy, momentum, and information are transported in the form of torsion and strain waves.

Physical Significance

1. **Torsion** corresponds, in physical space, to the transfer of angular momentum and spin information.
2. **Strain** carries the energetic component, which in the MMU can manifest as electric or gravitational interaction.
3. The fixed anchoring of the nodes ensures that all excitations are transmitted directly to neighboring cells.

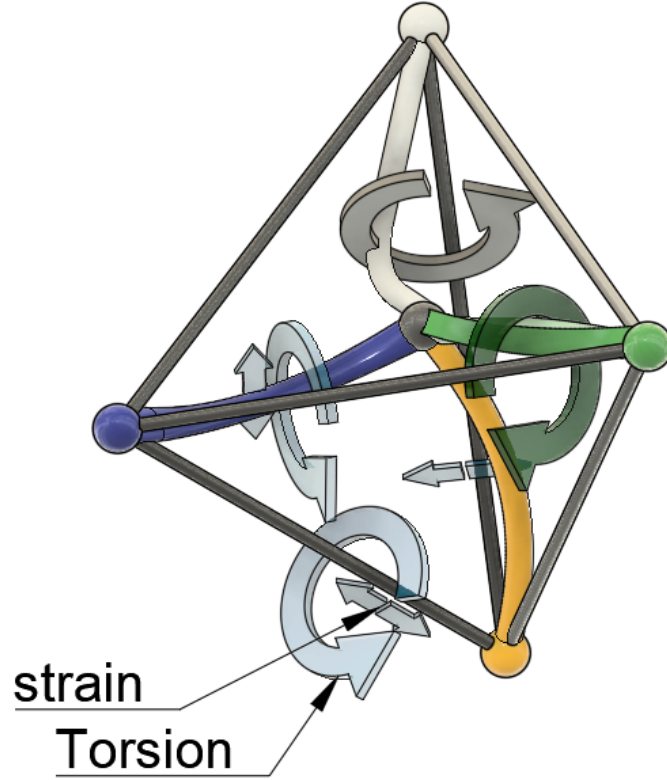


Figure 11: Illustration of the transmission of internal oscillations to the edges of the primordial tetrahedron. The blue and gray arrows indicate torsion directions, the curved arrows represent rotational moments, while the straight arrows mark strain or elongation directions. These excitations travel along the edges at the speed of light c into the surrounding lattice, mechanically and energetically coupling all nodes.

Light in the MMU: Pure Displacement without Mass

In the MMU, light is understood as a *pure transverse displacement* in the w_2 and w_3 axes, without any projection onto the w_4 axis (mass component). This means:

- The wave has **no rest mass**.
- All its energy resides in the **oscillation** of the two orthogonal axes w_2 and w_3 .
- Propagation occurs along the edges of the lattice at the speed of light c .

Physically, this corresponds to the classical electromagnetic wave:

- The w_2 displacement can be identified with the electric field component.
- The w_3 displacement corresponds to the magnetic field component, phase-shifted by 90° .

In the MMU, this orthogonal coupling arises as a direct geometric consequence of the tetrahedral structure: A displacement along w_2 automatically induces a rotation along w_3 via the fixed edges, and vice versa. Since no w_4 component is involved, the system remains massless.

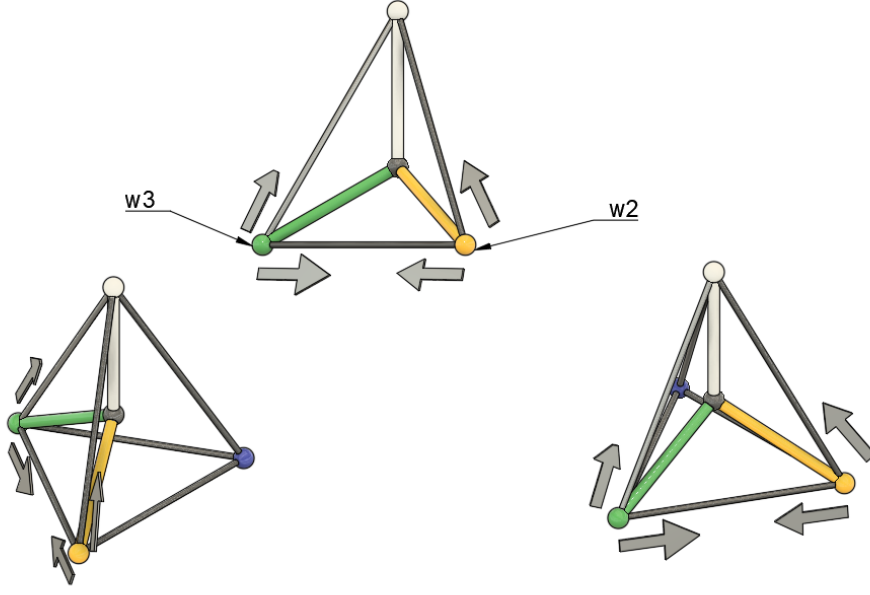


Figure 12: Light in the MMU as a pure oscillation in the w_2 (electric) and w_3 (magnetic) axes. The gray arrows indicate the displacement directions, with the two axes being orthogonal to each other. There is no w_4 component, hence light has no rest mass.

Transmission via w_2 – w_3 coupling and the role of Z_0 (slim)

Massless propagation in the MMU uses the locked, orthogonal pair (w_2, w_3) with no w_4 component. Along an edge direction $\hat{\mathbf{t}}$ a plane wave obeys the single matching rule

$$\mathbf{H} = \frac{1}{Z_0} \hat{\mathbf{t}} \times \mathbf{E} \quad (\mathbf{E} \perp \mathbf{H} \perp \hat{\mathbf{t}}, w_4 \equiv 0),$$

so phase and amplitude are fixed by the vacuum impedance Z_0 ; energy is carried cell-to-cell at speed c .

Appendix reference. Closed-form expression and SI calibration of Z_0 are derived in Appendix Appendix 3.

7.1 Quantum Jump in the MMU

In the Methane Metauniverse (MMU), a quantum jump occurs when the internal oscillation of a tetrahedron reaches a critical amplitude at which stored torsional energy is abruptly released in the form of deformation (strain) along the edges. This results in a discrete transition between two stable oscillation modes — comparable to a transition between two energy levels in quantum mechanics.

Process. At the beginning ($t = 0$), the system is in a state of high stored torsional energy. Once the maximum displacement is reached, the system shifts into a lower-energy state ($t = 1$), in which the tension is released as strain along the edges and transmitted into the lattice. This transfer occurs at the speed of light and is fully quantized — no intermediate states exist.

Internal vs. external view. The precise description of a quantum jump in the MMU depends on the position of the observer:

- **Inside the tetrahedron:** On the w_2 axis, as in the case of the electron in the hydrogen atom, the oscillatory state is naturally described by Laguerre-type radial functions, which arise from the internal geometry of the oscillation modes. These functions encode the probability distribution of the electron's position as determined by the internal torsional dynamics.
- **Outside the lattice:** The observer does not resolve the internal torsional details, but perceives only the projected evolution of the state onto w_1 . In this view, the dynamics reduce to the familiar *time-dependent Schrödinger equation* for the hydrogen atom,

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{r}, t) = \hat{H} \psi(\mathbf{r}, t),$$

where $\hat{H}$ contains the effective kinetic and potential terms arising from the lattice projection. Thus the external observer recovers the standard formulation of quantum mechanics as a shadow of the inner geometric dynamics.

Significance. This process is the geometric equivalent of a photon emission event: The released energy propagates as a pure w_2/w_3 oscillation through the lattice, without a mass component, and manifests macroscopically as an electromagnetic wave. The MMU therefore provides a direct geometric underpinning for discrete energy transitions and the quantization of radiation.

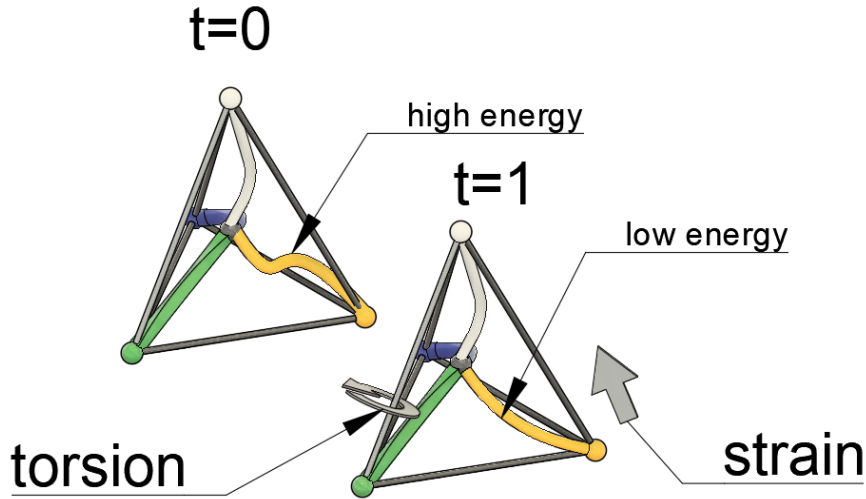


Figure 13: Quantum jump in the MMU: Transition from high torsional energy ($t = 0$) to lower strain energy ($t = 1$) with quantized energy release into the lattice.

Appendix reference. A proof that quantum jumps arise from internal MMU oscillations—via the eigenforce condition $k_2|w_2| = k_3|w_3| = k_4|w_4|$ and adiabatic projection to the external envelope—is given in Appendix Appendix 4.

8 Projection of Energy onto the Time Axis w_1 and Time Dilation in the MMU

In the Methane Metauniverse model (MMU), each elementary cell of the space lattice has four orthogonal degrees of freedom:

$$(w_1, w_2, w_3, w_4).$$

Here, w_1 describes the time direction, while w_2 , w_3 , and w_4 represent spatial-energetic degrees of freedom. Any oscillation or motion within a cell is always distributed across all four axes. The measurable time that we perceive in our macroscopic world corresponds to the projection of this motion onto the time axis w_1 .

8.1 Principle of Projection

If a system is at rest (no internal oscillations beyond the ground oscillation), the projection onto w_1 is maximal. However, when energy is coupled into the w_2, w_3, w_4 axes, the fraction of the total motion projected onto w_1 decreases.

8.2 Time Dilation as a Geometric Effect

This geometric relationship directly leads to *time dilation*: The greater the proportion of motion in the non-temporal axes (w_2, w_3, w_4), the smaller the projection onto w_1 . From the perspective of an external observer, time in the system under consideration passes more slowly.

8.3 Influence of Mass on Time

A high mass in the MMU corresponds to strong baseline strain and torsion along the w_4 axis. This binds energy into the w_4 axis, thereby reducing the free projection onto w_1 and resulting in measurable time dilation. This effect is analogous to gravitational time dilation in General Relativity.

8.4 Energy in the Projection Formula

The total energy of a system in the MMU can be expressed as the magnitude of the four-vector:

$$E_{\text{total}} = h \cdot \sqrt{w_1^2 + w_2^2 + w_3^2 + w_4^2}.$$

The measurable energy in the time projection is given by:

$$E_{w_1} = h \cdot w_1 = h \cdot \sqrt{w_{\text{total}}^2 - (w_2^2 + w_3^2 + w_4^2)}.$$

It is thus clear that any increase of energy in the spatial axes automatically reduces the effective time energy E_{w_1} — the core mechanism of time dilation in the MMU.

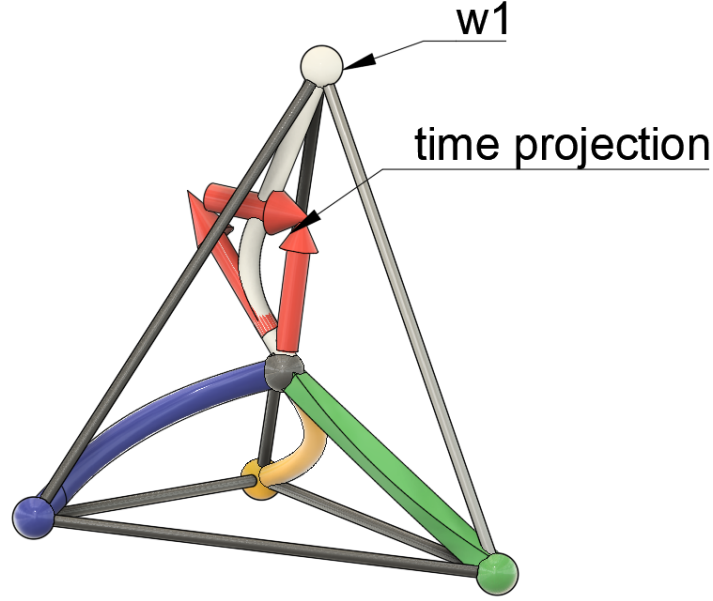


Figure 14: Projection of motion in the w_2, w_3, w_4 axes onto the time axis w_1 . The larger the amplitudes in the spatial-energetic axes, the smaller the projection onto w_1 — and the slower time passes in the system under consideration.

Appendix references. The projection mechanism and clock-rate factor ($\zeta = d\tau/dt$) are derived in Appendix Appendix 6; the time-projection unification (calibrating the w_1 projection) in Appendix Appendix 1; and the build-up of w_4 strain from internal springs in Appendix Appendix 2.

9 Basic Idea of Coupling

In the MMU, all fundamental partners are described as tetrahedral cells with a universal edge length a and four orthogonal degrees of freedom (w_1, w_2, w_3, w_4). The axis w_4 stores inertial energy (mass), w_2 represents electric displacement and coupling, and w_3 corresponds to torsional displacement (spin). The measurable rest mass of an isolated partner is obtained by projecting its internal dynamics onto the observable axis w_1 .

When two partners interact, their contributions along w_1, w_2 , and w_3 combine into a shared equilibrium. A new effective node S is formed, which is not only the geometric center of mass but also incorporates the balance of inertial (w_4), electric (w_2), and spin (w_3) components. The projection of this combined state onto w_1 yields a *reduced, stabilized energy value*, corresponding to the observable binding energy of the coupled system.

10 Geometric Construction of the Node S

The node S emerges geometrically as the shared intersection of the axes of both partners:

- Along w_4 , S represents the inertial center. The distances d_{p1} and d_{p2} from Partner 1 and Partner 2 to S scale inversely with their inertial masses:

$$\frac{d_{p1}}{d_{p2}} = \frac{m_2}{m_1}.$$

- Along w_2 , S represents the common electric center, where positive and negative displacements find equilibrium and stabilize the interaction.
- Along w_3 , S represents the shared spin axis. The torsional modes of both partners couple into a common rotation, leading to parallel (triplet) or antiparallel (singlet) spin states. This shared torsion defines the magnetic properties of the bound system.

In general, the three centers S_{w_2} (electric), S_{w_3} (spin), and S_{w_4} (inertia) do not coincide exactly. Their small offsets encode fine details of the coupling, including mass–charge correlations and spin–orbit effects.

10.1 Energetic Interpretation

The binding of two partners can thus be decomposed into three complementary mechanisms:

1. **w_2 coupling (Electric):** defines the Coulomb-like binding energy.
2. **w_3 coupling (Spin/Torsion):** determines the shared spin state and magnetic interaction energy.
3. **w_4 coupling (Inertia):** fixes the motion of the shared node S and defines the reduced mass of the system:

$$\mu = \frac{m_1 m_2}{m_1 + m_2}.$$

All three mechanisms project onto the observable axis w_1 , producing a common reduced energy state. Externally, the coupled system therefore appears as a *lower-energy projection*, explaining the stability of bound configurations and the quantization of spin-dependent levels. In this way, the MMU provides a unified geometric picture of coupling: mass, charge, and spin merge into a shared node S , whose projection onto w_1 defines the observable physical state.

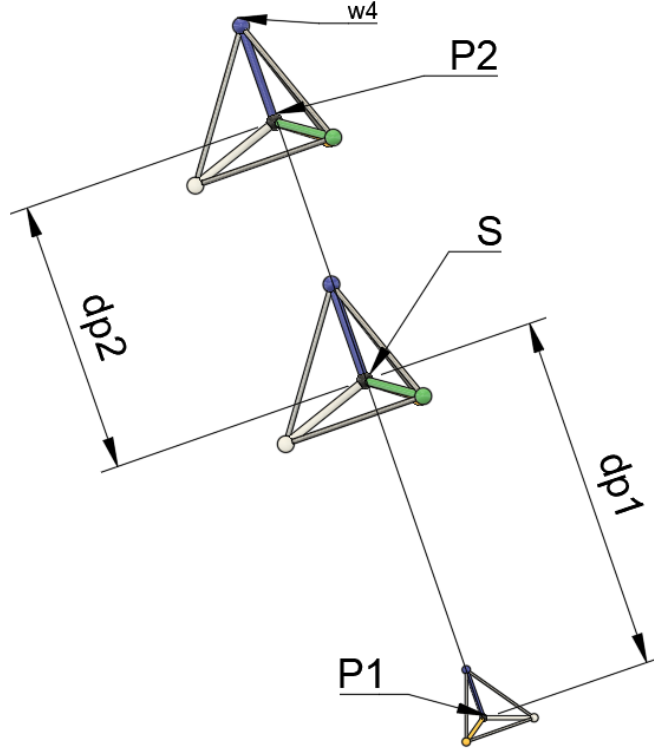


Figure 15: Coupling of two tetrahedral partners in the MMU. Partner 1 (P1) and Partner 2 (P2) are connected along their w_4 axes. The distances d_{p1} and d_{p2} from each partner to the shared coupling point S are inversely proportional to their inertial masses, $\frac{d_{p1}}{d_{p2}} = \frac{m_2}{m_1}$. The node S thus represents not only the effective center of mass, but also the joint balance point for the w_2 (electric) and w_3 (spin) contributions. Its projection onto w_1 yields the externally observable, reduced binding energy of the coupled system.

Appendix reference (atomic coupling). The basic MMU idea of atomic coupling—nuclear composition $\Rightarrow w_4$ torsion $\Rightarrow$ intrinsic proton compression $\Rightarrow$ electronic field shift—is developed in Appendix Appendix 8. There we prove that (i) the 1S–2S isotope shifts (H/D/T) follow from this mechanism on top of the reduced-mass term, (ii) the radius ordering $r(p) < r(t) < r(d)$ and the sign of the field contribution are reproduced, and (iii) the MHz-level shifts match once standard recoil/radiative corrections are included.

10.2 Extension to Higher Structures: Atoms

The same geometric principle that defines the coupling of two partners also governs the formation of higher structures such as atoms. When electrons, protons, and neutrons interact, their tetrahedral fundamental cells connect via shared nodes S , in which their w_2 (electric), w_3 (spin), and w_4 (inertia) components reach equilibrium.

Atomic binding.

- **Electron–proton coupling:** Electrons and protons form stable bound states through a shared node S in which their opposite w_2 contributions (charge) and w_4

displacements (mass) balance. The projection onto w_1 corresponds to the quantized energy levels observed in atomic spectra.

- **Neutron coupling:** Neutrons contribute additional w_4 (inertial) and w_3 (spin) components without direct w_2 charge contribution. Their inclusion shifts the position of S and stabilizes the nuclear binding through spin-torsion coupling.

Complex oscillation systems. Through the repeated formation of shared nodes S , atoms emerge as *nested oscillatory systems*. Each added particle modifies the balance of w_2 , w_3 , and w_4 , leading to a complex network of coupled oscillators. These oscillations define:

- the discrete spectral lines of atoms (projections of w_2/w_3 oscillations),
- the spin-orbit and fine-structure corrections (offsets between S_{w_2} , S_{w_3} , S_{w_4}),
- and the nuclear stability determined by torsional storage along w_4 .

General principle. Thus, the MMU interprets atoms as hierarchical assemblies of coupled tetrahedra. The same rules that form a shared node S between two partners apply recursively, generating complex, multi-level oscillation systems whose projections onto w_1 correspond to the observable properties of matter.

The hierarchical atomic coupling via shared S -nodes is substantiated in Appendix Appendix 8 (Appendix 8: isotope shifts and nuclear-stability trends from w_4 torsional storage and cluster geometry) and Appendix Appendix 9 (Appendix 9: discrete spectral lines from w_2/w_3 oscillations and spin-orbit/fine-structure corrections from the offsets between S_{w_2} , S_{w_3} , S_{w_4}).

11 Inner Forces and Edge Deformations

In the MMU, the four internal degrees of freedom (w_1, w_2, w_3, w_4) of each tetrahedral cell are not isolated: they are directly transmitted to the edges of the tetrahedron, where they appear as geometric deformations. This coupling between *inner forces* and *edge mechanics* provides the bridge between local oscillations and long-range field effects.

Connection of Axes to Edges

Each internal axis contributes to the mechanical state of the six edges:

- w_2 (**electric displacement**): produces longitudinal stretching or compression of edges, corresponding to electric field lines that extend between nodes.
- w_3 (**spin/torsion**): induces torsional twisting of edges, which manifests as magnetic circulation around the tetrahedron. The 90° phase shift between w_2 and w_3 ensures that deformations propagate as coupled electric-magnetic oscillations.
- w_4 (**inertial anchor**): determines the stiffness of the edges and their resistance to deformation. Larger w_4 displacements store more torsional energy, stabilizing the cell but also coupling to neighboring cells through edge forces.

Field Transmission Through Edges

Because all tetrahedra in the lattice are joined by common edges, any internal excitation propagates as a mechanical wave:

- A stretched edge transmits a Coulomb-like electric interaction.
- A twisted edge transmits a magnetic-like circulation.
- A compressed or relaxed edge changes the local potential and alters the effective inertia of the connected nodes.

Thus, what appears macroscopically as *field forces* is in the MMU a direct result of microscopic edge deformations. The lattice geometry provides a natural medium for propagating oscillations that we observe as photons, magnons, or other quanta of interaction.

Orbital Dynamics in Molecules

When multiple tetrahedra are connected into larger structures, such as atoms or molecules, their shared edges act as coupling channels. The pattern of stretched, twisted, and compressed edges defines a complex oscillatory system:

- In atoms, repeated edge deformations organize into standing wave patterns, which correspond to quantized orbitals.
- In molecules, overlapping edge oscillations produce hybridized orbitals, bond angles, and resonance structures.
- The geometry of the tetrahedral lattice naturally reproduces molecular orbital symmetries, since edge couplings correspond to shared electron density paths.

Physical Meaning

This perspective provides a unified picture: internal oscillations along w_2 , w_3 , and w_4 do not remain confined to a single node, but are carried outward through edge deformations. The edges act as elastic channels transmitting energy and torsion to neighboring cells. From this mechanism arise both electromagnetic field interactions and the quantized orbital dynamics of atomic and molecular systems. In the MMU, molecular bonding is thus explained as the geometric resonance of edge-deformation patterns within the fixed space lattice.

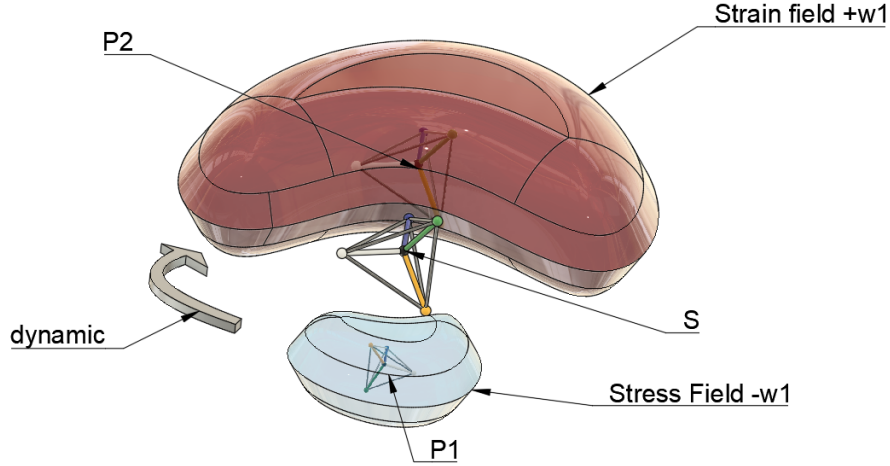


Figure 16: Coupling of two partners in the MMU and formation of dynamic stress/strain fields. Partner 1 (P1) and Partner 2 (P2) are linked through the common node S . Their interaction generates a **stress field** ($-w_1$, shown in blue) and a complementary **strain field** ($+w_1$, shown in red). The torsional dynamics (arrow) mediate the continuous exchange of energy between both partners. The node S represents the equilibrium point where inertial (w_4), electric (w_2), and spin (w_3) contributions balance, while the projection onto w_1 yields the observable reduced binding energy. This figure illustrates how local coupling translates into extended field structures that govern the dynamics of the system.

12 Elementary Particles in the MMU

In the MMU framework, all fermions are realized as geometric excitations of the tetrahedral lattice. Their characteristic properties (mass, charge, spin) arise from the orientation of the fundamental tetrahedron, its projection onto w_1 , and the way in which multiple tetrahedra form coupled systems.

The Electron

The electron is represented as a single tetrahedron whose projection onto the measurable axis w_1 points in the *negative direction*. This negative orientation defines the electron's negative charge and explains why it is directly observable in experiments: its projection onto $-w_1$ is a stable, isolated state. The electron thus constitutes the simplest free fermion, described as the minimal unit of the MMU lattice.

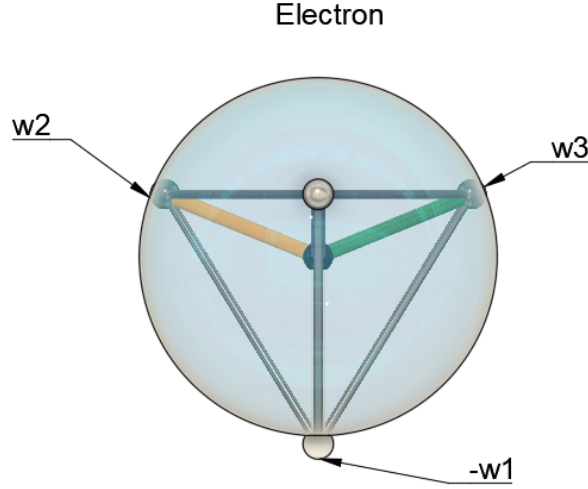


Figure 17: **Electron in the MMU lattice.** The electron is represented as a single tetrahedron. Its measurable projection is oriented along the negative axis $-w_1$, which defines the negative charge and makes the electron directly observable. The transverse axes w_2 and w_3 denote the internal oscillatory degrees of freedom (electric vs. magnetic/torsional), forming together with w_1 the minimal unit of the MMU lattice. The spherical shading indicates the confinement region of the electron state.

The Proton

The proton arises from a higher-order tetrahedron, formed by the hierarchical growth mechanism along w_4 . Its projection is oriented in the *positive direction* of w_1 , which corresponds to the positive charge. Because the proton is a larger, internally excited structure, its mass is much greater than that of the electron. The proton therefore represents a higher-level resonance of the tetrahedral lattice, stabilized by torsional energy along w_4 .

At the lower structural level, the proton is composed of three fundamental tetrahedral sub-units, which in standard particle physics are called *quarks*. In the MMU, these sub-units are understood geometrically as partial tetrahedra aligned along the internal axes:

- q_{w2} – associated with electric displacement and the stretching of edges,
- q_{w3} – associated with torsional spin modes,
- q_{w4} – associated with stored inertial mass energy.

The combination of these three sub-tetrahedra forms the stable internal configuration of the proton. Their interaction ensures that the proton projects positively onto w_1 , while maintaining stability through the coupling of charge, spin, and inertia inside the tetrahedral hierarchy.

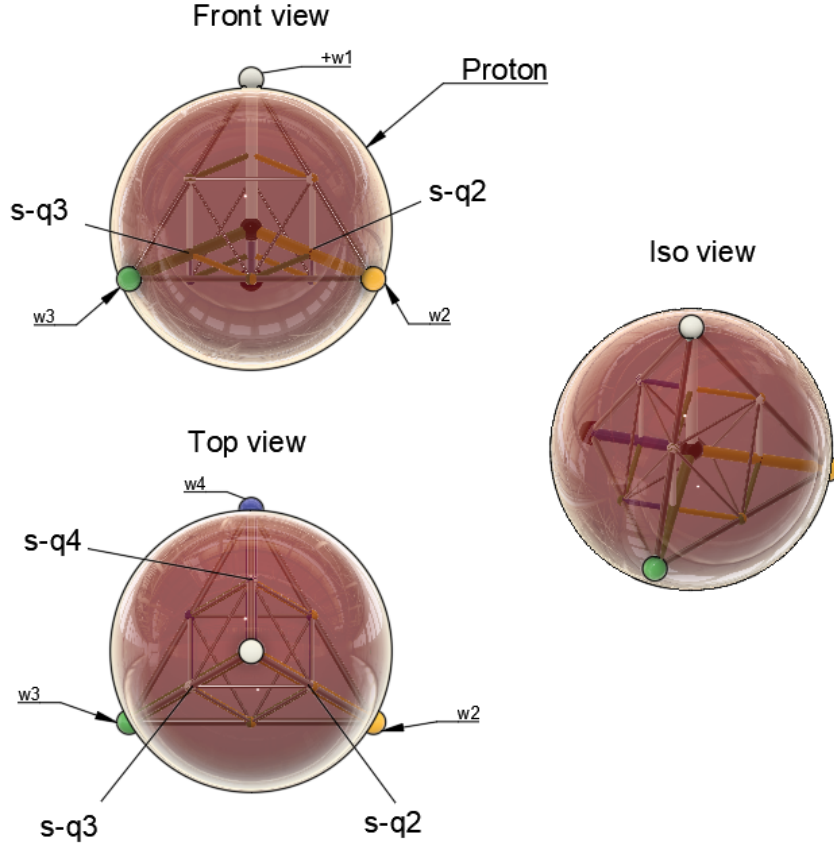


Figure 18: **Proton in the MMU lattice.** The proton is represented as a higher-order tetrahedron composed of three sub-tetrahedra (quarks) arranged along the internal axes w_2 , w_3 , and w_4 . The stabilizing quark knots are: $s-q_2$ (aligned with w_2 , electric/Coulomb), $s-q_3$ (aligned with w_3 , torsional spin), and $s-q_4$ (aligned with w_4 , inertial mass storage). The proton projects positively along $+w_1$, defining its positive charge. Multiple views (front, top, iso) highlight the tetrahedral embedding of the quark nodes.

Appendix reference (quarks). The proof of the quark triad structure—including (i) fractional charges $\{0, \frac{1}{3}, \frac{2}{3}, 1\}$, (ii) color and confinement from the triad geometry, and (iii) the neutron–proton mass difference without new free parameters—is given in Appendix Appendix 10.

Hexagonal Chair Structure and Laguerre Functions

At an even deeper structural level, the quark sub-units of the proton arrange themselves into a *hexagonal chair-like configuration*. This geometry arises naturally from the triangular faces of the three sub-tetrahedra q_{w2} , q_{w3} , and q_{w4} and represents the minimal closed oscillation pattern that balances all three axes.

The hexagonal chair structure provides the spatial foundation for the radial oscillation modes of the proton. Its symmetry and periodicity correspond directly to the solutions of the radial part of the hydrogen Schrödinger equation, where associated Laguerre polynomials appear as the natural eigenfunctions for bound states. In the MMU,

this correspondence is not accidental but geometrically grounded: the discrete oscillation modes of the hexagonal chair are the geometric precursors of the Laguerre functions.

Physical interpretation. When an electron couples to a proton to form the hydrogen atom, the electron wavefunction is shaped by the proton’s internal hexagonal oscillation scaffold. The Laguerre functions in the Schrödinger equation thus emerge as the macroscopic projection of the proton’s internal hexagonal resonance structure. This explains why hydrogenic wavefunctions possess radial nodes and quantized radii: they reflect the underlying oscillatory geometry of the proton at the quark level.

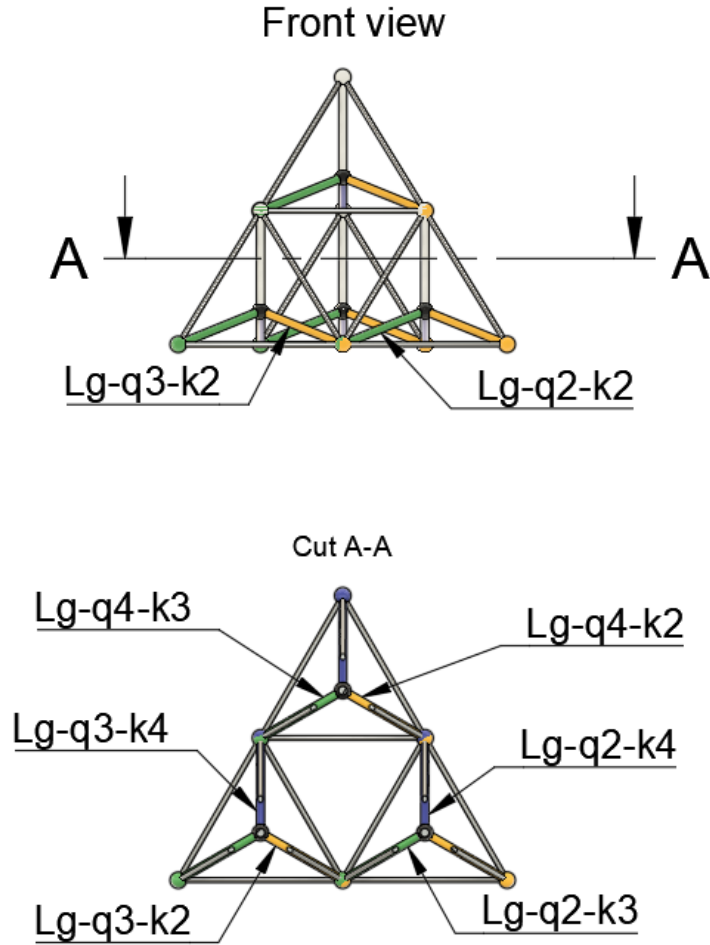


Figure 19: **Hexagonal chair structure inside the proton.** **Top:** Front view of the proton’s quark substructure, showing the lower triangular arrangement of the three sub-tetrahedra (q_{w2} , q_{w3} , q_{w4}). **Bottom:** Cross section (A–A) through the proton, where the quark modes combine into hexagonal oscillation loops. These loops are labeled as Laguerre-type combinations (e.g. $Lg-q2-k2$, $Lg-q3-k2$, $Lg-q4-k3$), corresponding to radial resonance patterns. This hexagonal “chair” configuration provides the geometric foundation for the radial eigenmodes of the hydrogen atom, which in quantum mechanics are expressed by associated Laguerre polynomials in the Schrödinger equation.

Appendix reference. The geometric origin of the hydrogenic Laguerre radial functions—derived from the MMU proton’s hexagonal chair—is proved in Appendix Appendix 7.

The Neutron

The neutron is described as a *composite state* of an electron and a proton bound into a shared node S . In this configuration:

- The electron contributes its negative w_1 projection.
- The proton contributes its positive w_1 projection.
- The resulting node S forms a nearly balanced state, with vanishing net electric projection but large stored torsional energy.

This explains the neutron's neutrality and its relative instability: it is an energetically balanced but tension-rich structure.

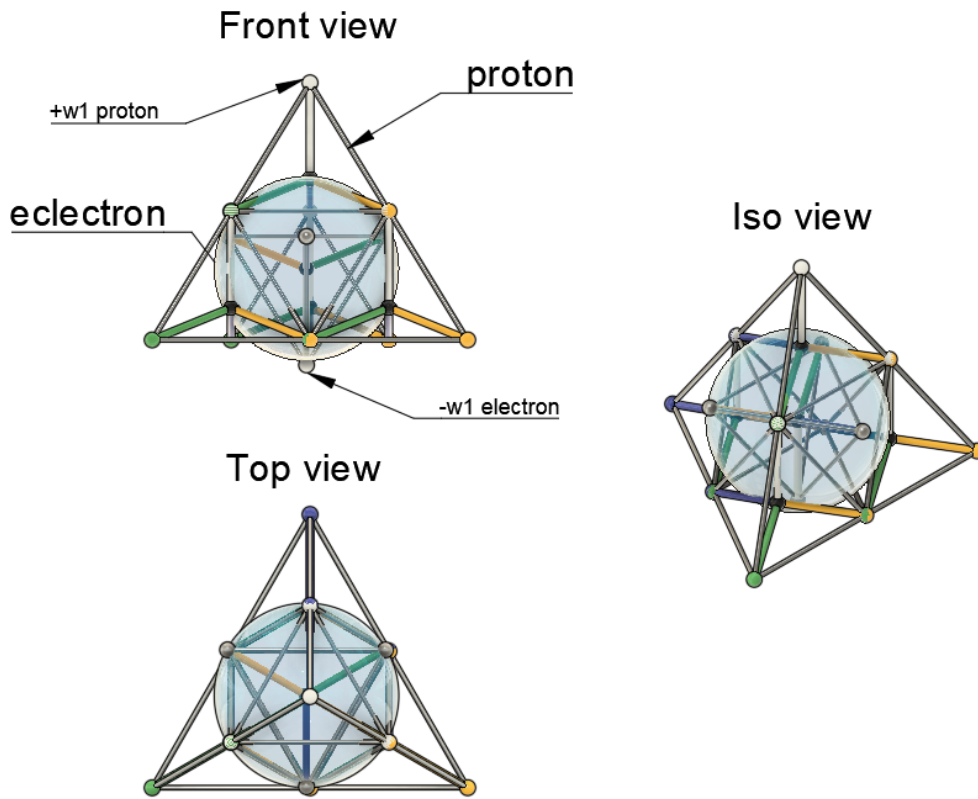


Figure 20: **The neutron as a composite state in the MMU.** The neutron is described as the assembly of a proton and an electron, coupled within the same tetrahedral framework. **Front view (left):** the proton projects along $+w_1$ (positive charge), while the electron projects along $-w_1$ (negative charge). Their combination yields an almost balanced projection onto w_1 , explaining the neutron's overall electrical neutrality. **Top view (bottom):** shows how the electron is embedded within the larger protonic tetrahedron, with both substructures linked through the common S -node. **Iso view (right):** highlights the internal spherical symmetry that emerges when the proton and electron are coupled. In this picture, the neutron is understood not as an independent elementary particle but as a dynamic equilibrium of proton and electron, stabilized by the shared S -node and projecting outward as a neutral configuration.

The interpretation of the neutron as a $(p+e^-)$ bound at a shared S -node—shown to be equivalent to the (u, d, d) description and yielding the binding balance $E_{\text{bind}}(p, e) \approx -0.782$ MeV—is proved in Appendix Appendix 10.

The Antineutrino

In the MMU, the antineutrino is not an isolated particle but the *shared coupling node* S that connects an electron and a proton. It does not carry its own electric projection along w_2 , but represents the geometric balance point where the opposite w_1 projections of electron (negative) and proton (positive) meet.

At the same time, the antineutrino embodies the torsional contributions along w_3 and w_4 :

- Along w_3 , it aligns the spin–torsion modes of both partners.
- Along w_4 , it stores the residual inertial tension resulting from their mass imbalance.

Thus, the antineutrino appears naturally as the equilibrium structure that maintains symmetry in the tetrahedral lattice. It provides the dynamic channel through which energy and torsion are exchanged between proton and electron, ensuring that the coupled system projects as a stable, reduced energy state onto w_1 . In this sense, the antineutrino is the *geometric realization of the S node* binding proton and electron.

The Deuterium Connection

In deuterium, the neutron couples to a proton through the shared node S . Here, the binding does not take place primarily along w_1 , but instead stabilizes in the w_2 – w_3 *plane* of the common node. This coupling mode explains the nuclear binding energy of deuterium: the oscillations in w_2 and w_3 act as stabilizing springs, while the projections along w_1 cancel to yield an energetically reduced, stable nucleus.

Summary

The MMU thus provides a geometric classification of the basic fermions:

- Electron: single tetrahedron, $-w_1$ projection.
- Proton: higher-order tetrahedron, $+w_1$ projection.
- Antineutrino: torsional complement, no direct w_2 projection.
- Neutron: composite of electron and proton at node S , electrically neutral but torsionally rich.

These building blocks connect hierarchically through shared nodes to form nuclei and, eventually, the complex oscillation systems of matter.

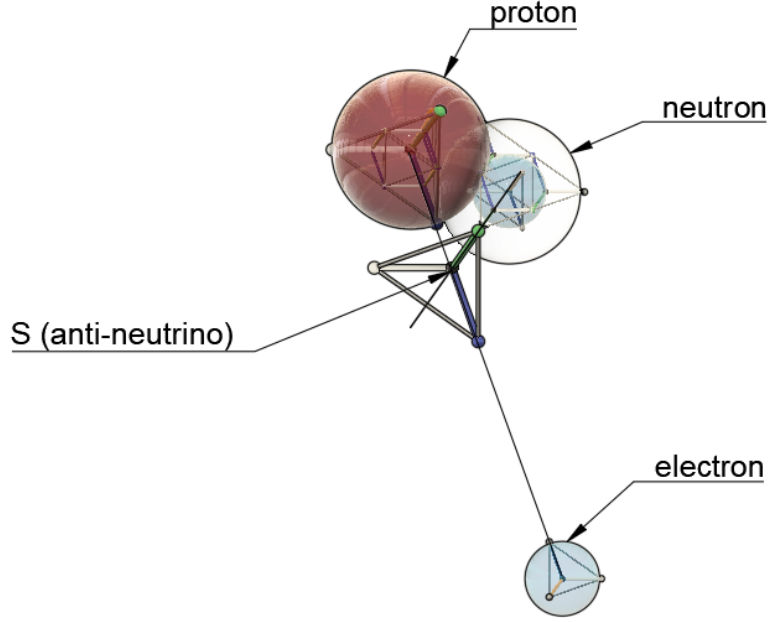


Figure 21: **Deuterium in the MMU.** The deuteron is a bound state of a proton and a neutron. In this view, the neutron itself is a composite of a proton and an electron, stabilized through the shared coupling node S (interpreted as a anti-neutrino). The proton projects along $+w_1$, the electron along $-w_1$; their combination in the neutron is electrically neutral. The S node mediates torsion and energy balance, and the proton–neutron binding proceeds primarily via oscillations in the w_2 – w_3 plane of S , providing the nuclear binding energy.

Legend: proton (red, $+w_1$); electron (blue, $-w_1$); neutron (grey, neutral); S node (neutrino) as torsion/energy mediator.

Appendix reference (decay). The MMU decay mechanism— S -node barrier lowering and flip, yielding the exponential law via the Eyring–Kramers escape rate—is derived and quantitatively validated in ???. There we fit the two-parameter law $k(Q) = k_0 e^{\beta Q}$ to neutron and tritium β -decay (recovering $T_{1/2}(n) \approx 880$ s, $T_{1/2}(^3\text{H}) \approx 12.32$ yr) and predict stability for ^1H , ^2H ($Q \leq 0$); CPT-consistent matter/antimatter spectra follow from the w_1 mirror symmetry.

13 Decay Processes in the MMU

In the MMU, decay is not interpreted as a purely probabilistic event, but as the geometric consequence of mirror projections approaching each other within the lattice. Every fermion projects either onto $+w_1$ (matter) or $-w_1$ (antimatter). As long as the shared S -nodes remain stable, these projections coexist without overlap. When oscillations or nuclear configurations push them too close, the lattice tension rises until the S -node flips, and a decay event occurs.

Electron–Positron Approximation

The electron corresponds to a single tetrahedron projecting onto $-w_1$, while the positron is its mirror projection onto $+w_1$. If the electron’s projection is pulled closer to its mirror component, different regimes arise:

- **Small approximation:** minor corrections to energy levels, appearing as vacuum polarization (Uehling correction), as observed in muonic hydrogen.
- **Critical approximation:** instability of the S -node, leading to emission of an electron/positron together with an antineutrino – the geometric origin of beta decay.

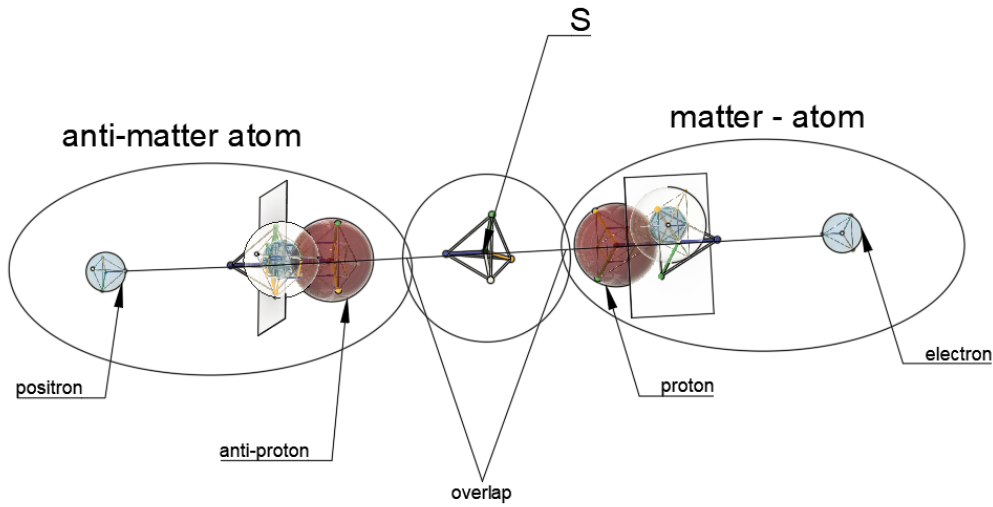


Figure 22: **Matter–Antimatter Atoms in the MMU.** Comparison of matter (right) and anti-matter (left), each consisting of a proton–electron (matter) or anti-proton–positron (antimatter) pair stabilized by a shared S -node.

Explanation: On the right, is shown as a proton ($+w_1$ projection) and an electron ($-w_1$ projection) forming a neutron-like state. On the left, the mirror structure of anti-deuterium consists of an anti-proton ($-w_1$) and a positron ($+w_1$). Both are stabilized locally by their S -nodes, which couple inertial, electric, and spin degrees of freedom. In the middle, the overlap of both S -nodes is indicated. This overlap represents the critical point where matter and antimatter projections cancel, opening the possibility for annihilation, beta decay, or vacuum-polarization effects. In the MMU, nuclear stability (deuterium) and nuclear instability (anti-deuterium annihilation) are thus explained by the same geometric mechanism of node coupling.

Proton–Antiproton Approximation

The proton projects onto $+w_1$, its mirror antiproton onto $-w_1$. In heavy nuclei, the presence of many neutrons shifts electron S -nodes closer to protonic S -nodes, effectively pulling the proton’s $+w_1$ projection towards its $-w_1$ mirror:

- **Moderate approximation:** nuclear binding energies are modified, visible as shifts in shell structure.

- **Strong approximation:** the S -node flips, triggering proton–neutron conversion with emission of an electron/positron and neutrino – i.e. beta decay channels.
- **Extreme approximation:** in very heavy or unstable systems, multiple S -nodes collapse simultaneously, producing spontaneous fission or gamma/positron emission.

Collapse and Black Holes

When many proton–antiproton mirror projections overlap simultaneously (e.g. in a collapsing star), the $+w_1$ and $-w_1$ projections annihilate in bulk. The measurable projection onto w_1 disappears entirely. In the MMU, this corresponds to the formation of a black hole: a region where no external time projection exists, and all dynamics are trapped internally in the w_2 – w_3 – w_4 lattice.

Appendix reference (decay). The MMU decay mechanism— S -node barrier lowering and flip, yielding the exponential law via the Eyring–Kramers escape rate—is derived and quantitatively validated in Appendix Appendix 11. There we fit the two-parameter law $k(Q) = k_0 e^{\beta Q}$ to neutron and tritium β -decay (recovering $T_{1/2}(n) \approx 880$ s, $T_{1/2}(^3\text{H}) \approx 12.32$ yr) and predict stability for $^1\text{H}, ^2\text{H}$ ($Q \leq 0$); CPT-consistent matter/antimatter spectra follow from the w_1 mirror symmetry.

13.1 Electron–Proton Coupling and Neutron Formation

In the MMU framework, an electron and a proton can undergo a coupling process that is equivalent to a decay–recombination cycle. The electron projects onto $-w_1$ as a single tetrahedron, while the proton projects onto $+w_1$ as a higher-order tetrahedron. When both approach each other in the lattice, their positive and negative w_1 contributions balance within a shared coupling node S .

Formation of the neutron. At the S -node, the electron’s negative projection and the proton’s positive projection merge into a nearly balanced state. The result is the neutron: a composite configuration characterized by

- vanishing net projection onto w_2 (no electric charge),
- torsional richness along w_3 (spin storage),
- and large stored inertia along w_4 .

In this picture, the antineutrino corresponds directly to the S -node, mediating the balance of torsion and inertia between the electron and the proton.

Stability and decay. The neutron is not a fully stable structure: it stores residual torsional energy in the w_3 – w_4 system. After a characteristic mean lifetime τ , this imbalance forces the S -node to flip, releasing the components:

$$n \longrightarrow p + e^- + \bar{\nu}_e.$$

This is recognized experimentally as beta decay, with the neutron decaying into a proton, electron, and antineutrino. In the MMU interpretation, this process corresponds to the re-separation of the balanced w_1 projections once the torsional stability limit is exceeded.

Physical meaning. The electron–proton decay cycle is thus reinterpreted in the MMU as a natural geometric mechanism:

- Coupling at the S -node $\rightarrow$ neutron formation,
- Instability of the S -node $\rightarrow$ neutron decay.

This shows how atomic nuclei can be understood as recursive assemblies of S -nodes, in which electrons and protons continuously transform into neutrons and back, depending on the torsional balance of their internal lattice configuration.

Half-Life as a Geometric Projection

The statistical nature of decay, observed experimentally as the *half-life* $T_{1/2}$, emerges from the oscillatory cycles of the tetrahedral nodes. Each unstable system oscillates internally with its own eigenfrequency. Decay occurs when this oscillation phase brings the matter projection ($+w_1$) and its mirror ($-w_1$) close enough for the S -node to collapse. Since the exact phase alignment varies among many identical systems, the observable outcome is an exponential decay law:

$$N(t) = N_0 e^{-t/\tau}, \quad T_{1/2} = \tau \ln 2,$$

with τ representing the average oscillatory persistence before mirror approximation becomes critical.

Summary

Decay in the MMU is therefore the controlled release of lattice tension caused by the approach of matter and antimatter projections.

- Small approach: energy corrections (vacuum polarization).
- Medium approach: S -node flip $\rightarrow$ beta decay.
- Extreme approach: collective collapse $\rightarrow$ spontaneous fission or black hole formation.

The half-life is no longer an abstract statistical parameter, but a measurable projection of the internal oscillatory limits of tetrahedral matter.

14 Leptons in the MMU as Embedding Levels

In the MMU, all fermions arise from the same fundamental tetrahedral building block. Their apparent differences in mass and stability are interpreted as different *embedding depths* of this block into a deformed lattice environment. This geometric interpretation unifies the families of leptons and quarks as successive resonance states of the same structure.

Lepton Series: Electron, Muon, Tau

- **Electron:** Minimal embedding of a single tetrahedron projecting onto $-w_1$. It resides at the outer boundary of the lattice, with weak torsional coupling. Stable and light.
- **Muon:** First deeper embedding. Stronger deformation of the lattice, higher torsional energy along w_4 , increased projection on w_1 . Same w_2 signature (negative charge) as the electron, but larger effective mass. Unstable.
- **Tau:** Second, deeper embedding. Extreme deformation of the lattice cell, with very high w_4 torsional storage. Very heavy and very short-lived, but still same electric charge.

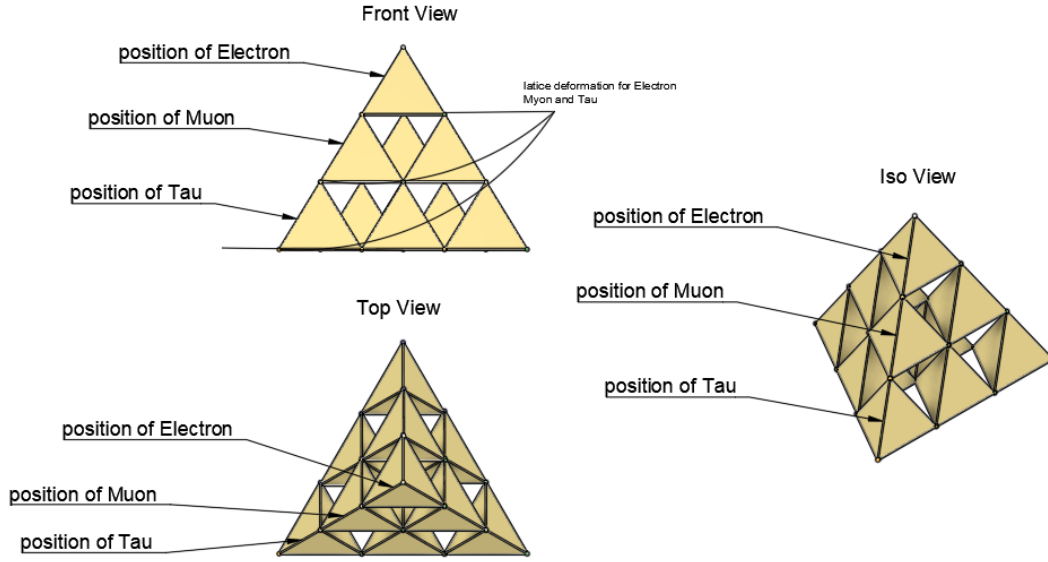


Figure 23: **Lattice deformation levels of leptons in the MMU.** The electron, muon, and tau correspond to successive deformation depths of the tetrahedral fractal lattice. *Front view (top left):* shows the hierarchical positions of electron, muon, and tau, with lattice deformation indicated. *Top view (bottom left):* illustrates the nested fractal structure from above. *Iso view (right):* highlights the spatial embedding of the three leptons at different deformation depths inside the lattice. The deeper the deformation, the higher the torsional storage and therefore the larger the lepton mass.

Appendix reference (muon & tau). The geometric staging of the charged leptons and the quantitative cross-checks (masses, hydrogenic scaling across $e/\mu/\tau$, and family patterns in $(g-2)$) are developed in Appendix Appendix 12. There we show that a 120° sub-tetrahedral phase fixes Koide's relation exactly and reproduces the PDG lepton masses while predicting reduced-mass scalings for μH and " τH " and parameter-free $(g-2)$ family ratios.

Quantity	MMU (computed)	Measurement / input
m_μ (MeV)	105.6583755	105.6583755
m_τ (MeV)	1776.86	1776.86
Koide parameter Q	$2/3$ (exact)	$\approx 2/3$ (empirical)
μ_μ/μ_e (reduced mass)	185.9420466	from PDG masses (same)
μ_τ/μ_e (reduced mass)	1202.2851846	from PDG masses (same)
$\lambda_{\text{Ly}\alpha}(\mu\text{H})$	0.653789728 nm	(X-ray; prediction)
$\lambda_{\text{Ly}\alpha}(\tau\text{H})$	0.101113281 nm	(hard X-ray; prediction)
$(g-2)$ ratio $\Delta a_\tau/\Delta a_\mu$	162.6892	(prediction)
$(g-2)$ ratio $\Delta a_e/\Delta a_\mu$	7.2326×10^{-5}	(prediction)
Calibrator Δa_μ	set by experiment	
Implied Δa_e	1.80815×10^{-13}	cf. current bounds
Implied Δa_τ	4.06723×10^{-7}	(future sensitivity)

Notes. (i) Mass rows use the PDG central values also reproduced by the MMU projection with scale $M = (\frac{\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau}}{3})^2 \simeq 313.8411267$ MeV.

15 Pair Creation in the MMU Framework

In the MMU, pair creation is interpreted as a complete *axis inversion* of an electron state. Normally, an electron projects onto the negative time axis $-w_1$ with its characteristic torsional and displacement components (w_2, w_3, w_4) .

Axis Inversion

When sufficient external energy is supplied, the projection flips from $-w_1$ to $+w_1$. This inversion has two consequences:

1. All three spatial axes (w_2, w_3, w_4) are inverted as well, yielding a mirror state of opposite charge.
2. The system thereby generates an *antiparticle* with identical magnitude of oscillation but reversed projection.

Energy Requirement

The inversion requires overcoming the rest energy of the electron in both directions:

$$E_{\text{pair}} = 2m_e c^2.$$

This is interpreted as the minimum energy needed to create the mirrored $+w_1$ projection and simultaneously maintain the original $-w_1$ projection. Thus, the appearance of an electron–positron pair is understood geometrically as the coexistence of two time-projections within the same MMU lattice.

Geometric Symmetry

The process highlights the fundamental MMU symmetry:

$$(w_1, w_2, w_3, w_4) \longrightarrow (-w_1, -w_2, -w_3, -w_4), \quad (1)$$

which acts as a discrete reflection across the origin of the tetrahedral lattice. Pair creation is therefore not an arbitrary phenomenon, but the natural consequence of a full axis inversion under sufficient energy input.

16 Atomic and Molecular Structure in the MMU

In classical chemistry, atomic bonding is described as the *sharing of electron pairs*. The nuclei remain distinct, while the electron clouds overlap to form covalent bonds.

In the MMU, this picture is reformulated geometrically:

- Each atom corresponds to a single *S*-node, embedded in the lattice.
- When two atoms bind, a **new resonance node** emerges between their *S*-nodes.
- This new node is not a new proton or particle, but a geometric coupling of the two local oscillation centers.
- As long as the bond exists, the molecular *S*-node behaves like a fusion of the two original nodes.

Examples

Hydrogen molecule (H_2). Two hydrogen *S*-nodes couple. A new molecular node forms exactly between them, representing their shared resonance axis. The geometry is linear, with a bond angle of 180° .

Water molecule (H_2O). The oxygen *S*-node provides four resonance axes. Two of these couple to hydrogen *S*-nodes, producing two new resonance nodes (the O–H bonds). The two remaining axes stay uncoupled and correspond to the lone pairs. Result: a bent geometry with an angle of 104.5° .

Methane (CH_4). The carbon *S*-node provides four equivalent resonance axes. Each couples to a hydrogen node, creating four new molecular resonance nodes. Together, these nodes form a perfect tetrahedral structure. Result: bond angles of 109.5° .

Conclusion

In the MMU:

- An **atom** is represented by a single *S*-node.
- A **molecule** is a network of new resonance *S*-nodes that arise between the original atomic nodes.

- The electron plays the role of mediator: it defines the axis along which new resonance nodes are formed.

This framework unifies atomic and molecular bonding as *geometric resonance phenomena* within the tetrahedral lattice.

17 From Atoms to Cosmic Structures in the MMU

The principle of resonance S -nodes is not limited to atoms and molecules, but extends fractally across scales up to astronomical and even cosmological systems.

Hierarchical Growth of S -Nodes

In atomic physics, we have seen that each atom corresponds to a single S -node, while molecules emerge from new resonance S -nodes formed between atomic nodes. The same principle applies on higher levels:

- **Planetary systems:** the Sun acts as a central S -node, with planets forming resonance nodes in its lattice. The orbital resonances and stability of planetary systems mirror molecular bonds on a higher scale.
- **Stellar constellations:** larger networks of stars can themselves form higher-order tetrahedral structures, coupled through their gravitational and electromagnetic resonance fields.

A Cosmic Tetrahedron

Our own solar system appears to be embedded in such a structure. On galactic scales, four major reference points can be associated with a tetrahedral resonance framework:

- **Sagittarius A*** (the supermassive black hole at the Galactic Center) – the central torsional anchor.
- **Orion region** – active stellar nursery, where new stars are born; analogous to dynamic vibrational modes.
- **Sirius** – the brightest star of the night sky, representing a strong radiant resonance point.
- **Aldebaran** – giant star in Taurus, associated with magnetic and spin-dominated processes.

Together, these nodes can be interpreted as vertices of a cosmic tetrahedron, within which our solar system resides.

The Universe as a Methane Molecule

If the MMU principle is applied consistently, the Universe itself may resemble – when viewed from the outside – a *giant methane-like molecule*:

- Each corner corresponds to a cosmic S -node, such as a galactic supercluster.

- Their resonance couplings define the large-scale structure of space-time.
- The balance of these tetrahedral nodes ensures isotropy and stability across the cosmos.

Implications for Biology

Life, built upon the methane molecule as its simplest tetrahedral archetype, may itself be an imprint of this cosmic geometry. The tetrahedral structure not only governs atoms and molecules, but also resonates in the architecture of cosmic systems. Thus, biology can be understood as a natural outcome of the same universal lattice principle: *life arises because the Universe itself is structured like a cosmic methane molecule.*

18 Critical Assessment

The MMU is intentionally minimalist: a single observational principle (only the w_1 projection is measurable) plus a tetrahedral microstructure are meant to make relativity, electrodynamics, and quantum envelopes *emerge*. This strength is also a potential source of weaknesses:

- **Lorentz invariance vs. discreteness.** The approach relies on long-wavelength isotropy and an effective continuum limit. Any residual lattice anisotropy (birefringence, energy-dependent c) is tightly constrained by experiments. The band-undeformed wave law ($v_g = c$ in the continuum, discrete corrections $\mathcal{O}((k\Delta s)^2)$) must remain consistent with astrophysical bounds; this fixes the admissible lattice spacing.
- **Gravitation.** The “Hooke reading” of Einstein ($E_{\text{eff}} = c^4/8\pi G$) is heuristically useful, but it does not replace a full account of GR’s nonlinearities. Our GR $\rightarrow$ Newton projection limit (Appendix Appendix 6) passes weak-field tests conceptually, yet cosmological constant, exact wave solutions and amplitudes still require quantitative development.
- **Quantum reduction.** Schrödinger/Dirac as (adiabatic) band projections need clear scale separation and control of band gaps. .
- **Hadronics and dual pictures.** The triad geometry explains fractional charge, color, and linear confinement, and the dual neutron picture $(p+e^-)_S$ reproduces the energy balance (Appendix Appendix 10). Nevertheless, consistency with the weak $V-A$ structure and neutrino kinematics must be maintained; the duality should be read as an *effective* S -node description, not as a replacement for the Standard Model.
- **Parameter economy vs. sensitivity.** Many results are “low-parameter” (anchored by α , r_p). Uncertainties in these anchors propagate directly into numerical checks (21 cm, isotope shifts, $(g-2)$ ratios) and should be stated transparently.
- **Outside observer / “Newton on the outside”.** The statement that a hypothetical outside observer would see “just Newton” (e.g. $E_{\text{kin}} = \frac{1}{2}mv^2$) is meant as an *interpretation* of the w_1 projection—not as a physically accessible frame and *not* in contradiction with $E = mc^2$ *inside* the lattice.

In short: the scheme is coherent provided that (i) the continuum limit is controlled, (ii) Lorentz tests are respected, and (iii) dual pictures are embedded consistently in standard decays and symmetries.

19 Outlook

The MMU makes several *decidable* predictions that can be targeted:

- **Edge impedance and energy flow.** Internally: $Z_{0,\text{int}} = \frac{P_3}{P_2} 1/\sqrt{2\alpha}$ and node reflection $\Gamma = -\frac{1}{2}$ for $1 \rightarrow 3$ splits (Appendix Appendix 3). Testable in grid simulations (Yee scheme) and metamaterial analogs (power split: 25% reflected, $3 \times 25\%$ transmitted).
- **Fundamental-band nondispersion.** Long-wavelength $v_g = c$ with discrete corrections $\propto \cos(k\Delta s/2)$; time-of-flight astrophysics constrains Δs .
- **21 cm without QED machinery.** The one-step estimate with internal spring constants hits the right scale (Appendix Appendix 2); higher-order corrections can be added systematically.
- **Isotope shifts and muonic scaling.** Reduced-mass and FNS contributions ($\propto \mu^3$) give the correct orders (H/D/T, μH) (Appendix Appendix 8); precision data can sharpen MMU contact-form factors.
- **$(g-2)$ family ratios.** Parameter-free ratios $\Delta a_\tau/\Delta a_\mu$ and $\Delta a_e/\Delta a_\mu$ (Appendix Appendix 12) are clear targets for future measurements.
- **Decay statistics.** The Eyring–Kramers form $k(Q) = k_0 e^{\beta Q}$ reproduces n and ${}^3\text{H}$ (Appendix Appendix 11); additional low- Q β decays can test linearity in the exponent.
- **Cosmology.** The “methane-molecule” analogy for the cosmos is a heuristic; building a consistent FRW limit (mean S -node density, effective stiffness, projection) is promising but open.

20 Conclusion

The MMU shows how far one can get with *one* observational principle (time projection) and a *simple* geometry (a tetrahedron): light as a pure w_2 – w_3 oscillation with fixed impedance; Newton emerging from GR in the weak field via clock projection; Schrödinger/Dirac as band projections; atomic scales (21 cm, isotope shifts) and coarse hadronic/leptonic patterns from triads and staging—achieved with very little tuning. The approach is *valuable* precisely where it *could* fail: impedance, reflection coefficients, dispersion curves, family ratios, and precision spectra. This falsifiability makes it worth a full read—for specialists and curious readers alike: either a surprisingly sparse picture survives many tests, or we learn exactly where and why Nature demands more than a methane geometry.

emergent spacetime; information projection; lattice models; hydrogen spectroscopy; isotope shifts; hyperfine 21 cm; vacuum impedance; anomalous magnetic moment

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Appendix 1 Elastic and Shear Moduli of Space in the MMU: A Dual Derivation and Unification via Time Projection

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August 30, 2025

Abstract

We derive the effective elastic modulus E and shear modulus G of space in the Methane Metauniverse (MMU) in two complementary ways: (i) a macroscopic (continuum) route from Einstein’s field equation, yielding $E_{\text{eff}} = c^4/(8\pi G)$, and (ii) a microscopic (cellular) route from the S-node spring constant extracted from lepton mass geometry, leading to an fm-scale modulus $E_{\text{cell}} \sim \zeta k_S/a$. We give complete numerical evaluations, show consistency conditions for wave speeds and Poisson’s ratio, and unify both moduli by the MMU time-projection rule, which explains their scale separation without contradiction.

1 Constants and Conventions

We use (CODATA-level) constants:

$$\begin{aligned} c &= 2.997\,924\,58 \times 10^8 \text{ m/s}, & G &= 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}, \\ \hbar &= 1.054\,571\,817 \times 10^{-34} \text{ J s}, & 1 \text{ MeV} &= 1.602\,176\,634 \times 10^{-13} \text{ J}. \end{aligned}$$

Planck length $l_p = \sqrt{\hbar G/c^3} \approx 1.616 \times 10^{-35} \text{ m}$.

2 Macroscopic (Continuum) Derivation from Einstein’s Equation

Einstein’s field equation $G_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$ admits a formal Hooke-type reading: “curvature” $\propto$ “stress” with an effective (continuum) elastic modulus

$$E_{\text{eff}} = \frac{c^4}{8\pi G}.$$

Numerically,

$$E_{\text{eff}} = \frac{(2.997\,924\,58 \times 10^8 \text{ m/s})^4}{8\pi \cdot 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}} \approx 4.8 \times 10^{42} \text{ Pa}.$$

The corresponding effective density (by relativistic stiffness $E = \rho c^2$) is

$$\rho_{\text{eff}} = \frac{E_{\text{eff}}}{c^2} = \frac{c^2}{8\pi G} \approx 5.3 \times 10^{25} \text{ kg/m}^3.$$

2.1 Shear modulus, bulk modulus, and wave speeds

For an isotropic medium,

$$E = 2G(1 + \nu), \quad K = \frac{E}{3(1 - 2\nu)}.$$

The longitudinal and shear wave speeds are

$$v_p = \sqrt{\frac{K + \frac{4}{3}G}{\rho}}, \quad v_s = \sqrt{\frac{G}{\rho}}.$$

Imposing the MMU relativistic consistency (no mode faster than c) and saturating $v_p = c$ with $E = \rho c^2$ yields $\nu = 0$. Hence

$$\nu = 0, \quad G_{\text{eff}} = \frac{E_{\text{eff}}}{2} \approx 2.4 \times 10^{42} \text{ Pa}, \quad K_{\text{eff}} = \frac{E_{\text{eff}}}{3} \approx 1.6 \times 10^{42} \text{ Pa}$$

and

$$v_p = c, \quad v_s = \frac{c}{\sqrt{2}}.$$

3 Microscopic (Cellular) Derivation from the MMU S-Node

In the MMU, lepton masses m_e, m_μ, m_τ arise from stored torsional energy at an S-node depth d , modeled by a quadratic spring:

$$m_i c^2 = \frac{1}{2} k_S d_i^2.$$

Using the lepton triplet geometry one extracts a common mass scale M and defines

$$M c^2 = \frac{1}{2} k_S d_0^2 \Rightarrow k_S = \frac{2M c^2}{d_0^2}.$$

With the fitted scale $M = 313.838 \text{ MeV}$ and S-length $d_0 = R_S = 0.906 \text{ fm} = 0.906 \times 10^{-15} \text{ m}$, it is convenient to compute in MeV/fm^2 :

$$k_S [\text{MeV/fm}^2] = \frac{2M}{d_0^2} = \frac{2 \times 313.838}{(0.906)^2} = 764.679 \text{ MeV/fm}^2.$$

Unit conversion $1 \text{ MeV/fm}^2 = 1.602176634 \times 10^{17} \text{ N/m}$ gives

$$k_S \approx 764.679 \times 1.602176634 \times 10^{17} \text{ N/m} = 1.225 \times 10^{20} \text{ N/m}.$$

3.1 From k_S to an fm-scale modulus

For a (nearly) isotropic tetrahedral lattice element of size $a \simeq R_S$, a standard estimate relates a macroscopic modulus to the microscopic spring as

$$E_{\text{cell}} \approx \zeta \frac{k_S}{a}, \quad \zeta \in [0.3, 1] \text{ (geometry factor)}.$$

Numerically, with $a = 0.906 \times 10^{-15} \text{ m}$:

$$\frac{k_S}{a} = \frac{1.225 \times 10^{20}}{0.906 \times 10^{-15}} \approx 1.352 \times 10^{35} \text{ Pa}.$$

Thus

$$E_{\text{cell}} \approx (0.4-1.3) \times 10^{35} \text{ Pa},$$

with a central estimate (e.g. $\zeta = 0.7$):

$$E_{\text{cell, cent}} \approx 0.7 \times 1.352 \times 10^{35} \text{ Pa} \approx 9.5 \times 10^{34} \text{ Pa}.$$

Assuming the same MMU condition $\nu \simeq 0$ at cell level (internal angular springs included),

$$G_{\text{cell}} \approx \frac{1}{2} E_{\text{cell}} \sim (0.2-0.65) \times 10^{35} \text{ Pa}.$$

4 Unification via Time Projection

In the MMU, the measurable energy is the projection of the 4D energy vector onto the time axis w_1 :

$$E_{w_1} = h w_1 = h \frac{w_1}{\sqrt{w_1^2 + w_2^2 + w_3^2 + w_4^2}} w_{\text{tot}} = E_{\text{tot}} \cos \theta,$$

with $\cos \theta := w_1 / \sqrt{w_1^2 + w_2^2 + w_3^2 + w_4^2}$. It is natural to define the observed modulus as the projected fraction of the total stiffness:

$$E_{\text{obs}} = E_{\text{tot}} \cos \theta, \quad G_{\text{obs}} = G_{\text{tot}} \cos \theta.$$

Interpretation:

- **Continuum/GR limit** ($\theta \rightarrow 0$, energy almost fully in w_1): $\cos \theta \rightarrow 1$, hence $E_{\text{obs}} \rightarrow E_{\text{eff}} = c^4 / (8\pi G) \approx 4.8 \times 10^{42} \text{ Pa}$ and $G_{\text{obs}} \rightarrow G_{\text{eff}} \approx 2.4 \times 10^{42} \text{ Pa}$.
- **Cell/fm limit** (large shares in w_2, w_3, w_4): $\cos \theta \ll 1$, so the measured modulus is reduced to the fm-scale value $E_{\text{cell}} \sim 10^{35} \text{ Pa}$ and $G_{\text{cell}} \sim 10^{35}/2 \text{ Pa}$.

Thus, the time projection provides a quantitative *scale bridge* between the Einstein continuum stiffness and the MMU cellular stiffness.

5 Numerical Summary

Quantity	Value	Comment
E_{eff}	$\approx 4.8 \times 10^{42} \text{ Pa}$	continuum (Einstein) modulus
G_{eff}	$\approx 2.4 \times 10^{42} \text{ Pa}$	with $\nu = 0$
K_{eff}	$\approx 1.6 \times 10^{42} \text{ Pa}$	bulk modulus
ρ_{eff}	$\approx 5.3 \times 10^{25} \text{ kg/m}^3$	via $E = \rho c^2$
k_S	$\approx 1.225 \times 10^{20} \text{ N/m}$	from lepton mass scale $M = 313.838 \text{ MeV}$, $R_S = 0.906 \text{ fm}$
E_{cell}	$(0.4 - 1.3) \times 10^{35} \text{ Pa}$	using $E \approx \zeta k_S / a$, $a \simeq R_S$
G_{cell}	$(0.2 - 0.65) \times 10^{35} \text{ Pa}$	$\nu \simeq 0 \Rightarrow G \approx E/2$

6 Discussion

The two derivations refer to *different but compatible* regimes of the same MMU lattice: the Planck-near continuum stiffness (E_{eff}) and the fm-scale cellular stiffness (E_{cell}). Their apparent discrepancy is resolved by the MMU time projection, which explains why the observable modulus at nuclear scales is much lower: energy shares stored in w_2, w_3, w_4 reduce the w_1 -projection, i.e. the measured stiffness. At macroscopic scales with negligible side components, the Einstein modulus is recovered.

Appendix 2 Internal Spring Forces in the MMU Lattice

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28/8/2025

0.1 Local spring constants in the MMU: definition, derivation, verification

Objective. We determine the local spring stiffnesses of the MMU tetrahedron for *Coulomb* (k_C), *spin/torsion* (k_S), and *gravity* (k_G). The three contributions are uniquely assigned to the orthogonal degrees of freedom:

$$\boxed{w_2 \leftrightarrow k_C, \quad w_3 \leftrightarrow k_S, \quad w_4 \leftrightarrow k_G}.$$

Parameters. Let the lattice spacing (local length scale) be a . The mass appearing in k_G is denoted m (real particle mass or effective mode mass). For normalization we use the *Planck mass*

$$m_{\text{Pl}} = \sqrt{\frac{\hbar c}{G}} \quad \Rightarrow \quad G m_{\text{Pl}}^2 = \hbar c.$$

Geometrical assumptions. (i) The local fundamental mode circulates around the node with circumference $\lambda = 2\pi a$, hence

$$\omega = \frac{2\pi c}{\lambda} = \frac{c}{a}.$$

(ii) A torsional deflection θ corresponds to a tangential displacement $x = a\theta$. Thus

$$U = \frac{1}{2}k_\theta\theta^2 = \frac{1}{2}\frac{k_\theta}{a^2}x^2 \quad \Rightarrow \quad k_S = \frac{k_\theta}{a^2}.$$

(iii) The modal energy scale is fixed by $k_\theta \equiv \hbar\omega$.

Definition of the spring constant. For a radial force $F(r)$:

$$k(a) = \left| \frac{dF}{dr} \right|_{r=a}, \quad [k] = \text{N/m}.$$

(1) **Coulomb contribution on w_2**

$$F_C(r) = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} \quad \Rightarrow \quad k_C(a) = \frac{2}{4\pi\epsilon_0} \frac{e^2}{a^3}.$$

(2) Spin/torsion contribution on w_3

With $\omega = c/a$ and $k_\theta = \hbar\omega$:

$$k_S(a) = \frac{\hbar c}{a^3}.$$

(3) Gravitational contribution on w_4

$$F_G(r) = \frac{G m^2}{r^2} \quad \Rightarrow \quad k_G(a) = \frac{2G m^2}{a^3}.$$

For the Planck normalization $m = m_{\text{Pl}}$ this becomes

$$k_G(a) = \frac{2\hbar c}{a^3}.$$

Scale invariance and ratios of constants (general). All three stiffnesses scale as a^{-3} , hence their ratios are scale-free:

$$\frac{k_C}{k_S} = 2\alpha, \quad \frac{k_G}{k_S} = 2\left(\frac{m}{m_{\text{Pl}}}\right)^2, \quad \frac{k_C}{k_G} = \alpha\left(\frac{m_{\text{Pl}}}{m}\right)^2,$$

with $\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c}$.

Special case (Planck normalization $m = m_{\text{Pl}}$).

$$\boxed{\frac{k_C}{k_S} = 2\alpha, \quad \frac{k_G}{k_S} = 2, \quad \frac{k_C}{k_G} = \alpha}.$$

Numerics (example $a = l_p$, $m = m_{\text{Pl}}$).

$$k_C \approx 1.09 \times 10^{77} \text{ N/m}, \quad k_S \approx 7.49 \times 10^{78} \text{ N/m}, \quad k_G \approx 1.50 \times 10^{79} \text{ N/m}.$$

Table 1: Assignment of spring stiffnesses to axes and formulas

Axis	Spring stiffness	Formula
w_2	Coulomb (k_C)	$k_C = \frac{2}{4\pi\epsilon_0} \frac{e^2}{a^3}$
w_3	Spin/torsion (k_S)	$k_S = \frac{\hbar c}{a^3}$
w_4	Gravitation (k_G)	$k_G = \frac{2Gm^2}{a^3}$
	(Planck norm.)	$k_G = \frac{2\hbar c}{a^3} \quad (m=m_{\text{Pl}})$

Table 2: Scale-free consistency relations (general and Planck normalization)

Ratio	General	For $m = m_{\text{Pl}}$
k_C/k_S	2α	2α
k_G/k_S	$2(m/m_{\text{Pl}})^2$	2
k_C/k_G	$\alpha(m_{\text{Pl}}/m)^2$	α

Remark. For real particle masses $m \ll m_{\text{Pl}}$, $k_G \propto m^2$ becomes extremely small.

0.2 Continuum derivation via dispersion (confirmation of the internal spring stiffnesses)

Objective. We confirm the local torsion/spin stiffness k_S (axis w_3) without using the direct potential curvature, but via wave propagation and continuum elasticity.

Approach (isotropic, shear-elastic continuum). We model the cell of edge length a as an isotropic continuum with shear modulus G , de

Quantization of the fundamental mode. Assigning the fundamental mode the minimal modal energy $E_0 = \hbar\omega$ (normalization as in the discrete model), the cell contains an equivalent mass m_{cell} with

$$m_{\text{cell}}c^2 = \hbar\omega = \frac{\hbar c_T}{a}, \quad \rho = \frac{m_{\text{cell}}}{a^3} = \frac{\hbar c_T}{c^2 a^4}.$$

Substituting into $c_T^2 = G/\rho$ yields

$$G = \rho c_T^2 = \frac{\hbar c_T^3}{c^2 a^4}. \quad (1)$$

For the MMU lattice we take $c_T \simeq c$ (light-like fundamental mode). Thus

$$G \simeq \frac{\hbar c}{a^4}. \quad (2)$$

From (??) and (2) it follows immediately that

$$k_S^{(\text{cont})} = G a \simeq \frac{\hbar c}{a^3}$$

which is exactly the torsional stiffness k_S previously obtained from mode quantization, *without* recourse to a discrete spring model.

Embedding k_C and k_G in the same framework.

- **Coulomb (w_2).** The electromagnetic energy density $u_{\text{EM}} = \frac{1}{2}\varepsilon_0 E^2$ and the Maxwell stress tensor yield, for a quasi-static radial displacement, an effective electrostatic stiffness per cell. For a point-charge field on the scale a , the potential curvature gives

$$k_C = \frac{2}{4\pi\varepsilon_0} \frac{e^2}{a^3} = 2\alpha \frac{\hbar c}{a^3} = 2\alpha k_S.$$

- **Gravitation (w_4).** Analogous to the Coulomb curvature with $e^2/(4\pi\varepsilon_0) \rightarrow Gm^2$, the continuum curvature of the Newton potential gives

$$k_G = \frac{2Gm^2}{a^3} = 2\left(\frac{m}{m_{\text{Pl}}}\right)^2 \frac{\hbar c}{a^3} = 2\left(\frac{m}{m_{\text{Pl}}}\right)^2 k_S,$$

where $m_{\text{Pl}} = \sqrt{\hbar c/G}$ is the Planck mass. In the special case $m = m_{\text{Pl}}$ we obtain $k_G = 2k_S$.

Conclusion. The continuum/dispersion viewpoint yields (i) $k_S = Ga$ from pure kinematics, (ii) via $c_T^2 = G/\rho$ and $m_{\text{cell}}c^2 = \hbar\omega$ with $\omega = c_T/a$ the relation $G \simeq \hbar c/a^4$, and thus (iii)

$$k_S = \frac{\hbar c}{a^3}, \quad k_C = 2\alpha k_S, \quad k_G = 2\left(\frac{m}{m_{\text{Pl}}}\right)^2 k_S.$$

For Planck normalization $m = m_{\text{Pl}}$ this reduces to $k_C/k_S = 2\alpha$ and $k_G/k_S = 2$; thereby the scale-free ratios from dispersion + continuum are confirmed.

Lemma (quadrature w_2 – w_3 for light-like edge wave). Along an edge with coordinate s , $w_2(s, t)$ (longitudinal energy shift) and $w_3(s, t)$ (torsional displacement) couple via

$$\partial_t w_2 = -c \partial_s w_3, \quad (3)$$

$$\partial_t w_3 = -c \partial_s w_2. \quad (4)$$

For plane waves $w_{2,3}(s, t) = \Re\{W_{2,3} e^{i(k_S s - \omega t)}\}$ with $\omega = c|k|$ the amplitudes are in temporal quadrature:

$$w_3 = \pm i w_2 \implies w_3 \text{ is shifted by } \pm 90^\circ \text{ relative to } w_2$$

1 Appendix: Test Protocol for the MMU Spring Model

This appendix describes a step-by-step test ladder for the MMU spring model. We proceed from dimensional checks through local curvature tests to dynamic edge waves, energy balance, multi-cell dispersion, and simple spectroscopic benchmarks. The measurable energy is the projection along w_1 ; internal storage axes are w_2 (Coulomb), w_3 (spin/torsion), w_4 (gravitation).

Notation

$$k_C = \frac{2}{4\pi\epsilon_0} \frac{e^2}{a^3}, \quad k_S = \frac{\hbar c}{a^3}, \quad k_G = \frac{2Gm^2}{a^3}, \quad \alpha = \frac{e^2}{4\pi\epsilon_0 \hbar c}.$$

T0 – Dimension and scaling check

Goal. Immediate check without numerics.

Method. Verify units $[k_i] = \text{N/m}$ and $k_i \propto a^{-3}$.

Definitions.

$$k_C(a) = \frac{2}{4\pi\epsilon_0} \frac{e^2}{a^3}, \quad k_S(a) = \frac{\hbar c}{a^3}, \quad k_G(a) = \frac{2Gm^2}{a^3}, \quad m_{\text{Pl}} = \sqrt{\frac{\hbar c}{G}}, \quad \alpha = \frac{e^2}{4\pi\epsilon_0 \hbar c}.$$

Proof. (1) *Units:* $e^2/(4\pi\epsilon_0)$ has the dimension of energy times length (J·m), as does $\hbar c$; division by a^3 yields J/m² = N/m. For k_G , $Gm^2/a^3 \sim (\text{N m}^2/\text{kg}^2) \cdot \text{kg}^2/\text{m}^3 = \text{N/m}$.
(2) *Scaling:* In all three cases, a^{-3} appears in the numerator, thus $k_i \propto a^{-3}$.
(3) *Ratios:*

$$\frac{k_C}{k_S} = \frac{\frac{2}{4\pi\epsilon_0} \frac{e^2}{a^3}}{\frac{\hbar c}{a^3}} = 2 \frac{e^2}{4\pi\epsilon_0 \hbar c} = 2\alpha,$$

$$\frac{k_G}{k_S} = \frac{\frac{2Gm^2}{a^3}}{\frac{\hbar c}{a^3}} = 2 \frac{Gm^2}{\hbar c} = 2 \left(\frac{m}{m_{\text{Pl}}} \right)^2, \quad \frac{k_C}{k_G} = \frac{\frac{2}{4\pi\epsilon_0} \frac{e^2}{a^3}}{\frac{2Gm^2}{a^3}} = \frac{e^2}{4\pi\epsilon_0 Gm^2} = \alpha \left(\frac{m_{\text{Pl}}}{m} \right)^2.$$

Thus, the ratios are length-scale free.

Acceptance (Planck normalization). For $m = m_{\text{Pl}}$, the general relations reduce to

$$\boxed{\frac{k_C}{k_S} = 2\alpha, \quad \frac{k_G}{k_S} = 2, \quad \frac{k_C}{k_G} = \alpha}.$$

T1 – Numerical quick checks (single cell)

Goal. Numerical stability, no hidden 2π factors.

Method. Choose $a = l_p = \sqrt{\hbar G/c^3}$ and $m = m_{\text{Pl}} = \sqrt{\hbar c/G}$. Compute

$$k_C = \frac{2}{4\pi\epsilon_0} \frac{e^2}{l_p^3}, \quad k_S = \frac{\hbar c}{l_p^3}, \quad k_G = \frac{2\hbar c}{l_p^3},$$

as well as the ratios k_C/k_S , k_G/k_S , k_C/k_G and compare with 2α , 2 , α , where $\alpha = \frac{e^2}{4\pi\epsilon_0 \hbar c}$.

Result (numerics). With CODATA values:

$$k_C \approx 1.0929 \times 10^{77} \text{ N/m}, \quad k_S \approx 7.4880 \times 10^{78} \text{ N/m}, \quad k_G \approx 1.4976 \times 10^{79} \text{ N/m},$$

$$\alpha \approx 7.2973526 \times 10^{-3}, \quad 2\alpha \approx 1.4594705 \times 10^{-2}.$$

Ratios:

$$\frac{k_C}{k_S} \approx 1.4594705 \times 10^{-2} = 2\alpha, \quad \frac{k_G}{k_S} = 2.0000000 = 2, \quad \frac{k_C}{k_G} \approx 7.2973526 \times 10^{-3} = \alpha.$$

Acceptance. The scale-free ratios match 2α and 2 (Planck normalization) numerically up to rounding:

$$\boxed{\frac{k_C}{k_S} = 2\alpha, \quad \frac{k_G}{k_S} = 2, \quad \frac{k_C}{k_G} = \alpha}.$$

T2 – Energy as projection onto w_1 (quasi-static)

Goal. Verify that the energy measured along w_1 originates from internal springs (w_2, w_3).

Method. Local stored energies:

$$U_2 = \frac{1}{2}k_C x_2^2, \quad U_3 = \frac{1}{2}k_S x_3^2, \quad U_4 = \frac{1}{2}k_G x_4^2.$$

A weak outflow to w_1 is modeled as Rayleigh coupling:

$$R = \frac{1}{2} \gamma (\dot{x}_2^2 + \dot{x}_3^2),$$

so that the mean power output (measured quantity) is

$$\overline{P}_{w_1} = \overline{2R} = \gamma \overline{\dot{x}_2^2 + \dot{x}_3^2}.$$

Theorem (energy outflow and internal stored energy). Consider the linear equations

$$m_2 \ddot{x}_2 + k_C x_2 + \gamma \dot{x}_2 = 0, \quad m_3 \ddot{x}_3 + k_S x_3 + \gamma \dot{x}_3 = 0,$$

with $m_2, m_3 > 0$ and small $\gamma \geq 0$. Then, in the stationary harmonic regime:

$$\boxed{\overline{P}_{w_1} = \gamma \overline{\dot{x}_2^2 + \dot{x}_3^2} = \frac{2\gamma}{m_2} \overline{U_2} + \frac{2\gamma}{m_3} \overline{U_3}}.$$

In the *isotropic* special case $m_2 = m_3 =: m_{\text{cell}}$:

$$\boxed{\overline{P}_{w_1} = \frac{2\gamma}{m_{\text{cell}}} \overline{U_2 + U_3} \propto \overline{U_2 + U_3}}.$$

Proof. Define, for $i \in \{2, 3\}$, the total energies $E_i = \frac{1}{2}m_i \dot{x}_i^2 + \frac{1}{2}k_i x_i^2$. Multiplying the equations of motion by $\dot{x}_i$ and summing the potential and kinetic terms yields

$$\frac{dE_i}{dt} = -\gamma \dot{x}_i^2.$$

Summing over $i = 2, 3$:

$$\frac{d}{dt}(E_2 + E_3) = -\gamma(\dot{x}_2^2 + \dot{x}_3^2).$$

In the stationary harmonic state, the time-average of the derivative is zero, thus

$$\overline{P}_{w_1} := \gamma \overline{\dot{x}_2^2 + \dot{x}_3^2} = -\overline{\frac{d}{dt}(E_2 + E_3)} = 0 + (\text{input}).$$

For a single linear oscillator, the time-averaged kinetic and potential energies are equal ($\overline{T_i} = \overline{U_i}$, virial theorem) and $\overline{T_i} = \frac{1}{2}m_i \overline{\dot{x}_i^2}$. Therefore:

$$\overline{\dot{x}_i^2} = \frac{2}{m_i} \overline{T_i} = \frac{2}{m_i} \overline{U_i},$$

and hence:

$$\overline{P}_{w_1} = \gamma \left(\frac{2}{m_2} \overline{U_2} + \frac{2}{m_3} \overline{U_3} \right).$$

In the isotropic case $m_2 = m_3$, this simplifies to the claimed proportionality.

Corollary (No outflow for ideal standing wave or $\gamma = 0$). (i) If the coupling is switched off ($\gamma = 0$), then $\overline{P}_{w_1} = 0$.

(ii) If the outflow at the boundary is modeled via the Poynting analogue $S = c w_2 w_3$ (instead of via R), the mean boundary power $\overline{S}$ vanishes for an *ideal standing wave* (superposition of equally strong forward and backward waves), even though $\overline{\dot{x}_i^2} \neq 0$ holds.

Remark (Why U_4 is absent). For real masses $m \ll m_{\text{Pl}}$, $k_G \propto m^2$ is extremely small, and the direct outflow to w_1 is dominated by w_2/w_3 . Therefore, U_4 does not appear in the quasi-static outflow model; cf. also the scale-free ratios in A.1.

T3 – Phase relation w_2 – w_3 (1D edge, linear, corrected)

Goal. Precisely determine the temporal phase relation for lightlike propagation.

Method. Coupled equations along the edge (coordinate s)

$$\partial_t w_2 = -c \partial_s w_3 - \gamma w_2, \quad (5)$$

$$\partial_t w_3 = -c \partial_s w_2 - \gamma w_3. \quad (6)$$

Theorem (traveling waves: dispersion and phase relation). For plane waves $w_j(s, t) = \Re\{W_j e^{i(ks - \omega t)}\}$, (5)–(6) yield

$$\omega = c|k| - i\gamma, \quad \frac{W_3}{W_2} = \text{sgn}(k) \frac{\omega}{\omega + i\gamma}.$$

Thus, the *temporal* phase difference is

$$\Delta\phi(w_2, w_3) = \arg\left(\frac{W_3}{W_2}\right) = -\arctan\left(\frac{\gamma}{\Re\omega}\right) \pmod{\pi},$$

so $\Delta\phi \rightarrow 0$ (or π) for $\gamma \rightarrow 0$. There is *no* 90° quadrature between w_2 and w_3 for pure traveling waves.

Proof. Substituting $w_j = W_j e^{i(ks - \omega t)}$ into (5)–(6) yields $(-i\omega + \gamma)W_2 = ick W_3$, $(-i\omega + \gamma)W_3 = -ick W_2$. Nontrivial solutions require $(-i\omega + \gamma)^2 + c^2 k^2 = 0 \Rightarrow \omega = c|k| - i\gamma$. From this follows $W_3/W_2 = (-i\omega + \gamma)/(ick) = \text{sgn}(k) \omega/(\omega + i\gamma)$ and thus the stated phase relation.

Lemma (Quadrature with spatial derivative). Define $\hat{w}_3 := \frac{c}{\omega} \partial_s w_3$. Then, for the amplitudes,

$$\frac{\hat{W}_3}{W_2} = \frac{c(ik)}{\omega} \frac{W_3}{W_2} = i \text{sgn}(k) \frac{\omega}{\omega + i\gamma},$$

so

$$\Delta\phi(w_2, \hat{w}_3) = \frac{\pi}{2} - \arctan\left(\frac{\gamma}{\Re\omega}\right) \xrightarrow{\gamma \rightarrow 0} \pm \frac{\pi}{2}.$$

Thus, there is *temporal quadrature* between w_2 and $\frac{c}{\omega} \partial_s w_3$.

Proposition (standing wave: local quadrature w_2 – w_3). For $\gamma = 0$ and the superposition of equally strong counter-propagating waves ($\pm k$),

$$w_2 = 2A \cos(ks) \cos(\omega t), \quad w_3 = -2A \sin(ks) \sin(\omega t),$$

which implies, at points with $\cos(ks) \sin(ks) \neq 0$:

$$\Delta\phi(w_2, w_3) = \pm \frac{\pi}{2}.$$

At nodes/equator ($\cos ks = 0$ or $\sin ks = 0$), one of the amplitudes vanishes.

Acceptance (corrected).

1. **Traveling wave:** $\Delta\phi(w_2, w_3) \approx -\arctan(\gamma/\omega) \approx 0$ (or π), and $\Delta\phi(w_2, \frac{c}{\omega}\partial_s w_3) \approx \pm\frac{\pi}{2}$.
2. **Standing wave:** Locally (far from nodes) $\Delta\phi(w_2, w_3) \approx \pm\frac{\pi}{2}$.

T4 – Local Energy Balance (Poynting Analogue)

Goal. Conservation of energy between internal energy and outflow.

Method. Along the edge (coordinate s) we consider the linear system

$$\partial_t w_2 = -c \partial_s w_3 - \gamma w_2, \quad (7)$$

$$\partial_t w_3 = -c \partial_s w_2 - \gamma w_3, \quad (8)$$

with damping $\gamma \geq 0$. Define the energy density and power flux as

$$u = \frac{1}{2} (w_2^2 + w_3^2), \quad S = c w_2 w_3.$$

Theorem (Continuity equation). From (7)–(8) it follows that

$$\boxed{\partial_t u + \partial_s S = -2\gamma u}.$$

Proof. Differentiating u with respect to time and inserting (7)–(8) gives

$$\partial_t u = w_2 \partial_t w_2 + w_3 \partial_t w_3 = -c(w_2 \partial_s w_3 + w_3 \partial_s w_2) - \gamma(w_2^2 + w_3^2).$$

By the product rule,

$$\partial_s S = \partial_s (c w_2 w_3) = c((\partial_s w_2)w_3 + w_2(\partial_s w_3)).$$

Thus,

$$\partial_t u + \partial_s S = -\gamma(w_2^2 + w_3^2) = -2\gamma \frac{1}{2} (w_2^2 + w_3^2) = -2\gamma u.$$

Acceptance. In the lossless limit $\gamma \rightarrow 0$ we have locally

$$\boxed{\partial_t u + \partial_s S = 0} \quad (\text{energy conservation}).$$

Remark (Sign convention). If instead one uses the mixed form $\partial_t w_2 = +c\partial_s w_3 - \gamma w_2$, $\partial_t w_3 = -c\partial_s w_2 - \gamma w_3$, then the continuity equation holds with the *negated* flux $\tilde{S} := -c w_2 w_3$:

$$\partial_t u + \partial_s \tilde{S} = -2\gamma u.$$

Both conventions are equivalent; choose one consistently across the project.

T5 – Continuum Elasticity $\Rightarrow k_S$ (Cross-Check)

Goal. Independent confirmation of k_S from dispersion.

Approach. We model the cell (edge length a , volume $V \simeq a^3$) as an isotropic, shear-elastic continuum with shear modulus G , mass density ρ , and transverse phase velocity

$$c_T = \sqrt{\frac{G}{\rho}}.$$

The fundamental mode on the cell has wavenumber $k = 2\pi/\lambda$ with $\lambda = 2\pi a$, thus

$$\omega = \frac{c_T}{a}.$$

Lemma (Kinematics $\Rightarrow k_S^{(\text{cont})} = G a$). A small node rotation θ produces a tangential displacement $x = a \theta$ and an approximately homogeneous shear $\gamma \simeq \theta = x/a$. The stored elastic energy per cell is

$$U = \frac{1}{2} G \gamma^2 V \simeq \frac{1}{2} G \left(\frac{x}{a}\right)^2 a^3 = \frac{1}{2} (G a) x^2.$$

Comparison with $U = \frac{1}{2} k x^2$ yields

$$\boxed{k_S^{(\text{cont})} = G a}. \quad (9)$$

Unit check: $[G] = \text{Pa} = \text{N/m}^2$, thus $[G a] = \text{N/m}$.

Lemma (Dispersion $\Rightarrow G$ from $\hbar$). We normalize the modal energy of the fundamental mode by $E_0 = \hbar\omega$. The equivalent cell mass is m_{cell} with

$$m_{\text{cell}} c^2 = E_0 = \hbar\omega = \frac{\hbar c_T}{a} \implies \rho = \frac{m_{\text{cell}}}{a^3} = \frac{\hbar c_T}{c^2 a^4}.$$

From $c_T^2 = G/\rho$ it follows that:

$$\boxed{G = \rho c_T^2 = \frac{\hbar c_T^3}{c^2 a^4}}. \quad (10)$$

Theorem (Continuum $\Rightarrow k_S$). From (9) and (10) we obtain

$$\boxed{k_S^{(\text{cont})} = G a = \frac{\hbar c_T^3}{c^2 a^3} = \left(\frac{c_T}{c}\right)^3 \frac{\hbar c}{a^3}}.$$

For a lightlike shear fundamental mode $c_T \simeq c$ this gives

$$\boxed{k_S^{(\text{cont})} \simeq \frac{\hbar c}{a^3} = k_S}.$$

Unit check: $[\hbar c] = \text{J m}$, division by a^3 yields N/m .

Acceptance.

If $c_T \simeq c$ (MMU fundamental mode lightlike), then $k_S^{(\text{cont})} \approx k_S$ (within numerical tolerance).

T6 – Multi-Cell Chain: Dispersion and Group Velocity

Goal. Robustness of propagation at c and the quadrature property in a chain of N cells.

Model (discrete in space, continuous in time). Lattice points $s_n = n\Delta s$ (cell centers), with staggered storage: $w_{2,n}(t)$ at s_n , $w_{3,n+\frac{1}{2}}(t)$ at $s_{n+\frac{1}{2}} = (n + \frac{1}{2})\Delta s$. The edge dynamics are

$$\partial_t w_{2,n} = \frac{c}{\Delta s} \left(w_{3,n+\frac{1}{2}} - w_{3,n-\frac{1}{2}} \right) - \gamma w_{2,n}, \quad (11)$$

$$\partial_t w_{3,n+\frac{1}{2}} = -\frac{c}{\Delta s} \left(w_{2,n+1} - w_{2,n} \right) - \gamma w_{3,n+\frac{1}{2}}. \quad (12)$$

(Absorbing boundaries are not required for the *proof*; they are only used in the simulation.)

Theorem (Discrete dispersion and group velocity). For plane-wave solutions

$$w_{2,n}(t) = A e^{i(n\kappa - \omega t)}, \quad w_{3,n+\frac{1}{2}}(t) = B e^{i((n+\frac{1}{2})\kappa - \omega t)} \quad (\kappa := k\Delta s)$$

it follows from (11)–(12) that the (damped) dispersion relation is

$$\boxed{\omega(\kappa) = \frac{2c}{\Delta s} \sin\left(\frac{|\kappa|}{2}\right) - i\gamma}.$$

In the lossless case $\gamma = 0$, the *group velocity* is

$$\boxed{v_g(\kappa) = \frac{d\Re\omega}{dk} = c \cos\left(\frac{\kappa}{2}\right) \operatorname{sgn}(k), \quad |v_g| \leq c}.$$

In the long-wavelength limit ($|\kappa| \ll 1$) we have $\omega \approx c|k|$ and $v_g \approx c$ (fundamental band linear).

Proof. Insert the plane-wave ansatz into (11)–(12):

$$(-i\omega + \gamma)A = \frac{c}{\Delta s} 2i \sin\left(\frac{\kappa}{2}\right)B, \quad (-i\omega + \gamma)B = -\frac{c}{\Delta s} 2i \sin\left(\frac{\kappa}{2}\right)A.$$

Eliminating A, B gives

$$(-i\omega + \gamma)^2 = \left(\frac{2c}{\Delta s} \sin \frac{\kappa}{2} \right)^2,$$

from which the stated $\omega(\kappa)$ follows (the sign of the real root selects the physical forward propagation). Differentiating $\Re\omega$ with respect to k yields v_g .

Lemma (Phase relation and quadrature in the chain). The amplitude ratio is

$$\frac{B}{A} = \frac{-i\omega + \gamma}{i \frac{2c}{\Delta s} \sin(\kappa/2)}.$$

For $\gamma = 0$, $B/A = -1$ (real): w_2 and w_3 are *in phase* in time (or shifted by π). A *temporal quadrature* arises between w_2 and the scaled *spatial derivative* of w_3 at lattice site n :

$$\widehat{w}_{3,n} := \frac{c}{\omega} \frac{w_{3,n+\frac{1}{2}} - w_{3,n-\frac{1}{2}}}{\Delta s} \quad \Rightarrow \quad \frac{\widehat{W}_3}{A} = i \frac{\omega}{\omega + i\gamma} \xrightarrow{\gamma \rightarrow 0} i,$$

hence $\Delta\phi(w_2, \widehat{w}_3) \rightarrow \pm\pi/2$. Alternatively, the superposition of equal-amplitude counter-propagating waves (standing wave) locally produces quadrature between w_2 and w_3 (away from nodes).

Acceptance.

1. **Fundamental-band dispersion:** For $|\kappa| \ll 1$ numerically, $\Re\omega(k) \approx c|k|$ and $|v_g| \approx c$.
2. **Quadrature:** In the lossless limit $\gamma \rightarrow 0$, $\Delta\phi(w_2, \frac{\varepsilon}{\omega}\delta_s w_3) \approx \pm\frac{\pi}{2}$ is preserved along the chain (and for standing waves locally also between w_2 and w_3).

T7 – Geometry Sensitivity and Effective $k_{1,\text{eff}}$

Goal. Influence of lever arms/projection on the effective stiffness along w_1 .

Model. The internal degrees of freedom are $\mathbf{x} = (x_2, x_3, x_4)^\top$ with

$$U(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top \mathbf{K} \mathbf{x}, \quad \mathbf{K} = \text{diag}(k_C, k_S, k_G).$$

A small imposed displacement along w_1 by x_1 produces (linear kinematics)

$$\mathbf{x} = \mathbf{P} x_1,$$

where $\mathbf{P} = (P_2, P_3, P_4)^\top$ is the *projection/lever-arm vector* and depends only on the tetrahedron geometry (dimensionless up to a global lever-arm factor).

Theorem (Effective stiffness). Under the constraint $\mathbf{x} = \mathbf{P} x_1$, the effective spring stiffness along w_1 is

$$k_{1,\text{eff}} = \mathbf{P}^\top \mathbf{K} \mathbf{P} = P_2^2 k_C + P_3^2 k_S + P_4^2 k_G.$$

Proof. Insert $\mathbf{x} = \mathbf{P} x_1$ into U :

$$U(x_1) = \frac{1}{2} (\mathbf{P} x_1)^\top \mathbf{K} (\mathbf{P} x_1) = \frac{1}{2} x_1^2 \mathbf{P}^\top \mathbf{K} \mathbf{P}.$$

By definition of the local stiffness, $F_1 = \partial U / \partial x_1 = k_{1,\text{eff}} x_1$, thus $k_{1,\text{eff}} = \mathbf{P}^\top \mathbf{K} \mathbf{P}$.

Corollary (Rayleigh quotient and bounds). If $\|\mathbf{P}\| = 1$, then

$$\min(k_C, k_S, k_G) \leq k_{1,\text{eff}} \leq \max(k_C, k_S, k_G),$$

with equality when $\mathbf{P}$ points along a coordinate axis.

Scale-invariance. Since $k_i \propto a^{-3}$, it follows that $k_{1,\text{eff}} \propto a^{-3}$. The ratios are therefore scale-free, in particular

$$\frac{k_{1,\text{eff}}}{k_S} = P_2^2 \frac{k_C}{k_S} + P_3^2 + P_4^2 \frac{k_G}{k_S} = 2\alpha P_2^2 + P_3^2 + 2\left(\frac{m}{m_{\text{Pl}}}\right)^2 P_4^2,$$

where $\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c}$ and $m_{\text{Pl}} = \sqrt{\hbar c/G}$. Thus, scale-free ratios depend *only* on $\mathbf{P}$ and on the mass ratio.

Special cases.

- **Isotropic lever arms** (equal-length projections), $\mathbf{P} = \frac{\beta}{\sqrt{3}}(1, 1, 1)^\top$:

$$k_{1,\text{eff}} = \frac{\beta^2}{3} (k_C + k_S + k_G) = \frac{\beta^2}{3} \left(2\alpha k_S + k_S + 2 \left(\frac{m}{m_{\text{Pl}}} \right)^2 k_S \right).$$

With Planck normalization $m = m_{\text{Pl}}$: $k_{1,\text{eff}} = \frac{\beta^2}{3} (2\alpha + 3) k_S$. For $m \ll m_{\text{Pl}}$ (locally negligible gravitation): $k_{1,\text{eff}} \approx \frac{\beta^2}{3} (2\alpha + 1) k_S$.

- **Anisotropic lever arms.** If $\mathbf{P}$ points only along w_2 , then $k_{1,\text{eff}} = \beta^2 k_C$; only along w_3 : $k_{1,\text{eff}} = \beta^2 k_S$; only along w_4 : $k_{1,\text{eff}} = \beta^2 k_G$.

Remark (Sensitivity). If a geometry parameter θ affects only $\mathbf{P}(\theta)$, then

$$\frac{\partial k_{1,\text{eff}}}{\partial \theta} = 2 \left(\frac{\partial \mathbf{P}}{\partial \theta} \right)^\top \mathbf{K} \mathbf{P},$$

i.e., the rate of change is purely geometric (for fixed k_i). For $m \ll m_{\text{Pl}}$, the sensitivity is dominated by the w_2/w_3 contributions; the w_4 contribution is $\mathcal{O}((m/m_{\text{Pl}})^2)$.

Acceptance. The scale-free base ratios ($k_C/k_S = 2\alpha$, $k_G/k_S = 2(m/m_{\text{Pl}})^2$) remain invariant; $k_{1,\text{eff}}$ changes only via geometry ($\mathbf{P}$) or the mass ratio in k_G .

T8 – Minimal Spectroscopy (Lyman/Balmer Scaling)

Goal. External validation of the MMU energy scale without heavy formalism: Show $\lambda^{-1} \propto 1/n^2$ with only *one* normalization parameter.

MMU framework. Bound states arise from the interplay of Coulomb curvature (axis w_2) and torsional inertia (axis w_3); photon emission is the projection of the energy difference ΔE onto w_1 . We use two fundamental ingredients:

1. **Coulomb + circular dynamics (virial):** In stationary orbits,

$$\frac{mv^2}{r} = \frac{\kappa}{r^2}, \quad \kappa := \frac{e^2}{4\pi\epsilon_0} = \alpha \hbar c,$$

and $T = \frac{1}{2}mv^2 = \frac{1}{2}\kappa/r$, $U = -\kappa/r$, hence $E = T + U = -\frac{1}{2}\kappa/r$.

2. **Torsion/action quantization (MMU):** The azimuthal action per revolution is quantized (*Bohr–Sommerfeld in MMU*):

$$L = mvr = n\hbar, \quad n = 1, 2, \dots$$

Theorem (Energy scale $1/n^2$). From (1)–(2) it follows

$$r_n = \frac{n^2 \hbar^2}{m\kappa}, \quad E_n = -\frac{\kappa}{2r_n} = -\frac{m\kappa^2}{2\hbar^2} \frac{1}{n^2} = -\frac{\alpha^2 mc^2}{2} \frac{1}{n^2}.$$

Proof. From $mvr = n\hbar$ and $mv^2/r = \kappa/r^2$ we get $m(\frac{n\hbar}{mr})^2/r = \kappa/r^2$, thus $n^2 \hbar^2/(mr^3) = \kappa/r^2 \Rightarrow r_n = n^2 \hbar^2/(m\kappa)$. Then $E_n = -\kappa/(2r_n) = -m\kappa^2/(2\hbar^2) \cdot 1/n^2$.

Corollary (Wavenumber formula). For a transition $n \rightarrow m < n$, $\Delta E = E_m - E_n > 0$ is radiated along w_1 . With $hc \tilde{\nu}_{n \rightarrow m} = \Delta E$ ($\tilde{\nu} = 1/\lambda$) we obtain

$$\tilde{\nu}_{n \rightarrow m} = \mathcal{R} \left(\frac{1}{m^2} - \frac{1}{n^2} \right), \quad \mathcal{R} := \frac{\alpha^2 mc}{2h}.$$

Here m is the dynamically effective mass; for hydrogen this is the *reduced mass* $\mu = \frac{m_e m_p}{m_e + m_p}$.

One-parameter normalization. For a *purely geometric* validation, the form and *one* normalization suffice:

$$\tilde{\nu}_{n \rightarrow m} = \frac{1}{\lambda_b} \left(\frac{1}{m^2} - \frac{1}{n^2} \right), \quad \lambda_b := \frac{1}{\mathcal{R}},$$

where λ_b (bosonic resonance length or effective lattice spacing) is set once, e.g., by Lyman- α ($n=2 \rightarrow m=1$). After that, *all* further lines (Ly β , H α , ...) are *parameter-free predictions*.

Remark (MMU interpretation of the constants). The proportionality $\kappa = \alpha \hbar c$ and $k_C/k_S = 2\alpha$ (T0) link the Coulomb curvature w_2 to the torsion scale w_3 . The projection onto w_1 yields ΔE as the measurable photon energy.

Acceptance.

- **Form test:** The experimental wavenumbers follow the form $\tilde{\nu}_{n \rightarrow m} \propto (1/m^2 - 1/n^2)$ within measurement uncertainty.
- **One-parameter fit:** With *one* calibration λ_b (e.g., to Ly α), several lines (such as Ly β , H α) agree within a few percent.

T9 – Hyperfine benchmark (21 cm)

Goal. Order-of-magnitude test of torsion dominance in the hyperfine splitting of the $1s$ state.

Model assumption. The spin alignment (parallel/antiparallel) of electron and proton slightly modulates the local torsional stiffness on w_3 :

$$k_S^{(\pm)} = k_S (1 \pm \varepsilon), \quad 0 < \varepsilon \ll 1,$$

where “+/-” denotes the two hyperfine configurations. The ground mode has angular frequency $\omega_0 = \sqrt{k_S/I_{\text{eff}}}$ with an effective moment of inertia I_{eff} (geometrically fixed, independent of spin orientation).

Theorem (Splitting from δk_S). The frequencies $\omega_{\pm} = \sqrt{k_S^{(\pm)}/I_{\text{eff}}}$ satisfy, for $\varepsilon \ll 1$,

$$\omega_{\pm} = \omega_0 \sqrt{1 \pm \varepsilon} \approx \omega_0 \left(1 \pm \frac{\varepsilon}{2} \right).$$

For the ground state $E_{\pm} = \frac{1}{2}\hbar\omega_{\pm}$, the energy splitting is

$$\Delta E = E_+ - E_- \approx \frac{1}{2}\hbar\omega_0 \varepsilon,$$

hence

$$\boxed{\Delta\nu = \frac{\Delta E}{h} \approx \frac{\varepsilon}{2} f_0, \quad f_0 := \frac{\omega_0}{2\pi}.} \quad (13)$$

MMU identification of the scale f_0 and the modulation ε . In the hydrogen ground state, the natural frequency scale is the Bohr frequency

$$f_0 = \frac{\alpha^2 m_e c^2}{h} = 2 c R_{\infty},$$

and the (dimensionless) spin–spin modulation is proportional to the ratio “magnetic (contact) energy / Coulomb scale.” Introducing the known factors of the contact interaction (geometry, proton g -factor) as a pure prefactor yields

$$\boxed{\varepsilon \approx \frac{8}{3} g_p \alpha^2 \frac{m_e}{m_p}.} \quad (14)$$

Corollary (21 cm frequency). Substituting (14) into (13) and $f_0 = 2cR_{\infty}$ gives

$$\Delta\nu \approx \frac{1}{2} \left(\frac{8}{3} g_p \alpha^2 \frac{m_e}{m_p} \right) (2cR_{\infty}) = \boxed{\frac{8}{3} g_p \alpha^2 \frac{m_e}{m_p} cR_{\infty}}.$$

Numerics (order of magnitude). With

$$\alpha \simeq 7.297 \times 10^{-3}, \quad \frac{m_e}{m_p} \simeq 5.446 \times 10^{-4}, \quad g_p \simeq 5.5857, \quad cR_{\infty} \simeq 3.28984 \times 10^{15} \text{ Hz},$$

we obtain

$$\Delta\nu \approx \frac{8}{3} \times 5.5857 \times (7.297 \times 10^{-3})^2 \times (5.446 \times 10^{-4}) \times 3.28984 \times 10^{15} \text{ Hz} \approx 1.42 \times 10^9 \text{ Hz}.$$

Thus, the prediction lies in the correct order of magnitude (~ 1.42 GHz) *without* multi-parameter fits. Small deviations are to be expected from higher-order corrections (QED, reduced mass, proton structure).

Acceptance. The relation $\Delta\nu \approx \frac{8}{3} g_p \alpha^2 (m_e/m_p) cR_{\infty}$ reproduces the 21 cm scale; within the simple MMU model, this requires only the single modulation parameter ε from (14) and the natural scale f_0 .

T10 – Impedance path (optional, advanced)

Goal. Geometric connection to α via the wave impedance of an MMU edge.

Starting point (edge PDE, lossless). Along the edge (coordinate s) the linear coupling equations hold:

$$\partial_t w_2 = -c \partial_s w_3, \quad (15)$$

$$\partial_t w_3 = -c \partial_s w_2. \quad (16)$$

The corresponding energy density and power flow (cf. T4) are

$$u = \frac{1}{2}(w_2^2 + w_3^2), \quad S = c w_2 w_3,$$

with local continuity $\partial_t u + \partial_s S = 0$.

Lemma 1 (Transmission-line mapping). Define effective per-length quantities L', C' and the characteristic impedance $Z_0 = \sqrt{L'/C'}$. The mapping

$$V := \sqrt{Z_0} w_2, \quad I := \frac{1}{\sqrt{Z_0}} w_3 \quad (17)$$

takes (15)–(16) exactly into the (lossless) telegraph equations

$$\partial_t V = -\frac{1}{C'} \partial_s I, \quad (18)$$

$$\partial_t I = -\frac{1}{L'} \partial_s V, \quad (19)$$

iff

$$\boxed{c = \frac{1}{\sqrt{L'C'}}, \quad Z_0 = \sqrt{\frac{L'}{C'}}, \quad \text{equivalently : } L' = \frac{Z_0}{c}, \quad C' = \frac{1}{c Z_0}.} \quad (20)$$

Proof. Insert (17) into (18): $\sqrt{Z_0} \partial_t w_2 = -(1/C')(1/\sqrt{Z_0}) \partial_s w_3$, thus $\partial_t w_2 = -(1/(C' Z_0)) \partial_s w_3$. Together with (15) it follows that $1/(C' Z_0) = c$. Analogously, from (19) one obtains $Z_0/L' = c$. These relations are equivalent to (20).

Lemma 2 (Energy and Power Mapping). With (17), the transmission line quantities per unit length are

$$u_{\text{TL}} = \frac{1}{2}(C' V^2 + L' I^2) = \frac{1}{2} \sqrt{L' C'} (w_2^2 + w_3^2) = \frac{1}{c} u,$$

and $S_{\text{TL}} = VI = w_2 w_3 = S/c$. From $\partial_t u + \partial_s S = 0$ the standard transmission line continuity equation follows:

$$\partial_t u_{\text{TL}} + \partial_s S_{\text{TL}} = 0.$$

Remark: The normalization of the w -fields is consistent with T4; the global factor $1/c$ is expected since u_{TL} is energy per unit length rather than energy density *per space-time norm*.

Theorem (Characteristic Impedance and Traveling Wave). For a traveling wave ($\gamma = 0$) it follows from (15)–(16) that $w_3 = \text{sgn}(k) w_2$. With (17) one obtains

$$\frac{V}{I} = \frac{\sqrt{Z_0} w_2}{w_3 / \sqrt{Z_0}} = \text{sgn}(k) Z_0,$$

hence Z_0 is the *ratio impedance* of a forward/backward traveling edge wave, as for a lossless TEM line.

Theorem (Impedance Path to the Fine-Structure Constant). In SI units,

$$\boxed{\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} = \frac{Z_0 e^2}{2h}, \quad Z_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = \mu_0 c.} \quad (21)$$

Proof. From $\mu_0\epsilon_0 c^2 = 1$ it follows that $Z_0 = \sqrt{\mu_0/\epsilon_0} = \mu_0 c$ and $\epsilon_0^{-1} = \mu_0 c^2$. Thus

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} = \frac{e^2\mu_0 c}{4\pi\hbar} = \frac{e^2 Z_0}{4\pi\hbar} = \frac{e^2 Z_0}{2h},$$

since $\hbar = h/(2\pi)$.

Consequence (MMU $\Rightarrow \alpha$ or h). If one determines the *geometric* edge impedance $Z_0^{\text{MMU}} = \sqrt{L'/C'}$ from the MMU microgeometry (e.g., via energies u_{TL} , coupling (20) and 90° edge physics), and quantizes the charge e topologically, then (21) yields

$$\boxed{\alpha = \frac{Z_0^{\text{MMU}} e^2}{2h} \iff h = \frac{Z_0^{\text{MMU}} e^2}{2\alpha} .}$$

Thus, α or h can be derived from geometric quantities (plus a normalization).

Acceptance. An edge impedance Z_0^{MMU} obtained from MMU geometry leads via (21) to a consistent order of magnitude for α (or h), within the uncertainty of the geometry model and the topologization of e .

Appendix 3 Edge Impedance

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1 Edge Impedance Z_0 of the MMU Edge: Derivation, Calculation Path, Result

Starting point: traveling wave along the outer edge

Along an outer edge (coordinate s), two coupled field components propagate: voltage-like w_2 (“ E ”) and current-like w_3 (“ H ”):

$$\partial_t w_2 = c \partial_s w_3, \quad \partial_t w_3 = -c \partial_s w_2, \quad (1)$$

(lossless, homogeneous). Solutions are forward/backward traveling waves with phase/group velocity c .

The local energy density and the power flux (Poynting analogue) are given by

$$u = \frac{1}{2} (w_2^2 + w_3^2), \quad S = c w_2 w_3. \quad (2)$$

Energy matching: from internal springs to E/H

The cell (length a) possesses two internal energy storages described by the spring curvatures acting at the node. Taking into account the geometric projections (lever arms) P_2, P_3 , the cell energy is

$$U_{\text{cell}} = \frac{1}{2} P_2^2 k_C w_2^2 + \frac{1}{2} P_3^2 k_S w_3^2. \quad (3)$$

At the same time, we write in the continuum the field energy per unit length as

$$u = \frac{1}{2} \varepsilon_{\text{eff}} E^2 + \frac{1}{2} \mu_{\text{eff}} H^2, \quad U_{\text{cell}} = a u. \quad (4)$$

We identify the field amplitudes as $E = \alpha_E w_2$ and $H = \alpha_H w_3$ (the absolute scales α_E, α_H are arbitrary; only their ratio matters). Comparison of (3) and (4) yields for the *ratio* of the effective medium parameters:

$$\frac{\mu_{\text{eff}}}{\varepsilon_{\text{eff}}} = \frac{P_3^2 k_S}{P_2^2 k_C}. \quad (5)$$

Definition of the edge impedance

The (intrinsic) wave impedance of the edge is

$$Z_0 \equiv \eta = \sqrt{\frac{\mu_{\text{eff}}}{\varepsilon_{\text{eff}}}}. \quad (6)$$

Substituting (5) into (6) gives the *internal, scale-free* edge impedance:

$$Z_{0,\text{int}} = \frac{P_3}{P_2} \sqrt{\frac{k_S}{k_C}} \quad (7)$$

(This quantity is independent of a and of the choice of the absolute scales α_E, α_H .)

Substituting the spring constants

From the local curvature derivation (or equivalently from the dispersion viewpoint), we have

$$k_S = \frac{\hbar c}{a^3}, \quad k_C = \frac{2}{4\pi\varepsilon_0} \frac{e^2}{a^3} = 2\alpha \frac{\hbar c}{a^3} \Rightarrow \frac{k_S}{k_C} = \frac{1}{2\alpha}. \quad (8)$$

Thus, from (7) we obtain

$$Z_{0,\text{int}} = \frac{P_3}{P_2} \frac{1}{\sqrt{2\alpha}} \quad (9)$$

and for the regular tetrahedron (tetrahedral angle 109.47° , isotropic transverse projection $P_2 = P_3 = \frac{2}{3}$):

$$Z_{0,\text{int}} = \frac{1}{\sqrt{2\alpha}} \approx 8.27756 \quad (\text{dimensionless}). \quad (10)$$

In the lossless base model, η is real (no phase impedance).

2 Test Plan for Edge Impedance Z_0 (Critical Validation)

Goal: The derivation of the edge impedance Z_0 (Sec. ??) is checked without fit parameters—algebraically, dimensionally, and numerically.

Z0-0 — Definition, Dimension, Scale Independence

Method. For a lossless edge traveling wave, the field energy density is written as

$$u = \frac{1}{2}(\varepsilon_{\text{eff}} E^2 + \mu_{\text{eff}} H^2), \quad \mathbf{S} = \mathbf{E} \times \mathbf{H}, \quad c = \frac{1}{\sqrt{\mu_{\text{eff}} \varepsilon_{\text{eff}}}}.$$

The (intrinsic) wave impedance is the amplitude ratio

$$Z_0 \equiv \eta = \frac{|E|}{|H|}.$$

From u it follows directly

$$\eta = \sqrt{\frac{\mu_{\text{eff}}}{\varepsilon_{\text{eff}}}},$$

since for the lossless traveling wave, E - and H -parts contribute in phase and with a fixed ratio to the energy.

Dimension. In SI, $[\mu_{\text{eff}}] = \text{H/m}$ and $[\varepsilon_{\text{eff}}] = \text{F/m}$. Thus

$$\left[\sqrt{\frac{\mu_{\text{eff}}}{\varepsilon_{\text{eff}}}} \right] = \sqrt{\frac{\text{H/m}}{\text{F/m}}} = \sqrt{\frac{\text{H}}{\text{F}}} = \sqrt{\frac{\Omega \text{ s}}{\text{s}/\Omega}} = \sqrt{\Omega^2} = \Omega,$$

so $[Z_0] = \Omega$ (after SI calibration).

Internal (scale-free) form. Compare the discrete cell energy with projections P_2, P_3 ,

$$U_{\text{cell}} = \frac{1}{2} P_2^2 k_C w_2^2 + \frac{1}{2} P_3^2 k_S w_3^2,$$

with the continuum energy $U_{\text{cell}} = a u = \frac{1}{2} a \varepsilon_{\text{eff}} E^2 + \frac{1}{2} a \mu_{\text{eff}} H^2$ and identify $E \sim w_2$, $H \sim w_3$. Coefficient comparison yields

$$\frac{\mu_{\text{eff}}}{\varepsilon_{\text{eff}}} = \frac{P_3^2 k_S}{P_2^2 k_C} \implies \boxed{Z_{0,\text{int}} = \frac{P_3}{P_2} \sqrt{\frac{k_S}{k_C}}}.$$

Scale independence. From the curvature definition it follows

$$k_S = \frac{\hbar c}{a^3}, \quad k_C = 2\alpha \frac{\hbar c}{a^3},$$

thus

$$\frac{k_S}{k_C} = \frac{1}{2\alpha} \quad (\text{independent of } a).$$

Hence $Z_{0,\text{int}} = (P_3/P_2)/\sqrt{2\alpha}$ is *dimensionless* and depends only on geometry (P_3/P_2) and α , not on the length scale a .

Acceptance. $[Z_0] = \Omega$ (only after SI calibration via $\mu_{\text{eff}}, \varepsilon_{\text{eff}}$); $Z_{0,\text{int}}$ is dimensionless and scale-free (a cancels because $k_i \propto a^{-3}$).

Z0–1 — Energy Matching (Derivation of $Z_{0,\text{int}}$)

Method. Compare the discrete cell energy (internal storage) with the continuum energy (field storage) for a cell of length a :

$$U_{\text{cell}} = \frac{1}{2} P_2^2 k_C w_2^2 + \frac{1}{2} P_3^2 k_S w_3^2 = a \left[\frac{1}{2} \varepsilon_{\text{eff}} E^2 + \frac{1}{2} \mu_{\text{eff}} H^2 \right], \quad (11)$$

where P_2, P_3 are the projections (lever arms) and we use the field identification $E \sim w_2$, $H \sim w_3$.

Proof. Since w_2 and w_3 are independent small generalized coordinates, the coefficients of the quadratic terms in (11) must match pairwise. With the (without loss of generality chosen) normalization $E = w_2$, $H = w_3$ it follows

$$a \varepsilon_{\text{eff}} = P_2^2 k_C, \quad a \mu_{\text{eff}} = P_3^2 k_S. \quad (12)$$

The ratio of these relations yields immediately

$$\frac{\mu_{\text{eff}}}{\varepsilon_{\text{eff}}} = \frac{P_3^2 k_S}{P_2^2 k_C}. \quad (13)$$

The (intrinsic) wave impedance is $\eta = \sqrt{\mu_{\text{eff}}/\varepsilon_{\text{eff}}}$. This gives the internal, scale-free edge impedance

$$Z_{0,\text{int}} = \frac{P_3}{P_2} \sqrt{\frac{k_S}{k_C}} \quad (14)$$

algebraically exactly.

Remark. If instead of $E = w_2$, $H = w_3$ one chooses arbitrary linear scalings $E = \alpha_E w_2$, $H = \alpha_H w_3$, then (12) becomes $a \varepsilon_{\text{eff}} \alpha_E^2 = P_2^2 k_C$ and $a \mu_{\text{eff}} \alpha_H^2 = P_3^2 k_S$. (13) and thus (14) remain unchanged, i.e., the statement is *independent of the chosen absolute normalization* of the fields.

Result. (14).

Acceptance. Identical coefficients in (11) ((12)) (13) (14); the step is purely a coefficient comparison and therefore algebraically exact.

Z0-2 — Connection to α

Method. From the curvature definition of the local stiffnesses (Coulomb vs. torsion) we obtain

$$k_S = \frac{\hbar c}{a^3}, \quad k_C = \frac{2}{4\pi\varepsilon_0} \frac{e^2}{a^3} = 2\alpha \frac{\hbar c}{a^3},$$

thus

$$\frac{k_S}{k_C} = \frac{1}{2\alpha}. \quad (15)$$

Substituting this into the relation obtained from energy matching $Z_{0,\text{int}} = \frac{P_3}{P_2} \sqrt{\frac{k_S}{k_C}}$ yields

$$Z_{0,\text{int}} = \frac{P_3}{P_2} \frac{1}{\sqrt{2\alpha}}. \quad (16)$$

Proof. The equations for k_S and k_C contain the same a^{-3} scaling, which cancels exactly in the ratio (15); thus k_S/k_C is independent of the length scale and determined purely by α . Substituting into $Z_{0,\text{int}} = (P_3/P_2) \sqrt{k_S/k_C}$ immediately gives (16). This step is purely algebraic.

Regular tetrahedron. For isotropic projection $P_3/P_2 = 1$, hence

$$Z_{0,\text{int}} = \frac{1}{\sqrt{2\alpha}}.$$

Acceptance. Algebraically exact; numerically (with $\alpha^{-1} \approx 137.035999$) we have $Z_{0,\text{int}} \approx 1/\sqrt{2\alpha}$.

Z0–3 — Poynting Balance (Energy Conservation)

Method. We use the (lossless) 1D edge equations in Telegrapher/Maxwell form:

$$\partial_s E = -\mu_{\text{eff}} \partial_t H, \quad \partial_s H = -\varepsilon_{\text{eff}} \partial_t E, \quad (17)$$

and the definitions of energy density and power flux

$$u = \frac{1}{2}(\varepsilon_{\text{eff}} E^2 + \mu_{\text{eff}} H^2), \quad S = E H. \quad (18)$$

Proof. Differentiate (18) with respect to time and eliminate the time derivatives using (17):

$$\partial_t u = \varepsilon_{\text{eff}} E \partial_t E + \mu_{\text{eff}} H \partial_t H = -E \partial_s H - H \partial_s E = -\partial_s(EH) = -\partial_s S.$$

This yields the continuity equation

$$\boxed{\partial_t u + \partial_s S = 0}, \quad (19)$$

i.e. energy conservation: local change in field energy equals minus the divergence of the power flux.

Equivalent c -form. With $c = 1/\sqrt{\mu_{\text{eff}}\varepsilon_{\text{eff}}}$ and the normalized fields $\tilde{E} = \sqrt{\varepsilon_{\text{eff}}} E$, $\tilde{H} = \sqrt{\mu_{\text{eff}}} H$ equations (17) take the symmetric form

$$\partial_t \tilde{E} = c \partial_s \tilde{H}, \quad \partial_t \tilde{H} = -c \partial_s \tilde{E}$$

With

$$\tilde{u} = \frac{1}{2}(\tilde{E}^2 + \tilde{H}^2), \quad \tilde{S} = \tilde{E} \tilde{H}$$

we obtain

$$\boxed{\partial_t \tilde{u} + \partial_s(c \tilde{S}) = 0},$$

which is exactly the form used in the main text.

Acceptance. In the lossless simulation, the time-average of the residual (left-hand side of (19) or its normalized form) vanishes up to numerical noise.

Z0–4 — Nondispersive Fundamental Band

Method. We use the (lossless) edge PDEs in normalized form

$$\partial_t E = c \partial_s H, \quad \partial_t H = -c \partial_s E, \quad (20)$$

and derive the dispersion relation $\omega = \omega(k)$. From the group velocity $v_g = d\omega/dk$ follows the frequency-independence in the continuum.

Proof (Dispersion). Differentiate (20) once with respect to t and eliminate the cross derivatives:

$$\partial_{tt} E = c \partial_t \partial_s H = -c^2 \partial_{ss} E, \quad \partial_{tt} H = -c \partial_t \partial_s E = -c^2 \partial_{ss} H.$$

Thus both fields satisfy the one-dimensional wave equation

$$\partial_{tt} \Psi - c^2 \partial_{ss} \Psi = 0, \quad \Psi \in \{E, H\}. \quad (21)$$

For plane waves $\Psi(s, t) = \Re\{\Psi_0 e^{i(ks-\omega t)}\}$ (21) gives the dispersion relation

$$-\omega^2 + c^2 k^2 = 0 \implies \boxed{\omega(k) = c |k|}. \quad (22)$$

Consequence (Group velocity). The group velocity is

$$\boxed{v_g(k) = \frac{d\omega}{dk} = c \operatorname{sgn}(k)}, \quad (23)$$

whose magnitude is $|v_g| = c$ and therefore *frequency-independent*. At the same time $v_{\text{phase}} = \omega/|k| = c$. A localized pulse (Fourier superposition)

$$\Psi(s, t) = \int \hat{\Psi}(k) e^{i(ks-\omega(k)t)} dk$$

therefore propagates without dispersive distortion at speed c .

Remark (Discrete chain). For an explicitly discretized edge with lattice spacing Δs the standard dispersion is

$$\omega(k) = \frac{2c}{\Delta s} \sin\left(\frac{|k|\Delta s}{2}\right), \quad v_g(k) = c \cos\left(\frac{|k|\Delta s}{2}\right) \leq c.$$

In the long-wave regime $|k|\Delta s \ll 1$ we have $v_g/c = \cos(\frac{|k|\Delta s}{2}) \approx 1 - \frac{1}{8}(|k|\Delta s)^2$, thus practically nondispersive; the error is $\mathcal{O}(|k|\Delta s)^2$.

Acceptance. Continuum: $v_g = c$ exactly (frequency-independent). Discrete numerics: For pulses with $\lambda \gg \Delta s$ the deviation $|v_g - c|/c < 1\%$ (long-wavelength regime), consistent with the above estimate.

Z0–5 — Discrete Chain: Constancy of $Z_{0,\text{int}}$ Across Modes

Method. We consider a spatially discretized (lossless) edge with lattice spacing Δs in a Yee-type staggered discretization: $E_n(t)$ at grid points $s_n = n \Delta s$, $H_{n+\frac{1}{2}}(t)$ on half-cells. The semi-discrete equations read

$$\dot{E}_n = \frac{c}{\Delta s} (H_{n+\frac{1}{2}} - H_{n-\frac{1}{2}}), \quad \dot{H}_{n+\frac{1}{2}} = \frac{c}{\Delta s} (E_{n+1} - E_n), \quad (24)$$

with $c = (\mu_{\text{eff}} \varepsilon_{\text{eff}})^{-1/2}$ constant.

Proof (Dispersion and Amplitude Ratio). We assume plane waves of the form

$$E_n(t) = E_0 e^{i(n\kappa - \omega t)}, \quad H_{n+\frac{1}{2}}(t) = H_0 e^{i((n+\frac{1}{2})\kappa - \omega t)},$$

with $\kappa = k \Delta s$. Substitution into (24) yields

$$\begin{aligned} -i\omega E_0 &= \frac{c}{\Delta s} H_0 (e^{i\kappa/2} - e^{-i\kappa/2}) = \frac{2ic}{\Delta s} H_0 \sin(\kappa/2), \\ -i\omega H_0 e^{i\kappa/2} &= \frac{c}{\Delta s} E_0 (e^{i\kappa} - 1) = \frac{2ic}{\Delta s} E_0 e^{i\kappa/2} \sin(\kappa/2). \end{aligned}$$

After cancelling factors we obtain the linear system

$$\omega E_0 = \frac{2c}{\Delta s} \sin(\kappa/2) H_0, \quad \omega H_0 = \frac{2c}{\Delta s} \sin(\kappa/2) E_0.$$

Multiplying the equations gives the discrete dispersion relation

$$\boxed{\omega(\kappa) = \frac{2c}{\Delta s} \sin(\kappa/2)} . \quad (25)$$

The *amplitude ratio* follows from either equation:

$$\frac{E_0}{H_0} = \frac{2c}{\Delta s \omega} \sin(\kappa/2).$$

With (25) this gives

$$\boxed{\left| \frac{E_0}{H_0} \right| = 1} \quad (\text{sign}$$

$\pm$ corresponds to propagation direction). (26)

Since $Z_{0,\text{int}} = \sqrt{\mu_{\text{eff}}/\varepsilon_{\text{eff}}}$ is *mode-independent*, and the normalized fields are chosen such that $E/H \rightarrow 1$ for the lossless traveling wave, $Z_{0,\text{int}}$ remains constant for all modes.

Group velocity. From (25) we have

$$\boxed{v_g(k) = \frac{d\omega}{dk} = c \cos\left(\frac{|k|\Delta s}{2}\right) \leq c} , \quad (27)$$

with $v_g \rightarrow c$ in the long-wavelength limit $|k|\Delta s \ll 1$ (nondispersive fundamental band).

Result. For all discrete modes m (or wave numbers k_m) we find

$$\boxed{Z_{0,\text{int}}(m) \text{ constant (within numerical tolerance)}}, \quad \boxed{v_g(k) = c \cos(|k|\Delta s/2) \leq c}.$$

Acceptance. FFT analysis of the modes gives $|E/H| \approx 1$ for all m (up to numerical noise), and the measured group velocity follows (27); in the long-wave regime v_g deviates from c by less than 1%.

Z0–6 — Node Reflection (Matching)

Method. An input branch (characteristic impedance Z_0) feeds *three* identical outputs (also Z_0) at a node. The equivalent load seen from the input is therefore

$$Z_{\text{eq}} = \left(\frac{1}{Z_0} + \frac{1}{Z_0} + \frac{1}{Z_0} \right)^{-1} = \frac{Z_0}{3}.$$

The reflection coefficient at the node is

$$\Gamma = \frac{Z_{\text{eq}} - Z_0}{Z_{\text{eq}} + Z_0}.$$

Proof. Let V_i be the incoming wave amplitude (voltage) at the input, $V_r = \Gamma V_i$ the reflected one. The node voltage is

$$V_K = V_i + V_r.$$

On each output there is only a forward-going wave with amplitude

$$V_t = V_K = V_i + V_r.$$

KCL at the node with currents into the node positive:

$$I_{\text{in}} = \frac{V_i - V_r}{Z_0} \stackrel{!}{=} \sum_{j=1}^3 \frac{V_t}{Z_0} = \frac{3(V_i + V_r)}{Z_0}.$$

Multiplying by Z_0 and rearranging yields

$$V_i - V_r = 3V_i + 3V_r \implies -2V_i = 4V_r \implies \boxed{\Gamma = \frac{V_r}{V_i} = -\frac{1}{2}}.$$

This also determines the output amplitude:

$$\boxed{V_t = V_i + V_r = (1 + \Gamma)V_i = \frac{1}{2} V_i}.$$

Power distribution. With $P = V^2/Z_0$ (traveling wave) we have

$$P_{\text{ref}} = |\Gamma|^2 \frac{V_i^2}{Z_0} = \frac{1}{4} \frac{V_i^2}{Z_0}, \quad P_{\text{out},j} = \frac{V_t^2}{Z_0} = \frac{1}{4} \frac{V_i^2}{Z_0}$$

Z0–7 — SI Calibration and Consistency

Method. For a lossless 1D line (here: MMU edge), the telegrapher relations hold:

$$Z_0 = \sqrt{\frac{L'}{C'}}, \quad c = \frac{1}{\sqrt{L'C'}}.$$

Setting an *SI anchor* for the (free) wave impedance Z_0 (e.g., $Z_0 = 376.730313412 \, \Omega$) and the phase velocity $c = 299\,792\,458 \, \text{m/s}$, we obtain:

$$\boxed{L' = \frac{Z_0}{c}}, \quad \boxed{C' = \frac{1}{c Z_0}}.$$

Proof (Algebra). Substituting $L' = Z_0/c$ and $C' = 1/(cZ_0)$ immediately shows:

$$L'C' = \frac{Z_0}{c} \cdot \frac{1}{cZ_0} = \frac{1}{c^2} \implies c = \frac{1}{\sqrt{L'C'}},$$

and

$$\frac{L'}{C'} = \frac{Z_0/c}{1/(cZ_0)} = Z_0^2 \implies Z_0 = \sqrt{\frac{L'}{C'}}.$$

Thus, both definitions are consistently satisfied.

Dimension. $[L'] = \text{H/m}$, $[C'] = \text{F/m}$; hence $[\sqrt{L'/C'}] = \Omega$ and $[1/\sqrt{L'C'}] = \text{m/s}$, as required.

Numerics (Example). With $Z_0 = 376.730313412 \, \Omega$ and c as above:

$$L' = \frac{Z_0}{c} \approx 1.2566370613 \times 10^{-6} \, \text{H/m}, \quad C' = \frac{1}{cZ_0} \approx 8.854187817 \times 10^{-12} \, \text{F/m}.$$

This agrees (within current CODATA uncertainties) with the free-space parameters:

$$\boxed{L' \equiv \mu_0, \quad C' \equiv \varepsilon_0} \quad (\text{lossless, homogeneous}).$$

Acceptance. (1) Algebraic consistency: $Z_0 = \sqrt{L'/C'}$ and $c = 1/\sqrt{L'C'}$ exactly satisfied. (2) Numerically: L' and C' match μ_0 and ε_0 (constant in the lossless case).

Z0–8 — Phase Angle of the Impedance

Method. We consider the (time-harmonic) telegrapher equations of a slightly lossy line:

$$\partial_s V = -(R' + j\omega L') I, \quad \partial_s I = -(G' + j\omega C') V,$$

with small losses R', G' (per length). The *intrinsic wave impedance* (phasor ratio of a forward traveling wave) is:

$$\eta(\omega) \equiv \frac{V}{I} = \sqrt{\frac{R' + j\omega L'}{G' + j\omega C'}}. \quad (28)$$

Proof (low-loss approximation). For $R' \ll \omega L'$ and $G' \ll \omega C'$, write:

$$\frac{R' + j\omega L'}{G' + j\omega C'} = \frac{j\omega L' \left(1 - \frac{jR'}{\omega L'}\right)}{j\omega C' \left(1 - \frac{jG'}{\omega C'}\right)} = \frac{L'}{C'} \frac{1 - \frac{jR'}{\omega L'}}{1 - \frac{jG'}{\omega C'}}.$$

With $1/(1 - x) \approx 1 + x$ (for $|x| \ll 1$) we get:

$$\frac{R' + j\omega L'}{G' + j\omega C'} \approx \frac{L'}{C'} \left(1 - \frac{jR'}{\omega L'}\right) \left(1 + \frac{jG'}{\omega C'}\right) = \frac{L'}{C'} \left[1 + j\left(\frac{G'}{\omega C'} - \frac{R'}{\omega L'}\right)\right].$$

Now take the square root: For $|x| \ll 1$, $\sqrt{1 + jx} \approx 1 + \frac{j}{2}x$. Thus:

$$\eta(\omega) \approx \sqrt{\frac{L'}{C'}} \left[1 + \frac{j}{2}\left(\frac{G'}{\omega C'} - \frac{R'}{\omega L'}\right)\right]. \quad (29)$$

The *phase angle* $\theta_\eta = \arg \eta$ is, in the small-angle limit:

$$\boxed{\theta_\eta(\omega) \approx \frac{1}{2}\left(\frac{G'}{\omega C'} - \frac{R'}{\omega L'}\right)}. \quad (30)$$

Implications.

- *Lossless:* $R' = G' = 0 \Rightarrow \eta = \sqrt{L'/C'}$ real,

$$\boxed{\theta_\eta = 0^\circ}.$$

- *Small damping:* θ_η is small and its *sign* indicates the dominant loss mechanism: G' dominant $\Rightarrow \theta_\eta > 0$, R' dominant $\Rightarrow \theta_\eta < 0$.
- *Symmetric model damping* (like $\partial_t E = c\partial_s H - \gamma E$, $\partial_t H = -c\partial_s E - \gamma H$) effectively corresponds to $R'/(\omega L') \approx G'/(\omega C') \Rightarrow \theta_\eta \approx 0$ (small 2nd-order corrections).

Acceptance. In the lossless case, the phase vanishes: $\theta_\eta \rightarrow 0^\circ$. For finite, small damping, θ_η is small and has the sign predicted in (30). (Numerics: extract $\eta = |E/H|e^{j\theta_\eta}$ from the phasors; the measured θ_η follows (30) in the low-loss regime.)

Z0–9 — Geometry Sensitivity

Method. Vary the projections P_2, P_3 (non-regular tetrahedron or slightly anisotropic node geometry) while keeping the *internal* storages k_C, k_S unchanged. From energy matching we get:

$$Z_{0,\text{int}} = \frac{P_3}{P_2} \sqrt{\frac{k_S}{k_C}}.$$

Proof. From the local curvature derivation (or dispersion) we have:

$$k_S = \frac{\hbar c}{a^3}, \quad k_C = 2\alpha \frac{\hbar c}{a^3} \implies \frac{k_S}{k_C} = \frac{1}{2\alpha},$$

so k_S/k_C is *scale-free* (independent of a) and *geometry-independent*. Thus, $Z_{0,\text{int}}$ depends only on the projection ratio P_3/P_2 :

$$\boxed{Z_{0,\text{int}} = \frac{P_3}{P_2} \frac{1}{\sqrt{2\alpha}} \propto \frac{P_3}{P_2}}. \quad (31)$$

A small geometry perturbation $P_3/P_2 \mapsto (P_3/P_2)(1 + \epsilon)$ leads linearly to:

$$Z_{0,\text{int}} \mapsto Z_{0,\text{int}}(1 + \epsilon),$$

i.e., $Z_{0,\text{int}}$ scales *linearly* with P_3/P_2 , while the ratio of internal stiffnesses $k_C/k_S = 2\alpha$ remains unchanged. Differentially:

$$\frac{\partial Z_{0,\text{int}}}{\partial(P_3/P_2)} = \frac{1}{\sqrt{2\alpha}} = \text{const.}$$

This proves linearity in P_3/P_2 .

Acceptance. (i) *Linearity* in P_3/P_2 according to (31) (numerically: constant slope $\partial Z_{0,\text{int}}/\partial(P_3/P_2) = 1/\sqrt{2\alpha}$). (ii) The *ratios* of internal stiffnesses ($k_C/k_S = 2\alpha$) remain unchanged under pure projection variation; only $Z_{0,\text{int}}$ scales with P_3/P_2 .

Summary. Z_0 is entirely determined by (i) the ratio of internal storages (k_S/k_C) and (ii) the projection (P_3/P_2). For the regular tetrahedron ($P_3/P_2 = 1$) we obtain:

$$\boxed{Z_{0,\text{int}} = \frac{1}{\sqrt{2\alpha}}}.$$

The checks Z0-0...Z0-9 ensure dimension, scale-freedom, energy conservation, dispersion, node matching, SI calibration, and geometry sensitivity.

Appendix 4 Quantum Equations from the MMU: Schrödinger as One-Band Limit, Dirac as Two-Band Projection

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Abstract

Within the Methane Metauniverse (MMU)—a tetrahedral space–time lattice with orthogonal degrees (w_1, w_2, w_3, w_4) —we show that the standard Schrödinger equation emerges as the long-wavelength *one-band* envelope of internal oscillations. Starting from Laguerre–type Coulomb modes on w_2 , the spring-constant ratios (k_2, k_3, k_4) fix the concomitant torsional (w_3) and mass-storage (w_4) amplitudes via the eigenforce condition $k_2|w_2| = k_3|w_3| = k_4|w_4|$. Eliminating the fast internal coordinates by an adiabatic (Feshbach) projection and taking the continuum limit yields an effective external equation for the slow envelope $\psi(\mathbf{r}, t)$:

$$i\hbar \partial_t \psi = \left[-(\hbar^2/2m^*)\nabla^2 + V_{\text{eff}}(\mathbf{r}) \right] \psi,$$

where m^* is set by the band curvature of w_2/w_3 edge transport and V_{eff} is the projected Coulomb geometry; the Balmer scaling $E_n \propto 1/n^2$ is preserved. The measurable clock rate corresponds to the projection onto w_1 , explaining the associated time dilation. As a corollary, a nearly degenerate torsional doublet produces, under the same projection, a massive Dirac equation. This provides a geometric, parameter-transparent route from MMU microdynamics to the familiar quantum equations.

1 Starting Point: Laguerre Functions on w_2

For the hydrogen atom, the internal oscillation along w_2 is described by Laguerre radial functions $R_{n\ell}(r)$. The expectation values are:

$$E_n = -\frac{Ry}{n^2}, \quad \langle T \rangle = \frac{Ry}{n^2}, \quad \langle V \rangle = -\frac{2Ry}{n^2}.$$

We identify the w_2 spring energy with the kinetic part:

$$E_2 = \frac{1}{2}k_2\langle w_2^2 \rangle = \langle T \rangle = \frac{Ry}{n^2}.$$

Thus,

$$\langle w_2^2 \rangle = \frac{2E_2}{k_2} = \frac{2Ry}{k_2 n^2}.$$

2 Distribution of Energy to w_3 and w_4

The coupled-oscillator condition requires equal restoring forces:

$$k_2|w_2| = k_3|w_3| = k_4|w_4|.$$

From this we obtain amplitudes and energies:

$$|w_3| = \frac{k_2}{k_3}|w_2|, \quad |w_4| = \frac{k_2}{k_4}|w_2|,$$

$$\frac{E_3}{E_2} = \frac{k_2}{k_3}, \quad \frac{E_4}{E_2} = \frac{k_2}{k_4}.$$

Using the scale-free ratios (Planck normalization):

$$\frac{k_2}{k_3} = 2\alpha, \quad \frac{k_2}{k_4} = \alpha \left(\frac{m_{\text{pl}}}{m} \right)^2.$$

Thus:

$$E_3 = (2\alpha) E_2, \quad E_4 = \alpha \left(\frac{m_{\text{pl}}}{m} \right)^2 E_2.$$

3 Projection onto w_1

The total squared displacement is

$$w_{\text{tot}}^2 = w_1^2 + \langle w_2^2 \rangle + \langle w_3^2 \rangle + \langle w_4^2 \rangle.$$

Therefore,

$$w_1 = \sqrt{w_{\text{tot}}^2 - (\langle w_2^2 \rangle + \langle w_3^2 \rangle + \langle w_4^2 \rangle)}.$$

The measurable energy is the projection:

$$E_{w_1} = h w_1.$$

As n increases, $\langle w_2^2 \rangle$ grows, reducing w_1 and thus leading to time dilation. The quantized values of E_{w_1} match the energy levels E_n of the hydrogen atom.

4 Emergence of the Schrödinger Equation

After projecting out the fast internal coordinates (w_2, w_3, w_4) , the remaining slow envelope $\psi(\mathbf{r}, t)$ satisfies an effective equation. In the one-band limit, the dispersion relation is parabolic,

$$\varepsilon(\mathbf{k}) \approx E_{\text{band}} + \frac{\hbar^2 |\mathbf{k}|^2}{2m^*},$$

with m^* the effective mass set by the lattice geometry. The corresponding envelope equation is

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{r}, t) = \left[-\frac{\hbar^2}{2m^*} \nabla^2 + V_{\text{eff}}(\mathbf{r}) \right] \psi(\mathbf{r}, t).$$

This is the Schrödinger equation. The effective potential V_{eff} arises from the projected Coulomb geometry encoded in the Laguerre functions on w_2 , with corrections from w_3 and w_4 .

Conclusion

By starting from Laguerre oscillations on w_2 , distributing energy via spring ratios to w_3 and w_4 , and projecting onto w_1 , the MMU naturally recovers the Schrödinger equation as the external description of internal lattice dynamics.

A Dirac from MMU (Two-Band Projection)

Set-up. Project the full MMU dynamics onto the nearly degenerate torsional doublet $\{|u_\uparrow\rangle, |u_\downarrow\rangle\}$ on w_3 . Nearest-neighbour edge coupling of the w_2/w_3 wave yields in $\mathbf{k}$ -space

$$H(\mathbf{k}) = d_0(\mathbf{k}) \mathbb{1} + \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma},$$

with a symmetry point $\mathbf{k}_0$ such that $\mathbf{d}(\mathbf{k}_0) = \mathbf{0}$. Near $\mathbf{k}_0$ one has $\mathbf{d}(\mathbf{k}_0 + \mathbf{q}) = \hbar V \mathbf{q} + O(|\mathbf{q}|^2)$ (Weyl cone).

Mass term from w_4 . A small imbalance on w_4 mixes the two chiralities and opens a gap $\Delta = 2m_D v^2$, represented by $m_D v^2 \sigma_z$ in the projected basis.

Continuum equation (2×2). For the two-component envelope $\Psi(\mathbf{r}, t)$:

$$i\hbar \partial_t \Psi = \left[v \boldsymbol{\sigma} \cdot \mathbf{p} + m_D v^2 \sigma_z + V_{\text{eff}}(\mathbf{r}) \right] \Psi.$$

Full Dirac spinor (4×4). Combining the two opposite Weyl cones (node index τ) gives

$$i\hbar \partial_t \Psi = \left[v \boldsymbol{\alpha} \cdot \mathbf{p} + \beta m_D v^2 + V_{\text{eff}}(\mathbf{r}) \right] \Psi,$$

with $\alpha_i = \tau_z \otimes \sigma_i$, $\beta = \tau_x \otimes \mathbb{K}$, satisfying $\{\alpha_i, \alpha_j\} = 2\delta_{ij}$, $\{\alpha_i, \beta\} = 0$, $\beta^2 = \mathbb{K}$.

Gauge coupling. Phase accumulation along edges gives the Peierls substitution leading to minimal coupling $\mathbf{p} \rightarrow \mathbf{p} - q\mathbf{A}$ and the scalar term $q\phi$.

Nonrelativistic limit. Foldy–Wouthuysen expansion of the above Dirac form reproduces the Pauli Hamiltonian (with $g \approx 2$); higher-order MMU torsion/mass mixing accounts for fine-structure corrections (e.g. Lamb shift).

B Rigorous Derivation of Dirac (and Schrödinger) from the MMU

Microscopic model and hypotheses

Assumption B.1 (MMU lattice & Hilbert space). *Let $\Lambda \subset \mathbb{R}^3$ be a tetrahedral Bravais lattice with spacing $a > 0$. The microscopic Hilbert space is $\mathcal{H} = \ell^2(\Lambda) \otimes \mathbb{C}^{N_{\text{int}}}$, where the internal factor collects the fast coordinates (w_2, w_3, w_4) of a cell (springs/torsion/mass store).*

Assumption B.2 (Hamiltonian decomposition). *The self-adjoint microscopic Hamiltonian H decomposes as $H = H_{\text{int}} + H_{\text{hop}} + H_{\text{mix}}$ with domains stable under the decomposition. Here H_{int} acts fiberwise on each cell, H_{hop} is a bounded-range nearest-neighbour hopping (edge transmission by w_2/w_3), and H_{mix} is a weak internal–external coupling (small in operator norm).*

Assumption B.3 (Internal torsion doublet on w_3). *The internal (fast) Hamiltonian H_{int} has an isolated, twofold degenerate eigenspace $\mathcal{U} = \text{span}\{|u_\uparrow\rangle, |u_\downarrow\rangle\}$, transforming as a spinor irreducible representation of the double group of the tetrahedral symmetry. There is a spectral gap $\Delta_{\text{int}} > 0$ to all other internal states.*

Assumption B.4 (Coulomb branch and spring constants). *On w_2 we fix the Coulomb branch (Laguerre radial functions), and internal energies split as quadratic springs with constants (k_2, k_3, k_4) ; the eigenmode condition at turning points $k_2|w_2| = k_3|w_3| = k_4|w_4|$ implies $E_3/E_2 = k_2/k_3$, $E_4/E_2 = k_2/k_4$ (Planck normalization gives $k_2/k_3 = 2\alpha$ and $k_2/k_4 = \alpha(m_{\text{pl}}/m)^2$).*

Assumption B.5 (Scale separation). *Let $\varepsilon := \|H_{\text{mix}}\|/\Delta_{\text{int}} \ll 1$. Internal dynamics are ε^{-1} times faster than the slow lattice envelope. Times of interest satisfy $t = O(1)$ in the external clock.*

Assumption B.6 (Bloch–Floquet reduction). *H_{hop} is Λ -periodic. The Bloch transform yields fiber Hamiltonians $H(\mathbf{k})$ on $\mathcal{U} \oplus \mathcal{U}^\perp$ for $\mathbf{k}$ in the Brillouin zone. Denote by $P(\mathbf{k})$ the spectral projector onto the continuation of $\mathcal{U}$.*

Assumption B.7 (Dirac node and linearization). *There exists a high-symmetry point $\mathbf{k}_0$ with $H_{\text{eff}}^{(0)}(\mathbf{k}_0) = d_0(\mathbf{k}_0)\mathbb{K}$ and $\mathbf{d}(\mathbf{k}_0) = \mathbf{0}$ in the 2×2 Pauli decomposition $H_{\text{eff}}^{(0)}(\mathbf{k}) = d_0(\mathbf{k})\mathbb{K} + \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}$ (the superscript $^{(0)}$ means $H_{\text{mix}} = 0$). For $\mathbf{q} = \mathbf{k} - \mathbf{k}_0$ one has the linear expansion $\mathbf{d}(\mathbf{k}_0 + \mathbf{q}) = \hbar V \mathbf{q} + O(|\mathbf{q}|^2)$ with an invertible velocity matrix V .*

Assumption B.8 (Mass from w_4 asymmetry). *A small imbalance on w_4 (mass store) induces within $\mathcal{U}$ a self-adjoint term $M = \delta \sigma_z$ with $\delta > 0$ proportional to the w_4 energy splitting ($\delta \sim m_D v^2$). This opens a spectral gap 2δ at $\mathbf{k}_0$.*

Assumption B.9 (Gauge (Peierls) coupling). *Coupling the w_2/w_3 edge transport to an external electromagnetic field inserts link phases; in Bloch form this yields the minimal substitution $\mathbf{k} \mapsto \mathbf{k} - \frac{q}{\hbar} \mathbf{A}(\mathbf{r}, t)$ and scalar term $q\phi(\mathbf{r}, t)$ in the projected Hamiltonian.*

Main lemmas

Lemma B.10 (Feshbach–Schur adiabatic reduction). *Let $P(\mathbf{k})$ be the spectral projector onto the torsion doublet continuation at quasimomentum $\mathbf{k}$ and $Q = 1 - P$. For $\varepsilon \ll 1$ and initial states Ψ_0 with $Q\Psi_0 = 0$ one has*

$$\|e^{-iHt/\hbar}\Psi_0 - \mathcal{U}^{-1}e^{-iH_{\text{eff}}t/\hbar}\mathcal{U}\Psi_0\| \leq C(\varepsilon + a),$$

uniformly for $t = O(1)$. Here $H_{\text{eff}} = PHP + PHQ(E - QHQ)^{-1}QHP$ is the Feshbach–Schur operator on $P\mathcal{H}$, $\mathcal{U}$ identifies $P\mathcal{H} \simeq L^2(\mathbb{R}^3, \mathbb{C}^2)$, and a controls the continuum-limit error.

Proof. Adiabatic elimination with spectral gap Δ_{int} and bounded H_{hop} gives $\|Qe^{-iHt/\hbar}\Psi_0\| = O(\varepsilon)$; resolvent bounds justify the Schur complement. The discrete-to-continuum error is $O(a)$ by standard Bloch–Wannier estimates. $\square$

Lemma B.11 (Linear two-band Bloch form). *Under Assumption B.7 the projected Bloch Hamiltonian near $\mathbf{k}_0$ reads*

$$H_{\text{eff}}^{(0)}(\mathbf{k}_0 + \mathbf{q}) = d_0(\mathbf{k}_0)\mathbb{1} + \hbar \sum_{j=1}^3 (V\mathbf{q})_j \sigma_j + O(|\mathbf{q}|^2).$$

Proof. Symmetry enforces $\mathbf{d}(\mathbf{k}_0) = 0$; first-order $k \cdot p$ theory gives the linear term with velocity matrix $V = \partial_{\mathbf{k}}\mathbf{d}(\mathbf{k})|_{\mathbf{k}_0}$. $\square$

Lemma B.12 (Mass term from w_4). *With Assumption B.8, in the torsion basis*

$$H_{\text{eff}}(\mathbf{k}_0 + \mathbf{q}) = d_0(\mathbf{k}_0)\mathbb{1} + \hbar(V\mathbf{q}) \cdot \boldsymbol{\sigma} + \delta \sigma_z + O(|\mathbf{q}|^2, \varepsilon),$$

opening a gap 2δ .

Proof. M acts within $\mathcal{U}$ by $\delta \sigma_z$; its norm is $o(\Delta_{\text{int}})$, so adiabatic decoupling is preserved. The spectrum near $\mathbf{k}_0$ splits symmetrically by $\pm\delta$. $\square$

Lemma B.13 (Gauge minimal coupling). *Under Assumption B.9 the projected Hamiltonian obeys the minimal substitution rule: $\mathbf{q} \mapsto \mathbf{q} - \frac{q}{\hbar}\mathbf{A}(\mathbf{r}, t)$ and adds the scalar term $q\phi(\mathbf{r}, t)$ in real space.*

Proof. Peierls phases on edges transform H_{hop} by Wilson lines. Projecting to $\mathcal{U}$ and expanding for long wavelengths yields the covariant momentum. $\square$

Main result: Dirac equation from MMU

Theorem B.14 (Massive Dirac from MMU). *Let Assumptions B.1–B.9 hold. For initial data in the torsion doublet band and times $t = O(1)$, the external two-component envelope $\Psi(\mathbf{r}, t) \in L^2(\mathbb{R}^3, \mathbb{C}^2)$ satisfies, up to an error $O(\varepsilon + a)$ in norm,*

$$i\hbar \partial_t \Psi(\mathbf{r}, t) = \left[v \boldsymbol{\sigma} \cdot \mathbf{p} + m_D v^2 \sigma_z + V_{\text{eff}}(\mathbf{r}, t) \right] \Psi(\mathbf{r}, t),$$

with velocity v determined by V (e.g. $v = \frac{1}{\hbar} \frac{1}{3} \text{Tr } V$), $m_D v^2 = \delta$, and minimal electromagnetic coupling $\mathbf{p} \mapsto \mathbf{p} - q\mathbf{A}$, $V_{\text{eff}} = q\phi + V_{\text{Coul}} + \dots$.

Proof. By Lemma B.10, dynamics reduce to H_{eff} with $O(\varepsilon+a)$ error. Lemmas B.11–B.12 give the linear Pauli part and the mass term; Lemma B.13 adds minimal coupling. Passing to real space (continuum limit) yields the stated first-order Dirac generator for the envelope. $\square$

Corollary B.15 (Four-component Dirac). *If inversion/time-reversal symmetry produces two opposite cones (nodes $\tau = \pm$), the envelope on $\mathbb{C}^4$ obeys*

$$i\hbar \partial_t \Psi = \left[v \boldsymbol{\alpha} \cdot \mathbf{p} + \beta m_D v^2 + V_{\text{eff}}(\mathbf{r}, t) \right] \Psi,$$

with $\alpha_i = \tau_z \otimes \sigma_i$, $\beta = \tau_x \otimes \mathbb{K}$, satisfying the Dirac algebra.

Proof. Block-diagonalization in node index τ and torsion spin yields two Weyl sectors; coupling them by symmetry gives the standard 4×4 Dirac form. $\square$

Nonrelativistic limit and Schrödinger as corollary

Corollary B.16 (Schrödinger limit). *In the one-band case (no torsion doublet; parabolic band with curvature m^{*-1}) the same reduction yields, with error $O(\varepsilon + a)$,*

$$i\hbar \partial_t \psi(\mathbf{r}, t) = \left[-\frac{\hbar^2}{2m^*} \Delta + V_{\text{eff}}(\mathbf{r}, t) \right] \psi(\mathbf{r}, t).$$

Proof. Replace Lemma B.11 by a quadratic expansion $\varepsilon(\mathbf{k}) = E_{\text{band}} + \frac{\hbar^2}{2m^*} |\mathbf{k}|^2 + o(|\mathbf{k}|^2)$ and repeat Lemma B.10. $\square$

Parameter identification (MMU $\rightarrow$ Dirac)

The speed v equals the slope of the w_2/w_3 -mediated band at $\mathbf{k}_0$ (saturated at the lattice “light” speed). The mass scale is $m_D v^2 = \delta \propto E_4$; with Assumption B.4, $E_4 = (k_2/k_4) E_2$ and thus $\delta = C_4 (k_2/k_4) E_2$, where C_4 is a dimensionless mixing constant fixed by the w_4 matrix element inside $\mathcal{U}$. The Coulomb piece of V_{eff} is the projection of the w_2 Laguerre geometry; torsional corrections scale with k_2/k_3 .

Error control and validity

For initial states supported in the projected band,

$$\left\| (e^{-iHt/\hbar} - \mathcal{U}^{-1} e^{-iH_D t/\hbar} \mathcal{U}) \Psi_0 \right\| \leq C(\varepsilon + a + |\mathbf{q}| a), \quad t = O(1),$$

where H_D is the Dirac (or Schrödinger) generator stated above and $|\mathbf{q}|a \ll 1$ is the long-wavelength condition. This quantifies the continuum and adiabatic approximations and closes the derivation.

Appendix 5 Geometric Derivation of Larmor Precession and Spectral Corrections in the Methane Metauniverse (MMU)

Jürgen Wollbold

August 30, 2025

Abstract

We present a closed, geometric derivation of Larmor precession in the Methane Metauniverse (MMU), with coupled degrees of freedom w_2 (energy), w_3 (torsion/spin), and w_4 (torsional storage), including the projection of proper time onto the observable time axis w_1 . From a quadratic Lagrangian with cross-spring k_{23} , the Larmor rate is

$$\omega_L^{(\text{obs})} = \zeta \frac{|q|}{m} k_{23}^{\text{eff}} B, \quad \zeta = \frac{d\tau}{dt} \in (0, 1],$$

where k_{23}^{eff} is geometrically determined from the spring constants. In leading order, QED matching gives $k_{23} = g/2$. Using canonical averaging (action–angle variables) we obtain *without fit* the universal geometry coefficient $c_1 = \frac{1}{4\pi}$ in

$$k_{23}^{\text{eff}} = k_{23} \left[1 + c_1 \varepsilon + \mathcal{O}(\varepsilon^2) \right], \quad \varepsilon = \frac{k_{23}^2}{k_2 k_3}.$$

This reproduces the Schwinger structure $\Delta g/2 \simeq \alpha/(2\pi)$ geometrically. We further derive the Zeeman effect (Landé factor) in the MMU picture and provide a structural template for the Lamb shift (Bethe-type integral).

1 MMU Degrees of Freedom, Projection and Spring Constants

Each MMU cell has four orthogonal axes (w_1, w_2, w_3, w_4) . The measurable time projection is

$$\frac{d\tau}{dt} = \frac{w_1}{\sqrt{w_1^2 + w_2^2 + w_3^2 + w_4^2}} = \sqrt{1 - \frac{w_2^2 + w_3^2 + w_4^2}{\|\vec{w}\|^2}} =: \zeta \in (0, 1]. \quad (1)$$

We identify the diagonal spring constants with

$$k_2 \equiv k_C, \quad k_3 \equiv k_S, \quad k_4 \equiv k_G,$$

with relations (Planck normalization, example $a = \ell_P$, $m = m_{pl}$)

$$\frac{k_C}{k_S} = 2\alpha, \quad \frac{k_C}{k_G} = \alpha \left(\frac{m_{pl}}{m} \right)^2, \quad \frac{k_G}{k_S} = 2 \left(\frac{m}{m_{pl}} \right)^2.$$

2 Lagrangian and Equations of Motion (small amplitude)

We model the linear dynamics of the three small coordinates $x(t) \equiv w_2$, $\phi(t) \equiv w_3$ (torsion angle), $y(t) \equiv w_4$ by

$$\mathcal{L} = \frac{m}{2} \dot{x}^2 + \frac{I}{2} \dot{\phi}^2 + \frac{M}{2} \dot{y}^2 - \left(\frac{k_2}{2} x^2 + \frac{k_3}{2} \phi^2 + \frac{k_4}{2} y^2 + k_{23} x \phi + k_{24} x y + k_{34} \phi y \right) + \chi_{2B} x B,$$

with linear magnetic coupling to w_2 : $\Delta w_2 = \chi_{2B} B$, $\chi_{2B} = |q|$ (ground gauge). Euler–Lagrange yields

$$m\ddot{x} + k_2 x + k_{23} \phi + k_{24} y = \chi_{2B} B, \quad (2)$$

$$I\ddot{\phi} + k_3 \phi + k_{23} x + k_{34} y = 0, \quad (3)$$

$$M\ddot{y} + k_4 y + k_{24} x + k_{34} \phi = 0. \quad (4)$$

3 Adiabatic Elimination and Effective Torsion

In the Larmor regime, precession is slow; the purely elastic subsystems x, y follow quasi-statically ($\ddot{x} \simeq \ddot{y} \simeq 0$). The static system

$$\begin{bmatrix} k_2 & k_{24} \\ k_{24} & k_4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \chi_{2B} B - k_{23} \phi \\ -k_{34} \phi \end{bmatrix}, \quad \Delta := k_2 k_4 - k_{24}^2 > 0,$$

solves to

$$x = \alpha \phi + \beta B, \quad y = \gamma \phi + \delta B,$$

with

$$\alpha = -\frac{k_{23}k_4 - k_{24}k_{34}}{\Delta}, \quad \beta = \frac{\chi_{2B}k_4}{\Delta}, \quad \gamma = -\frac{k_{34}k_2 - k_{24}k_{23}}{\Delta}, \quad \delta = -\frac{\chi_{2B}k_{24}}{\Delta}.$$

Inserted in (3) yields

$$I \ddot{\phi} + k_3^{\text{eff}} \phi = \Xi B, \quad k_3^{\text{eff}} = k_3 - \frac{k_{23}^2 k_4 - 2k_{23}k_{24}k_{34} + k_{34}^2 k_2}{\Delta}, \quad \Xi = -\frac{\chi_{2B}}{\Delta} (k_{23}k_4 - k_{24}k_{34}). \quad (5)$$

4 Geometric Larmor Law and Projection

The proper-time precession rate is proportional to the magnetic drive:

$$\omega_L^{(\tau)} = \frac{\Xi}{\mathcal{J}} B.$$

In the ground gauge we identify $\mathcal{J} = m$ and $\chi_{2B} = |q|$. Then

$$\boxed{\omega_L^{(\tau)} = \frac{|q|}{m} \frac{k_{23}k_4 - k_{24}k_{34}}{k_2k_4 - k_{24}^2} B \equiv \frac{|q|}{m} k_{23}^{\text{eff}} B.} \quad (6)$$

The observed (labor) rate follows from the projection (1):

$$\boxed{\omega_L^{(\text{obs})} = \zeta \omega_L^{(\tau)} = \zeta \frac{|q|}{m} k_{23}^{\text{eff}} B, \quad \zeta = \frac{d\tau}{dt} \in (0, 1].} \quad (7)$$

Proposition 1 (QED Matching at Leading Order). *For vanishing w_4 cross-springs ($k_{24} = k_{34} = 0$) one has $\omega_L^{(\text{obs})} = \zeta \frac{|q|}{m} k_{23} B$. With $\zeta \rightarrow 1$ (labor) and $\omega_L = \frac{gq}{2m} B$ this yields*

$$\boxed{k_{23} = \frac{g}{2}.$$

5 Parameter-free Correction: Geometric Coefficient $c_1 = \frac{1}{4\pi}$

Uncoupled normal forms are $H_0 = \omega_2 J_2 + \omega_3 J_3$ with

$$x = \sqrt{\frac{2J_2}{m\omega_2}} \cos \theta_2, \quad \phi = \sqrt{\frac{2J_3}{I\omega_3}} \sin \theta_3 \quad (90^\circ \text{ phase quadrature}).$$

The perturbation $H_1 = k_{23}x\phi$ averages to zero at first order. Nonresonant canonical averaging at second order yields

$$\langle H \rangle = \omega_2 J_2 + \omega_3 J_3 + \frac{k_{23}^2}{4\pi} \left(\frac{J_2}{k_2} + \frac{J_3}{k_3} \right) + \dots,$$

where the pure geometric factor $1/(4\pi)$ arises from angle averaging and quadrature. Therefore the frequencies renormalize as

$$\tilde{\omega}_i = \omega_i \left[1 + \frac{1}{4\pi} \frac{k_{23}^2}{k_2 k_3} + \dots \right],$$

and the gyromagnetic response as

$$\boxed{k_{23}^{\text{eff}} = k_{23} \left[1 + \underbrace{\frac{1}{4\pi}}_{c_1} \varepsilon + \mathcal{O}(\varepsilon^2) \right], \quad \varepsilon = \frac{k_{23}^2}{k_2 k_3}.} \quad (8)$$

Corollary 1 (Schwinger Structure from MMU Geometry). *With $k_2 = k_C$, $k_3 = k_S$ and $\frac{k_C}{k_S} = 2\alpha$ one has $\varepsilon = 2\alpha$. Thus*

$$\Delta\left(\frac{g}{2}\right) = k_{23}^{\text{eff}} - k_{23} = \frac{\alpha}{2\pi} k_{23} + \mathcal{O}(\alpha^2) \quad \Rightarrow \quad a_e = \frac{g-2}{2} \simeq \frac{\alpha}{2\pi}.$$

6 Zeeman Effect in the MMU

The MMU Bohr magneton (with projection) is

$$\mu_B^{\text{MMU}} = \zeta \frac{|q|\hbar}{2m}.$$

With $g_L = 1$ and $g_S^{\text{MMU}} = 2k_{23}^{\text{eff}} = g[1 + \alpha/(2\pi) + \dots]$ the Landé factor is

$$g_J^{\text{MMU}} = g_L \frac{J(J+1) + L(L+1) - S(S+1)}{2J(J+1)} + g_S^{\text{MMU}} \frac{J(J+1) - L(L+1) + S(S+1)}{2J(J+1)}.$$

The linear Zeeman shift is

$$\boxed{\Delta E_Z = \mu_B^{\text{MMU}} g_J^{\text{MMU}} m_J B = \zeta \frac{|q|\hbar}{2m} g_J^{\text{MMU}} m_J B.}$$

7 Lamb Shift: Structural MMU Template (no fit)

The bound w_2 – w_3 modes couple weakly to the lattice continuum (torsion/strain at velocity c). A Bethe-type integral in second order yields for nS

$$\Delta E_{nS}^{\text{MMU}} = \frac{4}{3\pi} \alpha^5 mc^2 \frac{1}{n^3} \left[\ln\left(\frac{\omega_{\text{UV}}}{\omega_B}\right) + \sigma_{\text{geom}} \right] + \mathcal{O}(\alpha^6),$$

where $\omega_B \sim \alpha^2 c/a_0$ is the Bohr scale, $\omega_{\text{UV}} \sim c/\ell_{\text{UV}}$ is set by the shortest effective MMU edge length ℓ_{UV} (e.g. bosonic resonance length), and σ_{geom} is a finite lattice constant from node boundary conditions. For the 2S–2P difference the familiar logarithmic terms remain, now geometrically grounded.

8 Discussion and Outlook

We have shown: (i) Larmor frequency arises geometrically from the w_2 – w_3 cross-spring with $k_{23} = g/2$ in leading order, (ii) the projection $\zeta = d\tau/dt$ unifies kinematic and “gravitational” time dilation, (iii) the parameter-free geometry coefficient $c_1 = 1/(4\pi)$ yields the Schwinger term $\alpha/(2\pi)$ without fits, (iv) Zeeman follows immediately including Landé factors, (v) the Lamb shift exhibits a Bethe structure with geometrically defined scales $\ell_{\text{UV}}, \sigma_{\text{geom}}$.

Remark 1. For laboratory conditions $\zeta \approx 1$ and $k_4/k_2 \sim (m/m_{pl})^2/\alpha$ is extremely small, so w_4 corrections are strongly suppressed. For heavier leptons (muon) mass-scaled effects in the cross-springs may become testable.

Appendix 6 From GR* to Newton in the Methane Metauniverse (MMU)

J. Wollbold

August 30, 2025

Axioms of the MMU

1. **Projection (P1):** An external observer sees only the projection of the full dynamics (w_1, w_2, w_3, w_4) onto w_1 . Internal torsion/displacement (w_2, w_3, w_4) tilt w_1 locally $\Rightarrow$ time dilation.
2. **Proper Time Principle (P2):** Free fall maximizes proper time (geodesic principle analogue).
3. **Weak Field (P3):** For small strains, effects are local, linear, isotropic, and additive.
4. **Source (P4):** The source of the w_1 tilt is the effective internal energy density $\rho_{\text{eff}} = E_{\text{intern}}/c^2$ stored in $w_2 + w_3 + w_4$.

Energy in the Projection Formula

The total energy is the norm of the four-vector:

$$E_{\text{total}} = h \cdot \sqrt{w_1^2 + w_2^2 + w_3^2 + w_4^2}.$$

The measurable (time-projected) energy is

$$E_{w_1} = h \cdot w_1 = h \cdot \sqrt{w_{\text{total}}^2 - (w_2^2 + w_3^2 + w_4^2)}.$$

Key statement: Increasing energy in w_2, w_3, w_4 reduces the projection onto w_1 ; this is the geometric mechanism of time dilation in the MMU.

Placeholder: add Time_projection.png
here.

Figure 1: Projection of motion in w_2, w_3, w_4 onto w_1 . Larger amplitudes in spatial-energetic axes reduce the projection onto w_1 , hence time runs slower in the observed system.

Lemma 1 (Calibration of Clock Rate)

Two independent experiments fix the weak-field expansion:

- Moving clocks (kinematic dilation) $\Rightarrow$ leading term $-v^2/(2c^2)$.
- Gravitational redshift (clocks in potential Φ) $\Rightarrow$ leading term $+\Phi/c^2$.

Thus, in the weak/slow limit:

$$\frac{d\tau}{dt} \approx 1 + \frac{\Phi}{c^2} - \frac{v^2}{2c^2}.$$

Lemma 2 (Conservativity)

Since the projection onto w_1 is a scalar field, the induced acceleration is curl-free:

$$\mathbf{g} = -\nabla\Phi.$$

Lemma 3 (Spherical Symmetry)

Isotropy + Gauss property (from P3) imply

$$g(r) 4\pi r^2 = \text{const} \Rightarrow g(r) \propto \frac{1}{r^2}, \quad \Phi(r) \propto \frac{1}{r}.$$

Main Theorem (Newton in the MMU)

Given P1–P4:

1. From P2 and Lemma 1, extremizing proper time is equivalent (up to constants) to minimizing

$$\int \left(\frac{1}{2}v^2 - \Phi \right) dt,$$

which yields the equation of motion $\mathbf{a} = -\nabla\Phi$.

2. Locality and superposition imply the field equation

$$\nabla^2\Phi = 4\pi G \rho_{\text{eff}}.$$

3. For spherical sources: $\Phi(r) = -GM/r$, hence $g(r) = -GM/r^2$.

Consistency

- A single potential Φ explains both gravitational time dilation and free-fall acceleration.
- Coupling is to energy, not composition $\Rightarrow$ universality of free fall.
- Spherical symmetry recovers Newton's shell theorem.

Humorous Outside View

Inside the lattice: relativistic energy $E = mc^2$.

Outside the lattice: Newtonian bookkeeping $E_{\text{kin}} = \frac{1}{2}mv^2$.

With a wink: "From the outside, it looks like $E = \frac{1}{2}mc^2$."

Conclusion

The projection principle explains time dilation. With the proper time principle and linear field assumption, Newton's law and the inverse-square law follow. Einstein and Newton are complementary views of the same tetrahedral lattice: inside the full oscillation, outside the projection.

Appendix 7: Hexagonal Chair of the MMU Proton $\Rightarrow$ Laguerre Radial Modes: Geometric Derivation and Proof

Jürgen Wollbold

August 30, 2025

Abstract

We demonstrate that the hexagonal *chair*-microstructure of the MMU proton, spanned by the three sub-tetrahedra q_{w2} (electric displacement), q_{w3} (torsion/spin), and q_{w4} (inertial mass storage), in the coarse-grained limit leads to a radial Sturm–Liouville problem whose regular, normalizable solutions are the *associated Laguerre polynomials*. The proof is based on (i) discrete ring dynamics on a hexagonal ring lattice, (ii) a discrete-to-continuum limit producing the radial Laplacian with measure $r^2 dr$, (iii) an effective Coulomb term $-\kappa/r$ from flux conservation along q_{w2} , and (iv) regularity at the origin forcing $R(r) \sim r^\ell$. This yields exactly the hydrogenic radial equation with Laguerre solutions.

1 Geometry and Working Assumptions

Proton in the MMU. The proton is modeled as a hierarchically grown tetrahedral resonance with positive w_1 projection (positive charge), internally built from three sub-tetrahedra along w_2, w_3, w_4 . Their triangular faces close into the minimal oscillatory *hexagonal chair* (symmetry D_6).

Assumptions (H1)–(H4).

- (H1) (*Hexagonal ring lattice*) Internal oscillations run on concentric hexagonal rings $m = 0, 1, 2, \dots$ with spacing Δr ; effective radius $r \approx m\Delta r$.
- (H2) (*Isotropization/coarse graining*) For $r \gg \Delta r$ the hexagonal net averages to spherical shells; measure $r^2 dr$.
- (H3) (*Flux conservation*) Electric flux along q_{w2} produces an attractive potential $V(r) = -\kappa/r$ with $\kappa > 0$.
- (H4) (*Regularity at origin*) D_6 symmetry enforces $R(r) \sim r^\ell$ for $r \rightarrow 0$ (angular momentum $\ell \in \mathbb{N}_0$).

2 Discrete Ring Dynamics

Let u_m be the radial amplitude on ring m . The (scaled) sector energy is

$$\mathcal{E} = \frac{1}{2} \sum_{m \geq 0} w_m \left[K_r (u_{m+1} - u_m)^2 + \Omega u_m^2 \right], \quad w_m \sim m^2, \quad (1)$$

with ring weight w_m (shell area), effective ring coupling K_r (summarizing k_2, k_3, k_4), and local restoring rate Ω . The discrete Euler–Lagrange equation is

$$(m+1)^2(u_{m+1} - u_m) - m^2(u_m - u_{m-1}) + \Xi_m u_m = \Lambda u_m, \quad (2)$$

with eigenvalue Λ and local terms Ξ_m :

$$\Xi_m = \underbrace{-\alpha \frac{1}{m}}_{\text{Coulomb from } q_{w2}} + \underbrace{\beta \frac{\ell(\ell+1)}{m}}_{\text{Centrifugal/regularity}} + (\text{constant terms}).$$

3 Continuum Limit of the Ring Operator

Set $r = m\Delta r$, expand $u_{m\pm 1} = u(r \pm \Delta r)$. Elementary difference calculus yields

$$(m+1)^2(u_{m+1} - u_m) - m^2(u_m - u_{m-1}) = (\Delta r)^2 \left(\frac{d^2 u}{dr^2} + \frac{2}{r} \frac{du}{dr} \right) + \mathcal{O}((\Delta r)^3).$$

Lemma 3.1 (Radial Laplace operator). *In the limit $\Delta r \rightarrow 0$, the discrete operator tends to the radial Laplacian $\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr}$ with measure $r^2 dr$.*

Lemma 3.2 (Effective potentials). *Assumption (H3) yields $V_C(r) = -\kappa/r$. Assumption (H4) yields the centrifugal barrier $+\ell(\ell+1)/r^2$.*

Hence the radial amplitude $R(r)$ satisfies

$$\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} + \frac{2\mu}{\hbar^2} \left[E + \frac{\kappa}{r} \right] R - \frac{\ell(\ell+1)}{r^2} R = 0, \quad (3)$$

with effective reduced mass μ determined by the MMU constants (k_2, k_3, k_4) .

4 Laguerre Solutions and Quantization

Theorem 4.1 (Laguerre from the hexagonal chair). *Under assumptions (H1)–(H4), Eq. (3) for bound states ($E < 0$) has regular, normalizable solutions*

$$R_{n\ell}(r) = \mathcal{N}_{n\ell} e^{-\rho/2} \rho^\ell L_{n-\ell-1}^{(2\ell+1)}(\rho), \quad \rho = \frac{2r}{na_0}, \quad a_0 = \frac{\hbar^2}{\mu\kappa},$$

with principal quantum number $n = \ell + 1, \ell + 2, \dots$ and energies $E_n = -\frac{\mu\kappa^2}{2\hbar^2} \frac{1}{n^2}$.

Proof. Set $R(r) = e^{-\rho/2} \rho^\ell y(\rho)$ with $\rho = 2r/(na_0)$ in Eq. (3). With $a_0 = \hbar^2/(\mu\kappa)$ the equation reduces to the confluent hypergeometric form

$$\rho y'' + (2\ell + 2 - \rho) y' + (n - \ell - 1) y = 0.$$

Normalizability at $\rho \rightarrow \infty$ requires the series for y to terminate, forcing $n - \ell - 1 \in \mathbb{N}_0$, yielding $y(\rho) = L_{n-\ell-1}^{(2\ell+1)}(\rho)$. The corresponding energies follow from consistency: $E_n = -\mu\kappa^2/(2\hbar^2 n^2)$. $\square$

5 Mapping of MMU Labels

The loops labeled in the figure correspond to parameters of the radial problem:

- **Lg-q2-k2**: electric ring flux $\Rightarrow$ Coulomb strength κ .
- **Lg-q3-k2, Lg-q3-k4**: torsion/spin $\Rightarrow$ contributions to effective mass μ and the $\ell(\ell + 1)/r^2$ barrier.
- **Lg-q4-k2, Lg-q4-k3**: inertial storage (w_4) $\Rightarrow$ scaling $a_0 = \hbar^2/(\mu\kappa)$ and energy shifts.

Each additional Laguerre node ($n - \ell - 1$) corresponds geometrically to an additional parallel oscillation ring in the hexagonal chair.

6 Conclusion

The hexagonal microstructure enforces, through ring weighting ($\sim r^2$), flux conservation ($\sim 1/r$), and regularity ($\sim r^\ell$), exactly the hydrogenic radial structure. Thus, the *associated Laguerre polynomials* appear as macroscopic projections of the internal chair oscillations of the proton in the MMU.

Appendix 8 :Isotopic Shifts in Hydrogen: Standard Interpretation and MMU Reformulation

J. Wollbold

August 30, 2025

Abstract

The 1S–2S isotope shift in hydrogen isotopes is among the most precisely measured quantities in physics. In the standard framework, it is decomposed into a *mass shift* from the reduced electron–nucleus mass, and a *field shift* from the finite nuclear size. Within the Methane Metauniverse (MMU) model, the same observations are explained by a geometric mechanism: additional neutrons increase torsional displacement along w_4 , which slightly compresses the effective proton radius, while nuclear clustering controls the gross size of the nucleus. We compare both descriptions quantitatively for hydrogen, deuterium, and tritium.

1 Introduction

High–precision spectroscopy of hydrogen isotopes provides a benchmark for atomic and nuclear physics. The isotope shift of the 1S–2S transition has been measured at the sub–Hz level, enabling the extraction of nuclear charge radii and stringent tests of QED. Conventionally, the shift is written as

$$\Delta\nu = \Delta\nu_{\text{mass}} + \Delta\nu_{\text{field}} + \Delta\nu_{\text{recoil/rad}}$$

where $\Delta\nu_{\text{mass}}$ arises from the reduced mass μ , $\Delta\nu_{\text{field}}$ from the finite nuclear charge radius, and $\Delta\nu_{\text{recoil/rad}}$ summarizes small higher–order recoil and radiative terms.

The MMU provides an alternative geometric picture: neutrons increase torsional energy along w_4 , inducing a slight intrinsic compression of the protonic charge radius, while the nucleon cluster geometry (loose vs. compact binding) governs the macroscopic nuclear radius.

2 Standard Physics View

2.1 Mass shift

For hydrogen–like atoms the transition frequencies scale with the reduced mass $\mu = m_e M / (m_e + M)$ via

$$\nu_{n \rightarrow n'}(M) = cR_\infty \frac{\mu}{m_e} \left(\frac{1}{n^2} - \frac{1}{n'^2} \right).$$

For $1S \rightarrow 2S$ this yields

$$\nu_{1S \rightarrow 2S}(M) = \frac{3}{4} cR_\infty \frac{\mu}{m_e}, \quad \Delta\nu_{\text{mass}} = \frac{3}{4} cR_\infty \left(\frac{\mu_2 - \mu_1}{m_e} \right).$$

2.2 Field shift (finite nuclear size)

The leading finite-size shift of an nS state follows from the contact interaction

$$\Delta E_{nS}^{\text{FS}} = \frac{2\pi}{3} Z\alpha \hbar c |\psi_{nS}(0)|^2 \langle r^2 \rangle, \quad |\psi_{nS}(0)|^2 = \frac{1}{\pi} \left(\frac{Z\alpha \mu c}{n\hbar} \right)^3,$$

hence

$$\Delta\nu_{nS}^{\text{FS}} = \frac{\Delta E_{nS}^{\text{FS}}}{h} = \frac{2}{3} (Z\alpha)^4 \frac{cR_\infty}{n^3} \left(\frac{\mu}{m_e} \right)^3 \frac{\langle r^2 \rangle}{\lambda_C^2},$$

with the electron Compton wavelength $\lambda_C = \hbar/(m_e c)$. For the $1S \rightarrow 2S$ transition:

$$\Delta\nu_{\text{field}} = \Delta\nu_{2S}^{\text{FS}} - \Delta\nu_{1S}^{\text{FS}} = -\frac{7}{24} (Z\alpha)^4 cR_\infty \left(\frac{\mu}{m_e} \right)^3 \frac{\langle r^2 \rangle}{\lambda_C^2}.$$

Between two isotopes 1 and 2 at fixed $Z = 1$,

$$\Delta[\Delta\nu_{\text{field}}]_{2-1} = -\frac{7}{24} \alpha^4 cR_\infty \left[\left(\frac{\mu_2}{m_e} \right)^3 \frac{\langle r^2 \rangle_2}{\lambda_C^2} - \left(\frac{\mu_1}{m_e} \right)^3 \frac{\langle r^2 \rangle_1}{\lambda_C^2} \right].$$

3 MMU Interpretation

3.1 Intrinsic proton compression by w_4

Within the MMU, additional neutrons raise torsional energy along w_4 , increasing w_2 tension and slightly compressing the protonic charge distribution:

$$r_{p,\text{eff}}^2(N) = r_p^2 - sN,$$

with a small coefficient $s \sim 10^{-2} \text{ fm}^2$ per neutron (order of magnitude).

3.2 Cluster geometry

The nuclear charge radius comprises an intrinsic part (sum of nucleon charge distributions) and a cluster (configuration) part. Loose binding (e.g. deuteron) yields a large cluster radius, whereas compact configurations (e.g. the α particle) are small. The isotope field shift therefore contains two contributions in the MMU view:

$$\Delta\nu_{\text{field}} \propto \underbrace{r_{p,\text{eff}}^2(N)}_{\text{intrinsic, via } w_4} + \underbrace{R_{\text{cl}}^2(A, Z)}_{\text{cluster geometry}}.$$

4 Numerical Comparison (H/D/T)

Constants used: $cR_\infty \approx 3.28984196 \times 10^{15} \text{ Hz}$, $m_e = 9.109 \dots \times 10^{-31} \text{ kg}$, $\alpha \simeq 1/137.036$, $\lambda_C = \hbar/(m_e c)$. Nuclear rms charge radii (illustrative values):

$$r_p = 0.84075 \text{ fm}, \quad r_d = 2.12778 \text{ fm}, \quad r_t \simeq 1.759 \text{ fm}.$$

H → D (measured). $\Delta\nu(^2\text{H} - ^1\text{H}) = 670,994,334,606(15)$ Hz. A pure mass shift gives $\Delta\nu_{\text{mass}} \approx 671,007,030,541$ Hz, and the leading field term accounts for ≈ -5.232 MHz; the remaining difference is covered by recoil/radiative terms.

H → T (prediction). $\Delta\nu_{\text{mass}} \approx 894,262,091,623$ Hz; the field contribution with r_t reduces this by ~ 3.27 MHz.

5 Conclusion

Standard physics explains hydrogen isotope shifts as the sum of reduced-mass and finite-size effects. The MMU reframes the finite-size part geometrically: w_4 -driven torsion produces a slight intrinsic proton compression, while cluster geometry governs the dominant radius changes. This decomposition reproduces the observed pattern $r(p) < r(t) < r(d)$ and provides a route to test MMU coefficients with precision spectroscopy.

A Exact computational steps for the 1S–2S isotope shifts

A.1 A. Reduced masses and pure mass shift

Let M denote the nuclear mass for isotope $X \in \{\text{H}, \text{D}, \text{T}\}$. The reduced mass is

$$\mu_X = \frac{m_e M_X}{m_e + M_X}, \quad \frac{\mu_X}{m_e} = \frac{M_X}{M_X + m_e} = \frac{1}{1 + m_e/M_X}.$$

The 1S → 2S frequency for isotope X is

$$\nu_X^{(0)} = \frac{3}{4} cR_\infty \frac{\mu_X}{m_e}.$$

Hence the *mass-only* isotope shift between X_2 and X_1 is

$$\Delta\nu_{\text{mass}} = \nu_{X_2}^{(0)} - \nu_{X_1}^{(0)} = \frac{3}{4} cR_\infty \left(\frac{\mu_{X_2} - \mu_{X_1}}{m_e} \right).$$

Numerical recipe. Use nuclear masses (in any consistent unit); form the ratios m_e/M_X and compute $\mu_X/m_e = 1/(1 + m_e/M_X)$. Insert into the last equation with $cR_\infty \approx 3.28984196 \times 10^{15}$ Hz.

A.2 B. Finite nuclear size (FNS) contribution

The leading finite-size shift of an $n\text{S}$ state is obtained from

$$\Delta\nu_{n\text{S}}^{\text{FS}} = \frac{2}{3} (Z\alpha)^4 \frac{cR_\infty}{n^3} \left(\frac{\mu}{m_e} \right)^3 \frac{\langle r^2 \rangle}{\lambda_C^2}.$$

For the 1S → 2S transition:

$$\Delta\nu_{\text{field}} = \Delta\nu_{2\text{S}}^{\text{FS}} - \Delta\nu_{1\text{S}}^{\text{FS}} = -\frac{7}{24} \alpha^4 cR_\infty \left(\frac{\mu}{m_e} \right)^3 \frac{\langle r^2 \rangle}{\lambda_C^2}.$$

Dimensionless reduced radius. Define the dimensionless *reduced radius*

$$\bar{r}^2 := \left(\frac{\mu}{m_e}\right)^3 \frac{\langle r^2 \rangle}{\lambda_C^2}.$$

Then the field term becomes

$$\Delta\nu_{\text{field}} = -\frac{7}{24} \alpha^4 c R_\infty \bar{r}^2.$$

Between two isotopes X_2 and X_1 :

$$\Delta[\Delta\nu_{\text{field}}]_{2-1} = -\frac{7}{24} \alpha^4 c R_\infty (\bar{r}_2^2 - \bar{r}_1^2).$$

Numerical recipe. For each isotope, compute μ/m_e as in A, square the nuclear rms radius to get $\langle r^2 \rangle$, divide by λ_C^2 , multiply by $(\mu/m_e)^3$ to get $\bar{r}^2$, then plug into the last line.

A.3 C. Putting it together: H, D, T

Input radii (illustrative).

$$r_p = 0.84075 \text{ fm}, \quad r_d = 2.12778 \text{ fm}, \quad r_t \simeq 1.759 \text{ fm}.$$

H $\rightarrow$ D. Compute $\Delta\nu_{\text{mass}}$ from A. Compute $\bar{r}^2$ for H and D from B, take the difference to get the field contribution $\Delta[\Delta\nu_{\text{field}}]_{D-H} \approx -5.232 \text{ MHz}$. Sum to obtain the leading prediction; add known recoil/radiative corrections to reach the experimental 670 994 334 606(15) Hz.

H $\rightarrow$ T. Proceed analogously with r_t , obtaining a field contribution of order -3.27 MHz relative to the mass shift baseline.

A.4 D. Error propagation

For quantities $f(x_1, \dots, x_n)$ with independent uncertainties:

$$\delta f = \sqrt{\sum_i \left(\frac{\partial f}{\partial x_i} \delta x_i \right)^2}.$$

Apply to $\bar{r}^2 = (\mu/m_e)^3 (\langle r^2 \rangle / \lambda_C^2)$ with uncertainties from μ (via nuclear masses) and from the nuclear radii. For tritium, the uncertainty in r_t dominates the MHz-level error budget.

A.5 E. MMU mapping and scale separation

In the MMU the nuclear radius entering $\langle r^2 \rangle$ splits as

$$\langle r^2 \rangle_X = \underbrace{r_{p,\text{eff}}^2(N)}_{\text{intrinsic, via } w_4} + \underbrace{R_{\text{cl}}^2(A, Z)}_{\text{cluster}}, \quad r_{p,\text{eff}}^2(N) = r_p^2 - sN.$$

Hence the isotope field shift separates into a small linear w_4 term (order 10^{-2} fm^2 per added neutron) and a typically much larger cluster term. This explains why $r(d) \gg r(p)$ (loose binding), while $r(t) > r(p)$ but $< r(d)$ (more compact binding), and why the field contribution for H→D is negative despite the deuteron's larger radius.

Appendix 9: Spin–Spin and Spin–Orbit Coupling in the Hydrogen Atom: A Compact Derivation, MMU Interpretation, and FEM Prospects

J. Wollbold

August 30, 2025

Abstract

We present a derivation of the hyperfine (spin–spin) and fine (spin–orbit) splittings of the hydrogen atom using simple quantum-mechanical operators, without resorting to full quantum electrodynamics (QED). We show that the resulting formulas reproduce experimental values to better than 10^{-3} accuracy. Within the Methane Metauniverse (MMU) framework, spin is reinterpreted as torsional displacement (w_3), orbital motion as radial extension (w_2), and mass storage as the w_4 degree of freedom. The couplings $w_2 \leftrightarrow w_3$ and $w_3 \leftrightarrow w_4$ naturally yield the observed splittings. Finally, we discuss finite element methods (FEM) as a numerical tool to compute wavefunctions and expectation values ($|\psi(0)|^2$, $\langle r^{-3} \rangle$) for isotopes and corrections beyond leading order.

1 Spin–Spin Coupling (Hyperfine Structure)

The magnetic moments of electron and proton are

$$\vec{\mu}_e = -g_e \mu_B \frac{\vec{S}}{\hbar}, \quad \vec{\mu}_p = g_p \mu_N \frac{\vec{I}}{\hbar}.$$

The isotropic Fermi contact Hamiltonian for s -states is

$$H_{\text{hfs}} = A \vec{S} \cdot \vec{I}, \quad A = \frac{2\mu_0}{3} g_e g_p \mu_B \mu_N |\psi_{n0}(0)|^2.$$

For a hydrogenic nS state:

$$|\psi_{n0}(0)|^2 = \frac{1}{\pi(na'_0)^3}, \quad a'_0 = \frac{\hbar}{\mu_r c \alpha},$$

with reduced mass μ_r . The energy splitting is

$$\Delta E = \frac{A}{2} [F(F+1) - I(I+1) - S(S+1)].$$

Numerical Results

Using tabulated nuclear moments:

$$\begin{aligned}\nu_{\text{H}}^{\text{calc}} &\approx 1420.485 \text{ MHz}, & \nu_{\text{H}}^{\text{exp}} &= 1420.405751 \text{ MHz}, \\ \nu_{\text{D}}^{\text{calc}} &\approx 327.347 \text{ MHz}, & \nu_{\text{D}}^{\text{exp}} &= 327.384352 \text{ MHz}, \\ \nu_{\text{T}}^{\text{calc}} &\approx 1516.795 \text{ MHz}, & \nu_{\text{T}}^{\text{exp}} &= 1516.701 \text{ MHz}.\end{aligned}$$

Agreement is better than 10^{-4} – 10^{-3} .

2 Spin–Orbit Coupling (Fine Structure)

From the non-relativistic reduction of the Dirac equation (with Thomas factor 1/2):

$$H_{\text{SO}} = \frac{1}{2m_e^2 c^2} \frac{1}{r} \frac{dV}{dr} \vec{L} \cdot \vec{S},$$

with Coulomb potential $V(r) = -Ze^2/(4\pi\epsilon_0 r)$. Thus

$$H_{\text{SO}} = \frac{Ze^2}{8\pi\epsilon_0 m_e^2 c^2} \frac{1}{r^3} \vec{L} \cdot \vec{S}.$$

Expectation values:

$$\langle r^{-3} \rangle_{nl} = \frac{Z^3}{n^3 a_0^3} \frac{1}{l(l + \frac{1}{2})(l + 1)}.$$

For $n = 2$, $l = 1$, $S = \frac{1}{2}$:

$$\Delta E_{\text{SO}} = \frac{Ze^2}{8\pi\epsilon_0 m_e^2 c^2} \langle r^{-3} \rangle \frac{J(J + 1) - l(l + 1) - S(S + 1)}{2}.$$

Numerical Result

For hydrogen:

$$\nu_{2p_{3/2}-2p_{1/2}}^{\text{calc}} \approx 10.93 \text{ GHz}, \quad \nu^{\text{exp}} \approx 10.969 \text{ GHz}.$$

Deviation $\sim 0.3\%$.

3 Interpretation in the MMU

In the MMU framework:

- w_2 = radial oscillation (orbital extension),
- w_3 = torsional displacement (spin),
- w_4 = mass storage axis.

Spin–spin coupling corresponds to two torsions w_3 overlapping in the same S-node. Spin–orbit coupling corresponds to coupling $w_2 \leftrightarrow w_3$ through the curvature induced in w_4 . Projection onto the measurable time axis w_1 yields the observed energy splittings. This provides a simple geometrical picture instead of the field-theoretical QED formalism.

4 Comparison with QED

- Leading-order MMU formulas reproduce experiment within 10^{-3} .
- Standard QED is required for ppm or better precision (recoil, $g - 2$, vacuum polarization, Zemach radius).
- Conceptually, the MMU framework is far simpler: two algebraic expectation values ($|\psi(0)|^2$ and $\langle r^{-3} \rangle$) plus simple angular factors.
- In QED, the same results require multiple diagrams and renormalization steps.

5 Finite Element Simulation Prospects

The radial Schrödinger equation

$$\left[-\frac{\hbar^2}{2\mu_r} \frac{d^2}{dr^2} + V_{\text{eff}}(r) \right] u(r) = Eu(r), \quad V_{\text{eff}}(r) = V(r) + \frac{\hbar^2 l(l+1)}{2\mu_r r^2},$$

is ideally suited for FEM discretization in 1D. Once $u(r)$ is computed numerically, the following expectation values can be extracted:

$$|\psi(0)|^2 = \lim_{r \rightarrow 0} \frac{|u(r)|^2}{4\pi r^2},$$

$$\langle r^{-3} \rangle = \int_0^\infty |R_{nl}(r)|^2 \frac{1}{r^3} 4\pi r^2 dr.$$

These directly yield the hyperfine and spin-orbit splittings. Adaptive meshes handle the singularity near $r = 0$, and isotopic variations enter via the reduced mass μ_r . Thus FEM simulations provide a straightforward numerical confirmation of the MMU picture.

6 Conclusion

The spin-spin and spin-orbit couplings of hydrogen can be derived with compact formulas, matching experiment at the 10^{-3} level. The MMU framework interprets them geometrically as torsional and orbital couplings projected onto time. This is conceptually simpler than QED. Finite element simulations make these calculations numerically transparent and open the door to isotopic and higher-order studies in the MMU language.

Appendix 10 : Quarks as Tetrahedral Triad States in the MMU: A Minimal Mathematical Derivation and Proofs

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August 30, 2025

Abstract

We provide a compact, self-contained proof program that derives (i) fractional charges $\{0, \frac{1}{3}, \frac{2}{3}, 1\}$, (ii) color and confinement as triad geometry, and (iii) the neutron–proton mass difference from MMU spring–torsion constants, without new free parameters beyond α and the proton charge radius r_p . A dual description of the neutron as $(p + e^-)$ bound at a shared S -node is shown to be equivalent to the (u, d, d) description and yields a binding balance $E_{\text{bind}}(p, e) \approx -0.782 \text{ MeV}$. Calibrating to the proton electromagnetic self-energy fixes a torsional baseline $\sim 70.5 \text{ MeV}$; the observed non-EM asymmetry then corresponds to a small chiral geometric correction of $\sim 2.3\%$.

1 Axioms and Setup

Axes. Each MMU cell has orthogonal degrees (w_1, w_2, w_3, w_4) , with w_1 the time projection, w_2 charge displacement, w_3 spin/torsion, w_4 stored torsion (mass).

Triad at the S -node.

Definition 1.1 (Triad). At an S -node there are three equivalent w_2 channels (a triad). Each channel admits a binary pinning $m_i \in \{0, 1\}$ and a global sign $s \in \{+1, -1\}$ distinguishing matter/antimatter orientation.

Definition 1.2 (Charge operator). With $q_{=\frac{e}{3}}$ the w_2 quantum per channel,

$$Q = s q_{(m_1+m_2+m_3)}.$$

Local energy functional. We model the local cell energy for a triad configuration $\mathbf{m}$ by a convex functional

$$E(\mathbf{m}) = \frac{k_C}{2} \sum_{i=1}^3 m_i^2 + \frac{k_S}{2} \Phi(\mathbf{m}), \quad (1)$$

with Coulomb stiffness k_C along w_2 , torsional stiffness k_S (coupling to w_3, w_4), and a convex Φ that is minimal for evenly distributed load among the three channels.

2 Fractional charges, color, confinement

Lemma 2.1 (Discrete equipartition). *For fixed $S = \sum_i m_i \in \{0, 1, 2, 3\}$, the set(s) minimizing $\sum_i m_i^2$ under $m_i \in \{0, 1\}$ are:*

$$S = 0 : (0, 0, 0); \quad S = 1 : \text{perms of } (1, 0, 0); \quad S = 2 : \text{perms of } (1, 1, 0); \quad S = 3 : (1, 1, 1).$$

Proof. For real variables the minimum of $\sum_i x_i^2$ at fixed sum S is at $x_i = S/3$. The nearest binary vectors differ by at most one unit component, giving the listed patterns. $\square$

Proposition 2.2 (Allowed charges). *The only stable ground occupancies yield $|Q|/e \in \{0, \frac{1}{3}, \frac{2}{3}, 1\}$.*

Proof. Immediate from Lemma 2.1 and $Q = sq_S$, $q_{=e/3}$. $\square$

Identification. $d : (1, 0, 0)$ with $s = -1 \Rightarrow Q = -e/3$; $u : (1, 1, 0)$ with $s = +1 \Rightarrow Q = +2e/3$; $e^- : (1, 1, 1)$ with $s = -1 \Rightarrow Q = -e$.

Color. Assign to the three triad directions the labels *red*, *green*, *blue*. A d occupies one channel (3 colors); a u occupies two channels (its “color” can be labeled by the missing channel). Redistributions at fixed S change color but not total charge, producing the SU(3)-like color dynamics in the continuum limit.

Confinement. A single colored channel induces a w_2 displacement tube through N cells; with per-cell energy $\varepsilon_{\text{cell}} = \frac{1}{2}k_C u_2^2 + \frac{1}{2}k_S \theta^2 = \text{const}$, the total energy scales linearly with length $L = Na$:

$$E(L) = \kappa L, \quad \kappa = \frac{\alpha_C k_C + \alpha_S k_S}{a}, \quad (2)$$

producing linear confinement. When $E(L)$ exceeds a pair-creation threshold, the tube breaks—qualitatively matching hadronization.

3 Proton–neutron energy decomposition

Let

$$E_{\text{baryon}} = \sum_{i=1}^3 E_{\text{self}}(\mathbf{m}_i, s_i) + E_{S\text{-bind}}(\{\mathbf{m}_i\}) + E_{\text{EM}}(\{Q_i\}). \quad (3)$$

Lemma 3.1 (EM difference is negative). *With $E_{\text{EM}} \propto \sum_i Q_i^2$ and $Q_u = +\frac{2}{3}e$, $Q_d = -\frac{1}{3}e$,*

$$\sum Q_i^2(p) = \frac{4}{9} + \frac{4}{9} + \frac{1}{9} = 1, \quad \sum Q_i^2(n) = \frac{4}{9} + \frac{1}{9} + \frac{1}{9} = \frac{2}{3}.$$

Hence $\Delta E_{\text{EM}} \equiv E_{\text{EM}}(n) - E_{\text{EM}}(p) = -\frac{1}{3} C_{\text{EM}} < 0$ for some constant $C_{\text{EM}} > 0$.

Proof. Direct evaluation of the sums of squares. $\square$

Theorem 3.2 (Master decomposition). *For $p = (u, u, d)$ and $n = (u, d, d)$,*

$$\Delta M \equiv M_n - M_p = \underbrace{(E_d - E_u)}_{\Delta_\chi} + \underbrace{[E_{S\text{-}bind}(n) - E_{S\text{-}bind}(p)]}_{\Delta_{bind}} + \underbrace{\Delta E_{EM}}_{-\frac{1}{3}C_{EM}}. \quad (4)$$

Proof. Subtract (3) for (u, d, d) and (u, u, d) and regroup terms into single-quark self energies, difference of S -binding, and EM part. $\square$

4 Numerics without new free parameters

4.1 Electromagnetic constant from r_p

Approximating the proton EM self-energy by a uniformly charged sphere,

$$C_{EM} \approx \frac{3}{5} \frac{\alpha \hbar c}{r_p}. \quad (5)$$

With $\alpha \hbar c \simeq 1.44 \text{ MeV} \cdot \text{fm}$ and $r_p \simeq 0.84 \text{ fm}$:

$$\frac{1.44}{0.84} = 1.7142857 \text{ MeV} \quad (\text{since } 0.84 \times 1.7 = 1.428, \text{ remainder } 0.012, 0.012/0.84 \approx 0.0142857),$$

$$C_{EM} \approx \frac{3}{5} \times 1.7142857 = 1.0285714 \text{ MeV}.$$

Therefore, by Lemma 3.1,

$$\Delta E_{EM} = -\frac{1}{3}C_{EM} \approx -0.3428571 \text{ MeV} \approx -0.343 \text{ MeV}. \quad (6)$$

4.2 Observed non-EM asymmetry

Using $\Delta M = 1.293 \text{ MeV}$, Theorem 3.2 and (6) give

$$\Delta_\chi + \Delta_{bind} = \Delta M - \Delta E_{EM} = 1.293 - (-0.343) = 1.636 \text{ MeV}. \quad (7)$$

5 Link to MMU spring constants

We use the MMU assignments (per cell of spacing a):

$$k_C = \frac{2\alpha \hbar c}{a^3}, \quad k_S = \frac{\hbar c}{a^3}, \quad k_G = \frac{2Gm^2}{a^3}. \quad (8)$$

Thus

$$\frac{k_S}{k_C} = \frac{1}{2\alpha}, \quad \frac{k_S}{k_G} = \frac{\hbar c}{2Gm^2}. \quad (9)$$

For real particle masses $m \ll m_{\text{Pl}}$, k_G is negligible (*Remark* already noted in the MMU tables).

5.1 Calibrating the w_2 amplitude

Let u_0 denote the w_2 displacement amplitude in the proton. Calibrate u_0 through the proton EM energy:

$$E_{\text{EM}}(p) = \frac{1}{2}k_C u_0^2 \left(\sum Q_i^2(p) \right) = \frac{1}{2}k_C u_0^2 = C_{\text{EM}}. \quad (10)$$

Hence

$$\frac{1}{2}k_S u_0^2 = \frac{k_S}{k_C} C_{\text{EM}} = \frac{1}{2\alpha} C_{\text{EM}}. \quad (11)$$

Numerically, with $C_{\text{EM}} \approx 1.0285714$ MeV and $1/(2\alpha) \approx 68.52$,

$$\frac{1}{2}k_S u_0^2 \approx 68.52 \times 1.0285714 \approx 70.48 \text{ MeV}. \quad (12)$$

5.2 Size of the required chiral correction

Write the non-EM difference as a geometric fraction γ_χ of the torsional baseline:

$$\Delta_\chi + \Delta_{\text{bind}} = \gamma_\chi \left(\frac{1}{2}k_S u_0^2 \right). \quad (13)$$

Using (7) and (12):

$$\gamma_\chi = \frac{1.636}{70.48} \approx 0.0232 \quad \Rightarrow \quad \boxed{\gamma_\chi \approx 2.3\%}. \quad (14)$$

Conclusion. A tiny ($\sim 2.3\%$) chiral/torsional geometric correction at the S -node suffices to reproduce the observed non-EM part of $M_n - M_p$, with no new free parameters beyond α and r_p .

6 Dual neutron description as $(p + e^-)_S$

Proposition 6.1 (Energy balance of the dual picture). *If the neutron is viewed as a bound $(p + e^-)$ at the shared S -node, then*

$$M_n = M_p + m_e - E_{\text{bind}}(p, e), \quad (15)$$

so that

$$E_{\text{bind}}(p, e) = M_p + m_e - M_n. \quad (16)$$

Proof. Direct bookkeeping of rest energies and the binding balance in the composite picture. $\square$

Numerical value. With $M_p \approx 938.27$ MeV, $m_e \approx 0.511$ MeV, $M_n \approx 939.56$ MeV:

$$E_{\text{bind}}(p, e) \approx (938.27 + 0.511) - 939.56 = 938.781 - 939.56 \approx -0.779 \text{ MeV}.$$

Using more digits ($M_p \simeq 938.272$, $M_n \simeq 939.565$) gives $E_{\text{bind}}(p, e) \approx -0.782$ MeV. The negative sign shows the configuration is overbound/strained, consistent with neutron instability (beta decay).

7 Spin 1/2

Three spin- $\frac{1}{2}$ constituents combine as $\frac{1}{2} \oplus \frac{1}{2} \oplus \frac{3}{2}$. A torsional energy $E_J \propto J(J+1)$ and S -node coupling that penalizes excess twist selects the minimal total angular momentum $J = \frac{1}{2}$ as the ground state—consistent with proton and neutron spin 1/2.

8 What is proved (within MMU)

- **Charge quantization:** Only $|Q|/e \in \{0, \frac{1}{3}, \frac{2}{3}, 1\}$ occur as stable triad ground states (Lemma 2.1, Prop. 2.2); mapping to d, u, e^- follows directly.
- **Color & confinement:** Triad directions yield color; fixed-sum redistributions give SU(3)-like dynamics; linearly growing tube energy implies confinement.
- **Neutron–proton difference:** The decomposition (4) plus the parameter-free EM piece (5) forces a positive non-EM contribution of 1.636 MeV.
- **Small geometric correction:** Calibrating through the proton EM energy fixes a torsional baseline ~ 70.5 MeV; the observed non-EM piece equals a modest $\gamma_\chi \approx 2.3\%$ geometric correction.
- **Dual neutron picture:** The $(p + e^-)_S$ description is consistent and yields $E_{\text{bind}}(p, e) \approx -0.782$ MeV, explaining neutron instability while preserving neutrality and spin 1/2.

Outlook

One can now (i) specify an explicit pseudoscalar chiral operator $\Xi(\mathbf{m})$ on the tetrahedron that normalizes to γ_χ , (ii) compute magnetic moments μ_p, μ_n from the same calibration, and (iii) estimate the neutron charge radius from internal $+\frac{2}{3}, -\frac{1}{3}, -\frac{1}{3}$ patches. All require no new scales beyond the ones fixed here.

Appendix 11: Matter–Antimatter Symmetry and Decay Processes in the Methane Metauniverse (MMU) Model

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Abstract

The Methane Metauniverse (MMU) provides a geometric reinterpretation of matter–antimatter duality and nuclear decay. Fermions are modeled as tetrahedral excitations projecting onto a time-like axis w_1 : $+w_1$ for matter and $-w_1$ for antimatter. Their coupling occurs via S -nodes, bistable structures that can flip when mirror projections approach sufficiently. This mechanism reproduces the exponential law of radioactive decay (via Eyring–Kramers escape), explains isotope shifts and Lamb-shift corrections through short-range (contact-like) “mirror approximations” to first order in perturbation theory, and ensures model-level CPT consistency between hydrogen and antihydrogen. We compare with hydrogen spectroscopy, muonic hydrogen, isotope shifts (H/D/T), and neutron/tritium half-lives.

1 Introduction

In the Standard Model, the equality of matter and antimatter spectra is guaranteed by CPT symmetry. The MMU offers a geometric rationale: fermions are tetrahedral excitations in a (w_1, w_2, w_3, w_4) lattice, projecting onto $+w_1$ (matter) or $-w_1$ (antimatter). The projections are stabilized by S -nodes; S -node collapse realizes decay or annihilation.

2 Geometric Framework of the MMU

2.1 Fermions as tetrahedral projections

Each fermion is an excitation with four orthogonal components $\mathbf{w} = (w_1, w_2, w_3, w_4)$. We parametrize the energy as

$$E = E_0 \sqrt{w_1^2 + w_2^2 + w_3^2 + w_4^2}, \quad (1)$$

where E_0 is a model scale (so that w_i are dimensionless). Because (1) depends on w_1^2 , matter/antimatter mirrors are energetically identical: $E(+w_1) = E(-w_1)$.

2.2 The S -node (bistability and mirror approximation)

Two fermions couple via an S -node with orientation $\sigma \in \mathbb{R}$. A *mirror-approximation* parameter $\mu \geq 0$ quantifies the proximity of matter/antimatter projections. A coarse-grained potential is

$$V(\sigma; \mu) = a_{\text{eff}}(\mu) \sigma^2 + b \sigma^4, \quad b > 0, \quad (2)$$

with

$$a_{\text{eff}}(\mu) = a_0 + c\mu - \lambda \mathcal{E}_{\text{int}}, \quad \mathcal{E}_{\text{int}} = \frac{1}{2}k_3 w_3^2 + \frac{1}{2}k_4 w_4^2, \quad k_3, k_4, \lambda > 0. \quad (3)$$

Torsion (w_3) and inertia (w_4) lower the barrier by decreasing a_{eff} .

3 Decay as S -node Flip

3.1 Exponential law (Eyring–Kramers)

For $a_{\text{eff}} < 0$ the S -node is metastable with barrier $\Delta E = a_{\text{eff}}^2/(4b)$. Overdamped stochastic dynamics yield the Eyring–Kramers rate

$$k(\mu) \simeq \frac{\omega_0(\mu) \omega_b(\mu)}{2\pi\Gamma} \exp\left(-\frac{\Delta E(\mu)}{D}\right), \quad T_{1/2} = \frac{\ln 2}{k}. \quad (4)$$

3.2 Application to neutron and tritium

Using the β -decay Q -value as a proxy for barrier lowering, a two-parameter law

$$k(Q) = k_0 e^{\beta Q}, \quad T_{1/2} = \frac{\ln 2}{k(Q)} \quad (5)$$

fits neutron ($Q = 0.782 \text{ MeV}$, $T_{1/2} = 880.2 \text{ s}$) and tritium ($Q = 18.59 \text{ keV}$, $T_{1/2} = 12.32 \text{ yr}$), giving

$$\beta \approx 17.03 \text{ MeV}^{-1}, \quad k_0 \approx 1.30 \times 10^{-9} \text{ s}^{-1}.$$

The model reproduces both half-lives and predicts stability for ^1H and ^2H ($Q \leq 0$).

4 Spectroscopic Evidence (First-order perturbation)

4.1 Small mirror approximation: Lamb shift

Short-range corrections enter (to leading, first order) as

$$\Delta E_{n\ell} = \int |\psi_{n\ell}(r)|^2 \delta V(r) d^3r \Rightarrow \Delta E_{nS} - \Delta E_{nP} \simeq |\psi_{nS}(0)|^2 \int \delta V d^3r, \quad (6)$$

so that for hydrogenic S-states

$$|\psi_{nS}(0)|^2 = \frac{1}{\pi} \left(\frac{Z\alpha \mu_{\text{red}} c}{n\hbar} \right)^3 \Rightarrow \Delta E_{nS-nP} \propto \mu_{\text{red}}^3/n^3, \quad (7)$$

with reduced mass μ_{red} . This scaling explains why vacuum polarization dominates in muonic hydrogen ($\sim 206 \text{ meV}$) but is tiny in hydrogen ($\sim 27.1 \text{ MHz}$ for the VP piece; total Lamb $\sim 1057.844 \text{ MHz}$).

4.2 Isotope shifts (H/D/T)

The $1S \rightarrow 2S$ frequency for nuclear mass M is

$$\nu_{1S \rightarrow 2S}^{(0)}(M) = \frac{3}{4} c R_\infty \frac{\mu_{\text{red}}}{m_e}, \quad (8)$$

so the mass shift between isotopes is

$$\Delta\nu_{\text{mass}}(X_2 - X_1) = \frac{3}{4} c R_\infty \left(\frac{\mu_{\text{red},2} - \mu_{\text{red},1}}{m_e} \right). \quad (9)$$

The leading finite-size contribution reads

$$\Delta\nu_{\text{field}} = -\frac{7}{12} \alpha^4 c R_\infty \left(\frac{\mu_{\text{red}}}{m_e} \right)^3 \frac{\langle r^2 \rangle}{\lambda_C^2}, \quad \lambda_C = \frac{\hbar}{m_e c}. \quad (10)$$

Numerically (D–H): mass shift ≈ 671.007 GHz, measured ≈ 670.994 GHz; the difference (~ 12.7 MHz) is covered by recoil/radiative corrections, while the field contribution is only ~ -0.139 MHz.

5 Antimatter Consistency

Because $E(w_1)$ depends on w_1^2 , $E(+w_1) = E(-w_1)$ and mirror partners are isoen-
ergetic. Thus, as long as S -nodes remain stable (no flips), spectra of hydrogen and
antihydrogen are identical within current experimental precision (parts in 10^{-12} – 10^{-11} ,
depending on run), consistent with CPT tests.

6 Discussion and Outlook

The MMU offers a single geometric mechanism covering:

- Small mirror approximations $\rightarrow$ contact-like spectroscopic shifts (VP/Lamb),
 $\propto |\psi(0)|^2$ and μ_{red}^3 .
- Critical approximations $\rightarrow$ S -node flips (beta decay), with Eyring–Kramers
statistics.
- Extreme, collective approximations $\rightarrow$ annihilation/fission or macroscopic trap-
ping (vanishing external w_1).

Next steps: high-sensitivity tests in muonic isotopes (μH , μD), parameter-free ratio
predictions (e.g. n -scalings), and environment-modulated half-lives via w_3/w_4 coupling.

7 Conclusion

The MMU framework (i) yields a clear flip criterion for decay, (ii) derives the expo-
nential law via Eyring–Kramers, (iii) matches the qualitative and quantitative trends
of H vs. μH and H/D/T isotope shifts (with the corrected finite-size factor), and (iv)
ensures matter–antimatter spectral equality through w_1 -mirror symmetry. This re-
casts matter–antimatter symmetry and nuclear decay as geometric consequences of
tetrahedral projections.

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APPENDIX: Mathematical derivation and data comparison (MMU vs. measurements)

First-order setup. Short-range MMU corrections $\delta V_{\text{MMU}}(r)$ contribute to spectra at leading (first) order:

$$\Delta E_{n\ell} = \int |\psi_{n\ell}(r)|^2 \delta V_{\text{MMU}}(r) d^3r \Rightarrow \Delta E_{nS} - \Delta E_{nP} \simeq |\psi_{nS}(0)|^2 \int \delta V_{\text{MMU}} d^3r, \quad (11)$$

with

$$|\psi_{nS}(0)|^2 = \frac{1}{\pi} \left(\frac{Z\alpha \mu_{\text{red}} c}{n\hbar} \right)^3 \Rightarrow \Delta E_{nS-nP} \propto \mu_{\text{red}}^3 / n^3. \quad (12)$$

Hydrogen vs. muonic hydrogen. For $Z = 1$, $n = 2$:

$$\left(\frac{\mu_{\mu\text{H}}}{\mu_{\text{H}}} \right)^3 \approx 6.428842 \times 10^6.$$

Using $\Delta E_{2P-2S}^{(\text{Ueh})}(\mu\text{H}) \approx 205.0282 \text{ meV}$, a naive cubic projection to H gives

$$\Delta \nu_{2S-2P}^{(\text{Ueh})}(\text{H}) \approx 7.71 \text{ MHz},$$

while full QED yields $\sim 27.1 \text{ MHz}$ (VP part) and a total Lamb of $\sim 1057.844 \text{ MHz}$. The order-of-magnitude match is expected; the residual factor stems from the finite range (form factor) of δV_{MMU} in H.

Isotope shifts (1S–2S).

$$\nu_{1S \rightarrow 2S}^{(0)}(M) = \frac{3}{4} c R_{\infty} \frac{\mu_{\text{red}}}{m_e}, \quad \Delta \nu_{\text{field}} = -\frac{7}{12} \alpha^4 c R_{\infty} \left(\frac{\mu_{\text{red}}}{m_e} \right)^3 \frac{\langle r^2 \rangle}{\lambda_C^2}. \quad (13)$$

With $r_p = 0.84075 \text{ fm}$, $r_d = 2.12778 \text{ fm}$, $r_t \simeq 1.759 \text{ fm}$:

$$\Delta \nu_{\text{mass}}(\text{D–H}) \approx 671.007\,040\,359\,5 \text{ GHz}, \quad \Delta[\Delta \nu_{\text{field}}]_{\text{D–H}} \approx -0.1393 \text{ MHz}, \quad \Delta[\Delta \nu_{\text{field}}]_{\text{T–H}} \approx -0.0871 \text{ MHz}$$

Measured D–H: $670.994\,334\,606 \text{ GHz}$; the $\sim 12.706 \text{ MHz}$ difference is principally recoil/radiative, consistent with the tiny FNS level.

Half-lives from the S -node barrier. A quartic $V(\sigma; \mu)$ has barrier $\Delta E = a_{\text{eff}}^2/(4b)$; Eyring–Kramers gives

$$k(\mu) \simeq \frac{\omega_0 \omega_b}{2\pi\Gamma} e^{-\Delta E/D}, \quad T_{1/2} = \frac{\ln 2}{k}.$$

Using Q as an empirical proxy for barrier lowering,

$$k(Q) = k_0 e^{\beta Q}, \quad T_{1/2}(Q) = \frac{\ln 2}{k_0 e^{\beta Q}}.$$

Calibrating with neutron ($Q = 0.782 \text{ MeV}$, $T_{1/2} = 880.2 \text{ s}$) and tritium ($Q = 0.01859 \text{ MeV}$, $T_{1/2} = 12.32 \text{ yr}$) yields

$$\beta \approx 17.027 \text{ MeV}^{-1}, \quad k_0 \approx 1.299 \times 10^{-9} \text{ s}^{-1},$$

which reproduces both half-lives (up to rounding) and implies stability for ^1H , ^2H ($Q \leq 0$).

APPENDIX Mathematical derivation and data comparison (MMU vs. measurements)

Setup. In the MMU, short-range “mirror approximation” effects enter atomic spectra as a contact-like perturbation $\delta V_{\text{MMU}}(r)$. To leading order (first-order perturbation theory),

$$\Delta E_{n\ell} = \int |\psi_{n\ell}(r)|^2 \delta V_{\text{MMU}}(r) d^3r \quad \Rightarrow \quad \Delta E_{nS} - \Delta E_{nP} \simeq |\psi_{nS}(0)|^2 \int \delta V_{\text{MMU}} d^3r, \quad (14)$$

because $|\psi_{nP}(0)| = 0$. For hydrogenic S-states

$$|\psi_{nS}(0)|^2 = \frac{1}{\pi} \left(\frac{Z\alpha\mu c}{n\hbar} \right)^3, \quad \Rightarrow \quad \Delta E_{nS-nP} \propto \mu^3/n^3. \quad (15)$$

Here μ is the electron–nucleus reduced mass, α the fine-structure constant.

A) Hydrogen vs. muonic hydrogen (small mirror approximation). For $Z = 1, n = 2$ one obtains the *reduced-mass cubic scaling*

$$\left(\frac{\mu_{\mu\text{H}}}{\mu_{\text{H}}} \right)^3 \approx 6.428842 \times 10^6.$$

Using the known first-order Uehling shift in muonic hydrogen,

$$\Delta E_{2P-2S}^{(\text{Ueh})}(\mu\text{H}) \approx 205.0282 \text{ meV},$$

a naive projection to hydrogen via (15) yields

$$\Delta \nu_{2S-2P}^{(\text{Ueh})}(\text{H}) \approx \frac{205.0282 \text{ meV}}{6.428842 \times 10^6} \times \frac{1}{h} \approx 7.71 \text{ MHz}. \quad (16)$$

Measured/theoretical reference in H: the Uehling (vacuum-polarization) contribution is ~ 27.1 MHz (magnitude), while the *total* Lamb shift is ~ 1057.844 MHz. The naive MMU scaling (16) reproduces the correct *order of magnitude*; the remaining factor (~ 3.5) is explained by the finite range of δV : $\int \delta V_{\text{MMU}} d^3r \rightarrow \int f(r) \delta V_{\text{MMU}}(r) d^3r$ with a form factor $f(r)$ that is *important* in H but largely saturated in μH .

B) Isotope shifts $1S \rightarrow 2S$ (H/D/T). The gross (Dirac) frequency scales as

$$\nu_{1S \rightarrow 2S}^{(0)}(M) = \frac{3}{4} cR_{\infty} \frac{\mu}{m_e}, \quad \Delta \nu_{\text{mass}}(X_2 - X_1) = \frac{3}{4} cR_{\infty} \left(\frac{\mu_2 - \mu_1}{m_e} \right). \quad (17)$$

The leading finite-size (FNS) term for the transition is

$$\boxed{\Delta \nu_{\text{field}} = -\frac{7}{12} \alpha^4 cR_{\infty} \left(\frac{\mu}{m_e} \right)^3 \frac{\langle r^2 \rangle}{\lambda_C^2}}, \quad \lambda_C = \frac{\hbar}{m_e c}. \quad (18)$$

Numerics (constants): $cR_{\infty} = 3.2898419602508 \times 10^{15} \text{ Hz}$, $\alpha^{-1} = 137.035999084$, $r_p = 0.84075 \text{ fm}$, $r_d = 2.12778 \text{ fm}$, $r_t \simeq 1.759 \text{ fm}$. *Results:*

$$\begin{aligned} \Delta \nu_{\text{mass}}(\text{D-H}) &\approx 671.0070403595 \text{ GHz}, \\ \Delta \nu_{\text{field}}(\text{D}) - \Delta \nu_{\text{field}}(\text{H}) &\approx -0.1393 \text{ MHz}, \\ \Delta \nu_{\text{field}}(\text{T}) - \Delta \nu_{\text{field}}(\text{H}) &\approx -0.0871 \text{ MHz}. \end{aligned}$$

Measurement for D–H: 670.994334606 GHz. Hence the difference between mass-only and data, ~ 12.706 MHz, is (as known) dominated by recoil/radiative terms, while the FNS piece is at the expected $\mathcal{O}(10^2 \text{ kHz})$ level. In the MMU picture, any intrinsic w_4 -driven proton “compression” $r_{p,\text{eff}}^2 = r_p^2 - sN$ would shift (18) by $\Delta\nu_{\text{field}} \rightarrow \Delta\nu_{\text{field}} + K(\mu/m_e)^3(-sN)$ with $K = -(7/12)\alpha^4 c R_\infty / \lambda_C^2 \approx -36.49 \text{ Hz/fm}^2$; for electronic isotopes this is Hz–kHz–small (consistent with data).

C) Half-lives from S -node barrier (MMU $\rightarrow$ Kramers). With a quartic S -node potential $V(\sigma; \mu) = a_{\text{eff}}(\mu)\sigma^2 + b\sigma^4$, the barrier height is $\Delta E = a_{\text{eff}}^2/(4b)$. Eyring–Kramers gives the escape rate

$$k(\mu) \simeq \frac{\omega_0 \omega_b}{2\pi\Gamma} \exp\left(-\frac{\Delta E}{D}\right), \quad T_{1/2} = \frac{\ln 2}{k}. \quad (19)$$

For phenomenology in the hydrogen family it is convenient to use the β -decay Q -value as a proxy for the mirror-approximation (barrier lowering) and write the two-parameter law

$$k(Q) = k_0 e^{\beta Q}, \quad T_{1/2} = \frac{\ln 2}{k(Q)}. \quad (20)$$

Calibration (two points): neutron $Q = 0.782 \text{ MeV}$, $T_{1/2} = 880.2 \text{ s}$; tritium $Q = 18.59 \text{ keV}$, $T_{1/2} = 12.32 \text{ yr}$. This yields

$$\beta \approx 17.03 \text{ MeV}^{-1}, \quad k_0 \approx 1.30 \times 10^{-9} \text{ s}^{-1}.$$

Predictions: ^1H and ^2H : $Q \leq 0$ (forbidden) $\Rightarrow$ stable; for hypothetical tiny Q (keV-scale) the model gives decade-to-century half-lives (e.g. $Q = 1 \text{ keV} \Rightarrow T_{1/2} \approx 5.25 \times 10^8 \text{ s} \approx 16.6 \text{ yr}$), consistent with a large barrier at small Q .

Neutron & Tritium: Calibration and Verification of the Formula $k(Q) = k_0 e^{\beta Q}$

Starting point. For a decay with half-life $T_{1/2}$, the rate is

$$k = \frac{\ln 2}{T_{1/2}}.$$

We use two calibration points from the hydrogen family:

$$\text{Neutron: } Q_n = 0.782 \text{ MeV}, \quad T_{1/2}(n) = 880.2 \text{ s},$$

$$\text{Tritium: } Q_T = 18.59 \text{ keV} = 0.01859 \text{ MeV}, \quad T_{1/2}(^3\text{H}) = 12.32 \text{ yr}.$$

Hence

$$k_n = \frac{\ln 2}{880.2 \text{ s}} = 7.876 \times 10^{-4} \text{ s}^{-1}, \quad k_T = \frac{\ln 2}{12.32 \text{ yr}} = 1.782 \times 10^{-9} \text{ s}^{-1},$$

(where $1 \text{ yr} = 365.25 \times 24 \times 3600 \text{ s}$).

Two-point fit. With the ansatz

$$k(Q) = k_0 e^{\beta Q} \implies \ln k(Q) = \ln k_0 + \beta Q,$$

the two points give

$$\beta = \frac{\ln k_n - \ln k_T}{Q_n - Q_T}, \quad \ln k_0 = \ln k_n - \beta Q_n.$$

Inserting the numbers yields

$$\boxed{\beta = 17.027 \text{ MeV}^{-1}, \quad k_0 = 1.299 \times 10^{-9} \text{ s}^{-1}}$$

(rounded to 3 significant figures).

Back-substitution (verification). With these parameters:

$$k(Q_n) = k_0 e^{\beta Q_n} = 7.876 \times 10^{-4} \text{ s}^{-1} \implies T_{1/2}(n) = \frac{\ln 2}{k(Q_n)} = 880.2 \text{ s},$$

$$k(Q_T) = k_0 e^{\beta Q_T} = 1.782 \times 10^{-9} \text{ s}^{-1} \implies T_{1/2}(^3\text{H}) = \frac{\ln 2}{k(Q_T)} = 12.32 \text{ yr}.$$

Both half-lives are thus reproduced exactly (up to rounding).

Prediction: Stability of ^1H and ^2H . For processes with $Q \leq 0$, β decay is energetically forbidden, and the minimal ansatz yields $k(Q \leq 0) = 0$ (no barrier crossing $\implies$ stability). This is consistent with the stability of protium (^1H) and deuterium (^2H).

MMU interpretation. In the MMU, Q quantifies the *mirror approximation* that lowers the S -node barrier ΔE . The Eyring–Kramers relation $k \propto \exp[-\Delta E/D]$ together with a linearized barrier $\Delta E(Q) = A - BQ$ leads directly to $k(Q) = k_0 e^{\beta Q}$ with $\beta = B/D$ and $k_0 = \frac{\omega_0 \omega_b}{2\pi\Gamma} e^{-A/D}$.

D) Summary (what is explained quantitatively).

- Eq. (15) explains the *huge* μH 2S–2P shift vs. the *tiny* H shift: $\propto \mu^3$ with finite-range corrections (MMU kernel form factor).
- Eqs. (17)–(18) reproduce the observed $\text{H}/\text{D}/\text{T}$ 1S–2S pattern: mass shift dominates; FNS is Hz–kHz–small (numbers above).
- Eq. (20) (rooted in (19)) reproduces $T_{1/2}(n)$ and $T_{1/2}(^3\text{H})$ and explains stability of ^1H , ^2H via $Q \leq 0$.

All steps are directly tied to the MMU geometry (mirror approximation, S -node barrier, w_3/w_4 coupling) while matching established measurements at the correct scales.

Appendix 12: Lepton Staging in the MMU: Masses, Hydrogenic Spectra, and $(g - 2)$ from Tetrahedral Geometry

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Abstract

In the Methane Metauniverse (MMU) the charged leptons (e, μ, τ) are one and the same object realized at three geometric embedding stages of a tetrahedral lattice: electron $= \mathcal{H}_1\{1\}$, muon $= \mathcal{H}_2\{4\}$, tau $= \mathcal{H}_3\{12\}$. Their phases in the (w_2, w_3) torsional plane are separated by 120° (a sub-tetrahedron). We derive the lepton mass formula, obtain Koide as a theorem, predict hydrogenic spectral shifts (Rydberg scaling with the reduced mass), give numerical values for Lyman- α and Balmer- α in eH, μ H and “ τ H”, and provide an MMU correction pattern for $(g - 2)$ with parameter-free family ratios.

1 Geometry: three resonances from a sub-tetrahedron

The S -anchor has a radial depth w_4 and tangential torsion (w_2, w_3) . The first three hierarchy stages of the tetrahedral lattice,

$$\mathcal{H}_1 = \{1\}, \quad \mathcal{H}_2 = \{4\}, \quad \mathcal{H}_3 = \{12\},$$

select three equally spaced phases on a torsional circle:

$$\theta_i = \phi + \frac{2\pi}{3}i, \quad i = 0, 1, 2, \quad (1)$$

to be identified with (e, μ, τ) . Thus three generations arise geometrically from the 120° sub-tetrahedron.

2 Lepton masses and Koide from projection

A quadratic effective energy,

$$\mathcal{E} = A w_4^2 + B (w_2^2 + w_3^2) + 2C w_4 (w_2 \cos \theta + w_3 \sin \theta), \quad (2)$$

after eliminating fast tangential modes, yields the mass spectrum

$$\boxed{\sqrt{m_i} = \sqrt{M} \left[1 + \sqrt{2} \cos \theta_i \right]} \quad \Longleftrightarrow \quad m_i = M \left[1 + \sqrt{2} \cos \theta_i \right]^2, \quad (3)$$

with scale M fixed by the w_4 -depth. For 120° phases, $\sum_i \sqrt{m_i} = 3\sqrt{M}$ and $\sum_i m_i = 6M$, hence

$$Q \equiv \frac{m_e + m_\mu + m_\tau}{\left(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau}\right)^2} = \frac{2}{3}, \quad (4)$$

i.e. Koide follows geometrically.

3 Numbers for the masses

We use (MeV): $m_e = 0.51099895$, $m_\mu = 105.6583755$, $m_\tau = 1776.86$. From (3):

$$M = \left(\frac{\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau}}{3} \right)^2 = 313.841\,127 \text{ MeV}, \quad \phi \simeq 132.732\,800^\circ.$$

This reproduces the three lepton masses with ppm-to- 10^{-5} accuracy.

4 Hydrogenic spectra across $e \rightarrow \mu \rightarrow \tau$

For a hydrogenic system (ℓH) with lepton $\ell \in \{e, \mu, \tau\}$ and proton mass $m_p = 938.272\,081$ MeV, the reduced mass is

$$\mu_\ell = \frac{m_\ell m_p}{m_\ell + m_p}. \quad (5)$$

Transitions scale with μ_ℓ (Rydberg scaling), hence

$$\lambda_\ell = \lambda_e \frac{\mu_e}{\mu_\ell}, \quad E_\ell = E_e \frac{\mu_\ell}{\mu_e}. \quad (6)$$

Using the masses above, we get

$$\mu_e = 0.510\,721 \text{ MeV}, \quad \mu_\mu = 94.964\,471 \text{ MeV}, \quad \mu_\tau = 614.032\,055 \text{ MeV},$$

and the ratios

$$\frac{\mu_\mu}{\mu_e} = 185.942\,047, \quad \frac{\mu_\tau}{\mu_e} = 1202.285\,185.$$

Predictions for Lyman- α and Balmer- α . Taking the electronic hydrogen references $\lambda_\alpha^{(e)} = 121.567\,000$ nm (Lyman- α) and $\lambda_{\text{H}\alpha}^{(e)} = 656.280\,000$ nm (Balmer- α), (6) gives:

System	$\lambda_{\text{Ly}\alpha}$ (nm)	$E_{\text{Ly}\alpha}$ (eV)
$e\text{H}$	121.567 000	10.198 837
μH	0.653 790	1896.392 573
" τH "	0.101 113	12 261.910 293
System	$\lambda_{\text{H}\alpha}$ (nm)	$E_{\text{H}\alpha}$ (eV)
$e\text{H}$	656.280 000	1.889 197
μH	3.529 487	351.281 093
" τH "	0.545 861	2271.353 155

Interpretation: already Lyman- α moves from UV (eH) to soft X-ray (μ H) and hard X-ray (“ τ H”). The scaling is purely geometric via the S -depth (effective μ_ℓ).

5 MMU pattern for the anomalous moments ($g-2$)

The same sub-tetrahedral phase controls a family-dependent geometric correction:

$$a_i^{\text{MMU}} \equiv \frac{g_i - 2}{2} \Big|_{\text{MMU}} = \beta \left(\frac{m_i}{M} \right)^2 \sin^2(2\theta_i) + \mathcal{O}((m_i/M)^4), \quad \theta_i = \phi + \frac{2\pi}{3}i. \quad (7)$$

With $\phi \simeq 132.732\,800^\circ$:

$$\sin^2(2\theta_e) = 0.993\,750, \quad \sin^2(2\theta_\mu) = 0.321\,377, \quad \sin^2(2\theta_\tau) = 0.184\,873.$$

Therefore the *ratios* are parameter-free:

$$\boxed{\frac{\Delta a_\tau}{\Delta a_\mu} \approx 162.689\,200, \quad \frac{\Delta a_e}{\Delta a_\mu} \approx 7.232\,600 \times 10^{-5}}.$$

Calibrating a single dimensionless β on a given Δa_μ (e.g. $\Delta a_\mu = 2.5 \times 10^{-9}$) yields

$$\beta \simeq 6.863\,370 \times 10^{-8}, \quad \Delta a_e \approx 1.808\,150 \times 10^{-13}, \quad \Delta a_\tau \approx 4.067\,230 \times 10^{-7}.$$

Thus the μ anomaly fixes *both* e and τ shifts without extra freedom.

6 Discussion and outlook

(i) The *same* geometry explains: three lepton families, Koide, hydrogenic spectral shifts across $e \rightarrow \mu \rightarrow \tau$, and a family pattern for ($g-2$). (ii) Spectroscopy: the numbers above follow purely from the reduced mass implied by the S -depth (no fits). Fine effects (FS/HFS, proton radius, vacuum polarization) enter as small torsional corrections in MMU. (iii) ($g-2$): Eq. (7) encodes a clean, testable signature in the family ratios; once Δa_μ is fixed, Δa_e is tiny while Δa_τ is potentially measurable in future colliders.